

NCIT SIS - Line-by-Line Code Explanation

Each row explains one line of source code in plain English.

File: app.py (Lines: 1788)

Line	Code	Explanation
1	from flask import Flask, render_template, request, redirect, url_for, session, flash	Import statement: loads modules or symbols needed in this file.
2	from werkzeug.security import generate_password_hash, check_password_hash	Import statement: loads modules or symbols needed in this file.
3	from dotenv import load_dotenv	Import statement: loads modules or symbols needed in this file.
4	import mysql.connector	Import statement: loads modules or symbols needed in this file.
5	from datetime import date, datetime, timedelta	Import statement: loads modules or symbols needed in this file.
6	import os	Import statement: loads modules or symbols needed in this file.
7		Blank line used to separate logical blocks of Python code.
8	load_dotenv()	Function/method call that performs an action at runtime.
9		Blank line used to separate logical blocks of Python code.
10	app = Flask(__name__)	Assignment statement: stores the expression result in a variable/field.
11	app.secret_key = os.getenv("SECRET_KEY", "dev_secret_key")	Assignment statement: stores the expression result in a variable/field.
12		Blank line used to separate logical blocks of Python code.
13	# =====	Comment line used to document intent or create section dividers.
14	# DATABASE CONNECTION	Comment line used to document intent or create section dividers.
15	# =====	Comment line used to document intent or create section dividers.
16		Blank line used to separate logical blocks of Python code.
17	def get_db():	Function definition `get_db` with no parameters.
18	return mysql.connector.connect(	Returns a value (or nothing) from the current function.
19	host=os.getenv("DB_HOST", "localhost"),	Assignment statement: stores the expression result in a variable/field.
20	user=os.getenv("DB_USER", "root"),	Assignment statement: stores the expression result in a variable/field.
21	password=os.getenv("DB_PASSWORD", ""),	Assignment statement: stores the expression result in a variable/field.
22	database=os.getenv("DB_NAME", "ncit_sis"),	Assignment statement: stores the expression result in a variable/field.
23	port=int(os.getenv("DB_PORT", "3306"))	Assignment statement: stores the expression result in a variable/field.
24	)	General Python statement that participates in program control flow.
25		Blank line used to separate logical blocks of Python code.
26	# =====	Comment line used to document intent or create section dividers.
27	# HELPER FUNCTIONS	Comment line used to document intent or create section dividers.
28	# =====	Comment line used to document intent or create section dividers.
29		Blank line used to separate logical blocks of Python code.
30	def is_password_hashed(value):	Function definition `is_password_hashed` with value.
31	if not value:	Starts an `if` block that runs only when its condition is true.
32	return False	Returns a value (or nothing) from the current function.
33	return (	Returns a value (or nothing) from the current function.

Line	Code	Explanation
34	value.startswith("pbkdf2:")	General Python statement that participates in program control flow.
35	or value.startswith("scrypt:")	General Python statement that participates in program control flow.
36	or value.startswith("argon2:")	General Python statement that participates in program control flow.
37	)	General Python statement that participates in program control flow.
38		Blank line used to separate logical blocks of Python code.
39	def verify_password(stored, provided):	Function definition `verify_password` with stored, provided.
40	if not stored:	Starts an `if` block that runs only when its condition is true.
41	return False	Returns a value (or nothing) from the current function.
42	if is_password_hashed(stored):	Starts an `if` block that runs only when its condition is true.
43	return check_password_hash(stored, provided)	Returns a value (or nothing) from the current function.
44	return stored == provided	Returns a value (or nothing) from the current function.
45		Blank line used to separate logical blocks of Python code.
46	def check_auth(role_required):	Function definition `check_auth` with role_required.
47	if 'user_id' not in session: return False	General Python statement that participates in program control flow.
48	if session.get('role') != role_required: return False	General Python statement that participates in program control flow.
49	return True	Returns a value (or nothing) from the current function.
50		Blank line used to separate logical blocks of Python code.
51	def get_notices(role):	Function definition `get_notices` with role.
52	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
53	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
54	cur.execute("SELECT * FROM notices WHERE target_role IN ('all', %s) ORDER BY date_posted DESC LIMIT 5", (role,))	Executes a SQL query/command using the active database cursor.
55	notices = cur.fetchall()	Reads query results from the cursor.
56	conn.close()	Closes a resource such as a cursor, connection, or file.
57	return notices	Returns a value (or nothing) from the current function.
58		Blank line used to separate logical blocks of Python code.
59	def calculate_grade_gpa(obtained, full):	Function definition `calculate_grade_gpa` with obtained, full.
60	if full == 0: return 'F', 0.0	General Python statement that participates in program control flow.
61	percentage = (obtained / full) * 100	Assignment statement: stores the expression result in a variable/field.
62		Blank line used to separate logical blocks of Python code.
63	if percentage >= 90: return 'A', 4.0	General Python statement that participates in program control flow.
64	elif percentage >= 85: return 'A-', 3.7	General Python statement that participates in program control flow.
65	elif percentage >= 80: return 'B+', 3.3	General Python statement that participates in program control flow.
66	elif percentage >= 75: return 'B', 3.0	General Python statement that participates in program control flow.
67	elif percentage >= 70: return 'B-', 2.7	General Python statement that participates in program control flow.
68	elif percentage >= 65: return 'C+', 2.3	General Python statement that participates in program control flow.
69	elif percentage >= 60: return 'C', 2.0	General Python statement that participates in program control flow.
70	elif percentage >= 55: return 'C-', 1.7	General Python statement that participates in program control flow.
71	elif percentage >= 50: return 'D+', 1.3	General Python statement that participates in program control flow.
72	elif percentage >= 45: return 'D', 1.0	General Python statement that participates in program control flow.
73	else: return 'F', 0.0	General Python statement that participates in program control flow.

Line	Code	Explanation
74		Blank line used to separate logical blocks of Python code.
75	# =====	Comment line used to document intent or create section dividers.
76	# AUTHENTICATION	Comment line used to document intent or create section dividers.
77	# =====	Comment line used to document intent or create section dividers.
78		Blank line used to separate logical blocks of Python code.
79	@app.route('/')	Flask route decorator: binds the next function to a URL endpoint.
80	def home():	Function definition `home` with no parameters.
81	if 'user_id' in session:	Starts an `if` block that runs only when its condition is true.
82	return redirect(url_for(f"{session['role']}_dashboard"))	Returns a value (or nothing) from the current function.
83	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
84		Blank line used to separate logical blocks of Python code.
85	@app.route('/login', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
86	def login():	Function definition `login` with no parameters.
87	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
88	email = request.form['email']	Assignment statement: stores the expression result in a variable/field.
89	password = request.form['password']	Assignment statement: stores the expression result in a variable/field.
90		Blank line used to separate logical blocks of Python code.
91	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
92	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
93	cur.execute("SELECT user_id, full_name, role, password FROM users WHERE email=%s", (email,))	Executes a SQL query/command using the active database cursor.
94	user = cur.fetchone()	Reads query results from the cursor.
95		Blank line used to separate logical blocks of Python code.
96	if user and verify_password(user['password'], password):	Starts an `if` block that runs only when its condition is true.
97	if not is_password_hashed(user['password']):	Starts an `if` block that runs only when its condition is true.
98	new_hash = generate_password_hash(password)	Assignment statement: stores the expression result in a variable/field.
99	cur.execute("UPDATE users SET password=%s WHERE user_id=%s", (new_hash, user['user_id']))	Executes a SQL query/command using the active database cursor.
100	conn.commit()	Commits the active database transaction.
101	session['user_id'] = user['user_id']	Assignment statement: stores the expression result in a variable/field.
102	session['name'] = user['full_name']	Assignment statement: stores the expression result in a variable/field.
103	session['role'] = user['role']	Assignment statement: stores the expression result in a variable/field.
104	conn.close()	Closes a resource such as a cursor, connection, or file.
105	return redirect(url_for(f"{user['role']}_dashboard"))	Returns a value (or nothing) from the current function.
106	else:	Fallback `else` branch for the current conditional.
107	conn.close()	Closes a resource such as a cursor, connection, or file.
108	flash("Invalid Credentials! Please try again.")	Function/method call that performs an action at runtime.
109		Blank line used to separate logical blocks of Python code.
110	return render_template('login.html')	Returns a value (or nothing) from the current function.
111		Blank line used to separate logical blocks of Python code.
112	@app.route('/logout')	Flask route decorator: binds the next function to a URL endpoint.

Line	Code	Explanation
113	def logout():	Function definition `logout` with no parameters.
114	session.clear()	General Python statement that participates in program control flow.
115	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
116		Blank line used to separate logical blocks of Python code.
117	# =====	Comment line used to document intent or create section dividers.
118	# PROFILE & SETTINGS (COMMON FOR ALL USERS)	Comment line used to document intent or create section dividers.
119	# =====	Comment line used to document intent or create section dividers.
120		Blank line used to separate logical blocks of Python code.
121	@app.route('/profile', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
122	def profile():	Function definition `profile` with no parameters.
123	if 'user_id' not in session:	Starts an `if` block that runs only when its condition is true.
124	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
125		Blank line used to separate logical blocks of Python code.
126	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
127	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
128		Blank line used to separate logical blocks of Python code.
129	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
130	full_name = request.form['full_name'].strip()	Assignment statement: stores the expression result in a variable/field.
131	email = request.form['email'].strip()	Assignment statement: stores the expression result in a variable/field.
132	contact_no = request.form.get('contact_no', '').strip()	Assignment statement: stores the expression result in a variable/field.
133	address = request.form.get('address', '').strip()	Assignment statement: stores the expression result in a variable/field.
134	gender = request.form.get('gender', '').strip()	Assignment statement: stores the expression result in a variable/field.
135	try:	Starts a `try` block for exception-safe execution.
136	cur.execute("""	Executes a SQL query/command using the active database cursor.
137	UPDATE users	General Python statement that participates in program control flow.
138	SET full_name = %s, email = %s, contact_no = %s, address = %s, gender = %s	General Python statement that participates in program control flow.
139	WHERE user_id = %s	General Python statement that participates in program control flow.
140	""", (full_name, email, contact_no, address, gender, session['user_id']))	General Python statement that participates in program control flow.
141	conn.commit()	Commits the active database transaction.
142	session['name'] = full_name	Assignment statement: stores the expression result in a variable/field.
143	flash('Profile updated successfully!', 'success')	Function/method call that performs an action at runtime.
144	except mysql.connector.Error as e:	Exception handler for errors raised in the matching `try` block.
145	conn.rollback()	General Python statement that participates in program control flow.
146	flash(f'Error updating profile: {e}', 'danger')	Function/method call that performs an action at runtime.
147	finally:	`finally` block: cleanup code that runs whether or not errors occur.
148	cur.close()	Closes a resource such as a cursor, connection, or file.
149	conn.close()	Closes a resource such as a cursor, connection, or file.
150		Blank line used to separate logical blocks of Python code.
151	return redirect(url_for('profile'))	Returns a value (or nothing) from the current function.
152		Blank line used to separate logical blocks of Python code.
153	# GET request – fetch user data	Comment line used to document intent or create section dividers.

Line	Code	Explanation
154	<code>cur.execute("SELECT * FROM users WHERE user_id = %s", (session['user_id'],))</code>	Executes a SQL query/command using the active database cursor.
155	<code>user = cur.fetchone()</code>	Reads query results from the cursor.
156	<code>cur.close()</code>	Closes a resource such as a cursor, connection, or file.
157	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
158		Blank line used to separate logical blocks of Python code.
159	<code>if not user:</code>	Starts an `if` block that runs only when its condition is true.
160	<code>flash('User not found.', 'danger')</code>	Function/method call that performs an action at runtime.
161	<code>return redirect(url_for('logout'))</code>	Returns a value (or nothing) from the current function.
162		Blank line used to separate logical blocks of Python code.
163	<code>return render_template('profile.html', user=user)</code>	Returns a value (or nothing) from the current function.
164		Blank line used to separate logical blocks of Python code.
165	<code>@app.route('/settings', methods=['GET', 'POST'])</code>	Flask route decorator: binds the next function to a URL endpoint.
166	<code>def settings():</code>	Function definition `settings` with no parameters.
167	<code>if 'user_id' not in session:</code>	Starts an `if` block that runs only when its condition is true.
168	<code>return redirect(url_for('login'))</code>	Returns a value (or nothing) from the current function.
169		Blank line used to separate logical blocks of Python code.
170	<code>conn = get_db()</code>	Assignment statement: stores the expression result in a variable/field.
171	<code>cur = conn.cursor(dictionary=True)</code>	Assignment statement: stores the expression result in a variable/field.
172		Blank line used to separate logical blocks of Python code.
173	<code>if request.method == 'POST':</code>	Starts an `if` block that runs only when its condition is true.
174	<code>current_pass = request.form['current_password']</code>	Assignment statement: stores the expression result in a variable/field.
175	<code>new_pass = request.form['new_password']</code>	Assignment statement: stores the expression result in a variable/field.
176	<code>confirm_pass = request.form['confirm_password']</code>	Assignment statement: stores the expression result in a variable/field.
177		Blank line used to separate logical blocks of Python code.
178	<code>cur.execute("SELECT password FROM users WHERE user_id = %s", (session['user_id'],))</code>	Executes a SQL query/command using the active database cursor.
179	<code>stored = cur.fetchone()</code>	Reads query results from the cursor.
180		Blank line used to separate logical blocks of Python code.
181	<code>if stored and verify_password(stored['password'], current_pass):</code>	Starts an `if` block that runs only when its condition is true.
182	<code>if new_pass == confirm_pass and new_pass.strip() != '':</code>	Starts an `if` block that runs only when its condition is true.
183	<code>try:</code>	Starts a `try` block for exception-safe execution.
184	<code>new_hash = generate_password_hash(new_pass)</code>	Assignment statement: stores the expression result in a variable/field.
185	<code>cur.execute("UPDATE users SET password = %s WHERE user_id = %s", (new_hash, session['user_id']))</code>	Executes a SQL query/command using the active database cursor.
186		General Python statement that participates in program control flow.
187	<code>conn.commit()</code>	Commits the active database transaction.
188	<code>flash('Password changed successfully!', 'success')</code>	Function/method call that performs an action at runtime.
189	<code>except mysql.connector.Error as e:</code>	Exception handler for errors raised in the matching `try` block.
190	<code>conn.rollback()</code>	General Python statement that participates in program control flow.
191	<code>flash(f'Error changing password: {e}', 'danger')</code>	Function/method call that performs an action at runtime.
192	<code>else:</code>	Fallback `else` branch for the current conditional.
193	<code>flash('New passwords do not match or are empty!', 'danger')</code>	Function/method call that performs an action at runtime.

Line	Code	Explanation
194	else:	Fallback `else` branch for the current conditional.
195	flash('Incorrect current password!', 'danger')	Function/method call that performs an action at runtime.
196		Blank line used to separate logical blocks of Python code.
197	cur.close()	Closes a resource such as a cursor, connection, or file.
198	conn.close()	Closes a resource such as a cursor, connection, or file.
199	return redirect(url_for('settings'))	Returns a value (or nothing) from the current function.
200		Blank line used to separate logical blocks of Python code.
201	cur.close()	Closes a resource such as a cursor, connection, or file.
202	conn.close()	Closes a resource such as a cursor, connection, or file.
203	return render_template('settings.html')	Returns a value (or nothing) from the current function.
204		Blank line used to separate logical blocks of Python code.
205	# =====	Comment line used to document intent or create section dividers.
206	# ADMIN DASHBOARD	Comment line used to document intent or create section dividers.
207	# =====	Comment line used to document intent or create section dividers.
208	@app.route('/admin')	Flask route decorator: binds the next function to a URL endpoint.
209	def admin_dashboard():	Function definition `admin_dashboard` with no parameters.
210	if not check_auth('admin'):	Starts an `if` block that runs only when its condition is true.
211	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
212		Blank line used to separate logical blocks of Python code.
213	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
214	cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
215		Blank line used to separate logical blocks of Python code.
216	# Dashboard statistics	Comment line used to document intent or create section dividers.
217	stats = {}	Assignment statement: stores the expression result in a variable/field.
218	queries = {	Assignment statement: stores the expression result in a variable/field.
219	's': "SELECT COUNT(*) FROM users WHERE role='student'",	General Python statement that participates in program control flow.
220	't': "SELECT COUNT(*) FROM users WHERE role='teacher'",	General Python statement that participates in program control flow.
221	'd': "SELECT COUNT(*) FROM departments",	General Python statement that participates in program control flow.
222	'c': "SELECT COUNT(*) FROM courses",	General Python statement that participates in program control flow.
223	'f': "SELECT COUNT(*) FROM fees WHERE status='Pending'",	General Python statement that participates in program control flow.
224	'e': "SELECT COUNT(*) FROM exam_schedule"	General Python statement that participates in program control flow.
225	}	General Python statement that participates in program control flow.
226		Blank line used to separate logical blocks of Python code.
227	for k, q in queries.items():	Starts a `for` loop over an iterable sequence.
228	cur.execute(q)	Executes a SQL query/command using the active database cursor.
229	stats[k] = cur.fetchone()[0]	Reads query results from the cursor.
230		Blank line used to separate logical blocks of Python code.
231	# Fetch recent notices (tuple format)	Comment line used to document intent or create section dividers.
232	cur.execute("""	Executes a SQL query/command using the active database cursor.
233	SELECT notice_id, title, content	General Python statement that participates in program control flow.
234	FROM notices	General Python statement that participates in program control flow.

Line	Code	Explanation
235	ORDER BY date_posted DESC	General Python statement that participates in program control flow.
236	LIMIT 3	General Python statement that participates in program control flow.
237	""")	General Python statement that participates in program control flow.
238	notices = cur.fetchall()	Reads query results from the cursor.
239		Blank line used to separate logical blocks of Python code.
240	conn.close()	Closes a resource such as a cursor, connection, or file.
241		Blank line used to separate logical blocks of Python code.
242	return render_template(	Returns a value (or nothing) from the current function.
243	'admin_dash.html',	General Python statement that participates in program control flow.
244	stats=stats,	Assignment statement: stores the expression result in a variable/field.
245	notices=notices	Assignment statement: stores the expression result in a variable/field.
246	)	General Python statement that participates in program control flow.
247		Blank line used to separate logical blocks of Python code.
248	# =====	Comment line used to document intent or create section dividers.
249	# ADMIN LIBRARY MODULE	Comment line used to document intent or create section dividers.
250	# =====	Comment line used to document intent or create section dividers.
251		Blank line used to separate logical blocks of Python code.
252	@app.route('/admin/library', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
253	def admin_library():	Function definition `admin_library` with no parameters.
254	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
255		Blank line used to separate logical blocks of Python code.
256	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
257	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
258	edit_book = None	Assignment statement: stores the expression result in a variable/field.
259		Blank line used to separate logical blocks of Python code.
260	try:	Starts a `try` block for exception-safe execution.
261	# --- 1. Handle POST Actions (Delete Book) ---	Comment line used to document intent or create section dividers.
262	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
263	if 'delete_book' in request.form:	Starts an `if` block that runs only when its condition is true.
264	book_id_del = request.form['book_id']	Assignment statement: stores the expression result in a variable/field.
265	# Check if currently borrowed (unreturned)	Comment line used to document intent or create section dividers.
266	cur.execute("SELECT * FROM borrows WHERE book_id = %s AND return_date IS NULL", (book_id_del,))	Executes a SQL query/command using the active database cursor.
267	if cur.fetchone():	Starts an `if` block that runs only when its condition is true.
268	flash("Cannot delete: Book is currently borrowed.", "danger")	Function/method call that performs an action at runtime.
269	else:	Fallback `else` branch for the current conditional.
270	cur.execute("DELETE FROM library_books WHERE book_id = %s", (book_id_del,))	Executes a SQL query/command using the active database cursor.
271	conn.commit()	Commits the active database transaction.
272	flash("Book deleted successfully.", "success")	Function/method call that performs an action at runtime.
273	# Redirect to clear POST data so refresh doesn't re-submit	Comment line used to document intent or create section dividers.
274	return redirect(url_for('admin_library'))	Returns a value (or nothing) from the current function.

Line	Code	Explanation
275		Blank line used to separate logical blocks of Python code.
276	# --- 2. Search Logic ---	Comment line used to document intent or create section dividers.
277	search_query = request.args.get('search_query')	Assignment statement: stores the expression result in a variable/field.
278	edit_book_id = request.args.get('edit_book_id', type=int)	Assignment statement: stores the expression result in a variable/field.
279		Blank line used to separate logical blocks of Python code.
280	if edit_book_id:	Starts an `if` block that runs only when its condition is true.
281	cur.execute("""	Executes a SQL query/command using the active database cursor.
282	SELECT book_id, title, author, category_id, isbn, copies_total	General Python statement that participates in program control flow.
283	FROM library_books	General Python statement that participates in program control flow.
284	WHERE book_id = %s	General Python statement that participates in program control flow.
285	""", (edit_book_id,))	General Python statement that participates in program control flow.
286	edit_book = cur.fetchone()	Reads query results from the cursor.
287	if not edit_book:	Starts an `if` block that runs only when its condition is true.
288	flash("Book not found for editing.", "danger")	Function/method call that performs an action at runtime.
289		Blank line used to separate logical blocks of Python code.
290	# --- 3. Fetch Categories (Always fetch all for the "Add Book" dropdown) ---	Comment line used to document intent or create section dividers.
291	cur.execute("SELECT * FROM book_categories ORDER BY name")	Executes a SQL query/command using the active database cursor.
292	categories = cur.fetchall()	Reads query results from the cursor.
293		Blank line used to separate logical blocks of Python code.
294	# --- 4. Fetch Books (Filtered) ---	Comment line used to document intent or create section dividers.
295	books_sql = ""	Assignment statement: stores the expression result in a variable/field.
296	SELECT b.book_id, b.title, b.author, c.name as category_name, b.isbn, b.copies_total	General Python statement that participates in program control flow.
297	FROM library_books b	General Python statement that participates in program control flow.
298	LEFT JOIN book_categories c ON b.category_id = c.category_id	General Python statement that participates in program control flow.
299	""	General Python statement that participates in program control flow.
300		Blank line used to separate logical blocks of Python code.
301	if search_query:	Starts an `if` block that runs only when its condition is true.
302	books_sql += ""	General Python statement that participates in program control flow.
303	WHERE b.title LIKE %s OR b.author LIKE %s OR c.name LIKE %s OR b.isbn LIKE %s	General Python statement that participates in program control flow.
304	""	General Python statement that participates in program control flow.
305	wc = f"%{search_query}%"	Assignment statement: stores the expression result in a variable/field.
306	params = (wc, wc, wc, wc)	Assignment statement: stores the expression result in a variable/field.
307	cur.execute(books_sql + " ORDER BY b.book_id DESC", params)	Executes a SQL query/command using the active database cursor.
308	else:	Fallback `else` branch for the current conditional.
309	cur.execute(books_sql + " ORDER BY b.book_id DESC")	Executes a SQL query/command using the active database cursor.
310		Blank line used to separate logical blocks of Python code.
311	books = cur.fetchall()	Reads query results from the cursor.
312		Blank line used to separate logical blocks of Python code.
313	# --- 5. Calculate Availability ---	Comment line used to document intent or create section dividers.
314	# We calculate this only for the books we fetched (whether filtered or all)	Comment line used to document intent or create section dividers.

Line	Code	Explanation
315	for book in books:	Starts a `for` loop over an iterable sequence.
316	cur.execute("""	Executes a SQL query/command using the active database cursor.
317	SELECT COUNT(*) as borrowed_count	General Python statement that participates in program control flow.
318	FROM borrows	General Python statement that participates in program control flow.
319	WHERE book_id = %s AND return_date IS NULL	General Python statement that participates in program control flow.
320	""", (book['book_id'],))	General Python statement that participates in program control flow.
321	borrowed = cur.fetchone()['borrowed_count']	Reads query results from the cursor.
322		Blank line used to separate logical blocks of Python code.
323	total = book['copies_total'] if book['copies_total'] else 0	Assignment statement: stores the expression result in a variable/field.
324	book['copies_available'] = total - borrowed	Assignment statement: stores the expression result in a variable/field.
325		Blank line used to separate logical blocks of Python code.
326	# --- 6. Fetch Borrow History (Filtered) ---	Comment line used to document intent or create section dividers.
327	borrows_sql = """	Assignment statement: stores the expression result in a variable/field.
328	SELECT br.*, b.title as book_title, u.full_name as student_name, u.semester	General Python statement that participates in program control flow.
329	FROM borrows br	General Python statement that participates in program control flow.
330	JOIN library_books b ON br.book_id = b.book_id	General Python statement that participates in program control flow.
331	JOIN users u ON br.student_id = u.user_id	General Python statement that participates in program control flow.
332	"""	General Python statement that participates in program control flow.
333		Blank line used to separate logical blocks of Python code.
334	if search_query:	Starts an `if` block that runs only when its condition is true.
335	borrows_sql += """	General Python statement that participates in program control flow.
336	WHERE b.title LIKE %s OR u.full_name LIKE %s	General Python statement that participates in program control flow.
337	"""	General Python statement that participates in program control flow.
338	wc = f"%{search_query}%"	Assignment statement: stores the expression result in a variable/field.
339	cur.execute(borrows_sql + " ORDER BY br.borrow_date DESC", (wc, wc))	Executes a SQL query/command using the active database cursor.
340	else:	Fallback `else` branch for the current conditional.
341	cur.execute(borrows_sql + " ORDER BY br.borrow_date DESC")	Executes a SQL query/command using the active database cursor.
342		Blank line used to separate logical blocks of Python code.
343	borrows = cur.fetchall()	Reads query results from the cursor.
344		Blank line used to separate logical blocks of Python code.
345	except Exception as e:	Exception handler for errors raised in the matching `try` block.
346	print(f"LIBRARY ERROR: {e}")	Function/method call that performs an action at runtime.
347	flash(f"System Error: {e}", "danger")	Function/method call that performs an action at runtime.
348	categories = []	Assignment statement: stores the expression result in a variable/field.
349	books = []	Assignment statement: stores the expression result in a variable/field.
350	borrows = []	Assignment statement: stores the expression result in a variable/field.
351	edit_book = None	Assignment statement: stores the expression result in a variable/field.
352		Blank line used to separate logical blocks of Python code.
353	finally:	`finally` block: cleanup code that runs whether or not errors occur.
354	cur.close()	Closes a resource such as a cursor, connection, or file.
355	conn.close()	Closes a resource such as a cursor, connection, or file.

Line	Code	Explanation
356		Blank line used to separate logical blocks of Python code.
357	return render_template(	Returns a value (or nothing) from the current function.
358	'admin_library.html',	General Python statement that participates in program control flow.
359	categories=categories,	Assignment statement: stores the expression result in a variable/field.
360	books=books,	Assignment statement: stores the expression result in a variable/field.
361	borrows=borrows,	Assignment statement: stores the expression result in a variable/field.
362	edit_book=edit_book	Assignment statement: stores the expression result in a variable/field.
363	)	General Python statement that participates in program control flow.
364		Blank line used to separate logical blocks of Python code.
365	@app.route('/admin/library/add_category', methods=['POST'])	Flask route decorator: binds the next function to a URL endpoint.
366	def add_category():	Function definition `add_category` with no parameters.
367	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
368	name = request.form.get('name')	Assignment statement: stores the expression result in a variable/field.
369	if name:	Starts an `if` block that runs only when its condition is true.
370	try:	Starts a `try` block for exception-safe execution.
371	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
372	cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
373	cur.execute("INSERT INTO book_categories (name) VALUES (%s)", (name,))	Executes a SQL query/command using the active database cursor.
374	conn.commit()	Commits the active database transaction.
375	conn.close()	Closes a resource such as a cursor, connection, or file.
376	flash('Category added successfully!', 'success')	Function/method call that performs an action at runtime.
377	except mysql.connector.Error:	Exception handler for errors raised in the matching `try` block.
378	flash('Category already exists!', 'danger')	Function/method call that performs an action at runtime.
379	return redirect(url_for('admin_library'))	Returns a value (or nothing) from the current function.
380		Blank line used to separate logical blocks of Python code.
381	@app.route('/admin/library/add_book', methods=['POST'])	Flask route decorator: binds the next function to a URL endpoint.
382	def add_book():	Function definition `add_book` with no parameters.
383	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
384		Blank line used to separate logical blocks of Python code.
385	book_id = request.form.get('book_id')	Assignment statement: stores the expression result in a variable/field.
386	title = request.form.get('title')	Assignment statement: stores the expression result in a variable/field.
387	author = request.form.get('author')	Assignment statement: stores the expression result in a variable/field.
388	category_id = request.form.get('category')	Assignment statement: stores the expression result in a variable/field.
389	isbn = request.form.get('isbn')	Assignment statement: stores the expression result in a variable/field.
390	copies = request.form.get('copies')	Assignment statement: stores the expression result in a variable/field.
391		Blank line used to separate logical blocks of Python code.
392	if title and author and category_id and isbn and copies:	Starts an `if` block that runs only when its condition is true.
393	try:	Starts a `try` block for exception-safe execution.
394	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
395	cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
396	if book_id:	Starts an `if` block that runs only when its condition is true.

Line	Code	Explanation
397	cur.execute("""	Executes a SQL query/command using the active database cursor.
398	UPDATE library_books	General Python statement that participates in program control flow.
399	SET title = %s, author = %s, category_id = %s, isbn = %s, copies_total = %s	General Python statement that participates in program control flow.
400	WHERE book_id = %s	General Python statement that participates in program control flow.
401	""", (title, author, category_id, isbn, copies, book_id))	General Python statement that participates in program control flow.
402	flash('Book updated successfully!', 'success')	Function/method call that performs an action at runtime.
403	else:	Fallback `else` branch for the current conditional.
404	cur.execute("""	Executes a SQL query/command using the active database cursor.
405	INSERT INTO library_books (title, author, category_id, isbn, copies_total)	General Python statement that participates in program control flow.
406	VALUES (%s, %s, %s, %s, %s)	General Python statement that participates in program control flow.
407	""", (title, author, category_id, isbn, copies))	General Python statement that participates in program control flow.
408	flash('Book added successfully!', 'success')	Function/method call that performs an action at runtime.
409	conn.commit()	Commits the active database transaction.
410	conn.close()	Closes a resource such as a cursor, connection, or file.
411	except mysql.connector.Error as err:	Exception handler for errors raised in the matching `try` block.
412	flash(f'Error adding book: {err}', 'danger')	Function/method call that performs an action at runtime.
413	else:	Fallback `else` branch for the current conditional.
414	flash('Please fill in all fields.', 'warning')	Function/method call that performs an action at runtime.
415		Blank line used to separate logical blocks of Python code.
416	return redirect(url_for('admin_library'))	Returns a value (or nothing) from the current function.
417		Blank line used to separate logical blocks of Python code.
418	@app.route('/admin/library/edit_borrow', methods=['POST'])	Flask route decorator: binds the next function to a URL endpoint.
419	def edit_borrow():	Function definition `edit_borrow` with no parameters.
420	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
421		Blank line used to separate logical blocks of Python code.
422	borrow_id = request.form.get('borrow_id')	Assignment statement: stores the expression result in a variable/field.
423	due_date = request.form.get('due_date')	Assignment statement: stores the expression result in a variable/field.
424	fine = request.form.get('fine')	Assignment statement: stores the expression result in a variable/field.
425	return_date = request.form.get('return_date')	Returns a value (or nothing) from the current function.
426		Blank line used to separate logical blocks of Python code.
427	if not return_date:	Starts an `if` block that runs only when its condition is true.
428	return_date = None	Returns a value (or nothing) from the current function.
429		Blank line used to separate logical blocks of Python code.
430	if borrow_id:	Starts an `if` block that runs only when its condition is true.
431	try:	Starts a `try` block for exception-safe execution.
432	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
433	cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
434	cur.execute("""	Executes a SQL query/command using the active database cursor.
435	UPDATE borrows	General Python statement that participates in program control flow.
436	SET due_date = %s, fine = %s, return_date = %s	General Python statement that participates in program control flow.
437	WHERE borrow_id = %s	General Python statement that participates in program control flow.

Line	Code	Explanation
438	""", (due_date, fine, return_date, borrow_id))	General Python statement that participates in program control flow.
439	conn.commit()	Commits the active database transaction.
440	conn.close()	Closes a resource such as a cursor, connection, or file.
441	flash('Borrow record updated!', 'success')	Function/method call that performs an action at runtime.
442	except Exception as e:	Exception handler for errors raised in the matching `try` block.
443	flash(f"Error updating: {e}", "danger")	Function/method call that performs an action at runtime.
444		Blank line used to separate logical blocks of Python code.
445	return redirect(url_for('admin_library'))	Returns a value (or nothing) from the current function.
446		Blank line used to separate logical blocks of Python code.
447		Blank line used to separate logical blocks of Python code.
448	# =====	Comment line used to document intent or create section dividers.
449	# OTHER ADMIN MODULES	Comment line used to document intent or create section dividers.
450	# =====	Comment line used to document intent or create section dividers.
451		Blank line used to separate logical blocks of Python code.
452	@app.route('/admin/departments', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
453	def manage_departments():	Function definition `manage_departments` with no parameters.
454	if 'user_id' not in session or session.get('role') != 'admin':	Starts an `if` block that runs only when its condition is true.
455	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
456		Blank line used to separate logical blocks of Python code.
457	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
458	cursor = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
459	edit_department = None	Assignment statement: stores the expression result in a variable/field.
460		Blank line used to separate logical blocks of Python code.
461	# --- Handle Adding New Department ---	Comment line used to document intent or create section dividers.
462	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
463	dept_name = request.form['dept_name']	Assignment statement: stores the expression result in a variable/field.
464	hod_name = request.form.get('hod_name') # Optional field	Assignment statement: stores the expression result in a variable/field.
465	edit_id = request.form.get('edit_id')	Assignment statement: stores the expression result in a variable/field.
466		Blank line used to separate logical blocks of Python code.
467	if edit_id:	Starts an `if` block that runs only when its condition is true.
468	cursor.execute(	Executes a SQL query/command using the active database cursor.
469	"UPDATE departments SET dept_name = %s, hod_name = %s WHERE dept_id = %s",	General Python statement that participates in program control flow.
470	(dept_name, hod_name, edit_id)	General Python statement that participates in program control flow.
471	)	General Python statement that participates in program control flow.
472	flash('Department updated successfully!', 'success')	Function/method call that performs an action at runtime.
473	else:	Fallback `else` branch for the current conditional.
474	cursor.execute(	Executes a SQL query/command using the active database cursor.
475	"INSERT INTO departments (dept_name, hod_name) VALUES (%s, %s)",	General Python statement that participates in program control flow.
476	(dept_name, hod_name)	General Python statement that participates in program control flow.
477	)	General Python statement that participates in program control flow.
478	flash('Department added successfully!', 'success')	Function/method call that performs an action at runtime.

Line	Code	Explanation
479		Blank line used to separate logical blocks of Python code.
480	conn.commit()	Commits the active database transaction.
481	conn.close()	Closes a resource such as a cursor, connection, or file.
482	return redirect(url_for('manage_departments'))	Returns a value (or nothing) from the current function.
483		Blank line used to separate logical blocks of Python code.
484	# --- Handle Search & Listing ---	Comment line used to document intent or create section dividers.
485	search_query = request.args.get('search_query')	Assignment statement: stores the expression result in a variable/field.
486		Blank line used to separate logical blocks of Python code.
487	if search_query:	Starts an `if` block that runs only when its condition is true.
488	# Search by Department Name OR HOD Name	Comment line used to document intent or create section dividers.
489	# Using % wildcard for partial matches	Comment line used to document intent or create section dividers.
490	query = """	Assignment statement: stores the expression result in a variable/field.
491	SELECT * FROM departments	General Python statement that participates in program control flow.
492	WHERE dept_name LIKE %s OR hod_name LIKE %s	General Python statement that participates in program control flow.
493	ORDER BY dept_id DESC	General Python statement that participates in program control flow.
494	"""	General Python statement that participates in program control flow.
495	like_val = f"%{search_query}%"	Assignment statement: stores the expression result in a variable/field.
496	cursor.execute(query, (like_val, like_val))	Executes a SQL query/command using the active database cursor.
497	else:	Fallback `else` branch for the current conditional.
498	# Default view: Show all departments	Comment line used to document intent or create section dividers.
499	cursor.execute("SELECT * FROM departments ORDER BY dept_id DESC")	Executes a SQL query/command using the active database cursor.
500		Blank line used to separate logical blocks of Python code.
501	departments = cursor.fetchall()	Reads query results from the cursor.
502	edit_id = request.args.get('edit_id', type=int)	Assignment statement: stores the expression result in a variable/field.
503	if edit_id:	Starts an `if` block that runs only when its condition is true.
504	cursor.execute(	Executes a SQL query/command using the active database cursor.
505	"SELECT dept_id, dept_name, hod_name FROM departments WHERE dept_id = %s",	General Python statement that participates in program control flow.
506	(edit_id,)	General Python statement that participates in program control flow.
507	)	General Python statement that participates in program control flow.
508	edit_department = cursor.fetchone()	Reads query results from the cursor.
509	if not edit_department:	Starts an `if` block that runs only when its condition is true.
510	flash("Department not found for editing.", "danger")	Function/method call that performs an action at runtime.
511		Blank line used to separate logical blocks of Python code.
512	conn.close()	Closes a resource such as a cursor, connection, or file.
513		Blank line used to separate logical blocks of Python code.
514	return render_template(	Returns a value (or nothing) from the current function.
515	'admin_departments.html',	General Python statement that participates in program control flow.
516	departments=departments,	Assignment statement: stores the expression result in a variable/field.
517	edit_department=edit_department	Assignment statement: stores the expression result in a variable/field.
518	)	General Python statement that participates in program control flow.
519		Blank line used to separate logical blocks of Python code.

Line	Code	Explanation
520	@app.route('/admin/delete_dept/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
521	def delete_department(id):	Function definition `delete_department` with id.
522	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
523	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
524	try:	Starts a `try` block for exception-safe execution.
525	cur.execute("DELETE FROM departments WHERE dept_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
526	conn.commit(); flash("Department Deleted")	Commits the active database transaction.
527	except: flash("Error: Cannot delete department. Dependent records exist.")	Exception handler for errors raised in the matching `try` block.
528	conn.close()	Closes a resource such as a cursor, connection, or file.
529	return redirect(url_for('manage_departments'))	Returns a value (or nothing) from the current function.
530		Blank line used to separate logical blocks of Python code.
531		Blank line used to separate logical blocks of Python code.
532		Blank line used to separate logical blocks of Python code.
533	@app.route('/admin/students', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
534	def manage_students():	Function definition `manage_students` with no parameters.
535	if not check_auth('admin'):	Starts an `if` block that runs only when its condition is true.
536	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
537		Blank line used to separate logical blocks of Python code.
538	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
539	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
540	edit_student = None	Assignment statement: stores the expression result in a variable/field.
541		Blank line used to separate logical blocks of Python code.
542	# --- 1. HANDLE ADDING NEW STUDENT (POST) ---	Comment line used to document intent or create section dividers.
543	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
544	edit_id = request.form.get('edit_id')	Assignment statement: stores the expression result in a variable/field.
545	try:	Starts a `try` block for exception-safe execution.
546	roll_no = request.form['roll_no'].strip().upper()	Assignment statement: stores the expression result in a variable/field.
547	contact_no = request.form['contact_no'].strip()	Assignment statement: stores the expression result in a variable/field.
548	gender = request.form['gender']	Assignment statement: stores the expression result in a variable/field.
549	address = request.form['address'].strip()	Assignment statement: stores the expression result in a variable/field.
550	enroll_date = request.form['enroll_date']	Assignment statement: stores the expression result in a variable/field.
551	full_name = request.form['full_name'].strip()	Assignment statement: stores the expression result in a variable/field.
552	email = request.form['email'].strip()	Assignment statement: stores the expression result in a variable/field.
553	dept_id = request.form['dept_id']	Assignment statement: stores the expression result in a variable/field.
554	semester = request.form['semester']	Assignment statement: stores the expression result in a variable/field.
555		Blank line used to separate logical blocks of Python code.
556	if edit_id:	Starts an `if` block that runs only when its condition is true.
557	cur.execute(	Executes a SQL query/command using the active database cursor.
558	"SELECT user_id FROM users WHERE user_id = %s AND role = 'student'",	General Python statement that participates in program control flow.
559	(edit_id,)	General Python statement that participates in program control flow.
560	)	General Python statement that participates in program control flow.

Line	Code	Explanation
561	if not cur.fetchone():	Starts an `if` block that runs only when its condition is true.
562	flash("Student not found.", "danger")	Function/method call that performs an action at runtime.
563	else:	Fallback `else` branch for the current conditional.
564	new_password = request.form.get('password', '').strip()	Assignment statement: stores the expression result in a variable/field.
565	if new_password:	Starts an `if` block that runs only when its condition is true.
566	password_hash = generate_password_hash(new_password)	Assignment statement: stores the expression result in a variable/field.
567	cur.execute("""	Executes a SQL query/command using the active database cursor.
568	UPDATE users	General Python statement that participates in program control flow.
569	SET roll_no = %s, full_name = %s, email = %s, password = %s,	General Python statement that participates in program control flow.
570	dept_id = %s, semester = %s, enroll_date = %s,	Assignment statement: stores the expression result in a variable/field.
571	contact_no = %s, gender = %s, address = %s	Assignment statement: stores the expression result in a variable/field.
572	WHERE user_id = %s AND role = 'student'	General Python statement that participates in program control flow.
573	""", (	General Python statement that participates in program control flow.
574	roll_no, full_name, email, password_hash, dept_id, semester,	General Python statement that participates in program control flow.
575	enroll_date if enroll_date else date.today(),	General Python statement that participates in program control flow.
576	contact_no, gender, address, edit_id	General Python statement that participates in program control flow.
577	))	General Python statement that participates in program control flow.
578	else:	Fallback `else` branch for the current conditional.
579	cur.execute("""	Executes a SQL query/command using the active database cursor.
580	UPDATE users	General Python statement that participates in program control flow.
581	SET roll_no = %s, full_name = %s, email = %s,	General Python statement that participates in program control flow.
582	dept_id = %s, semester = %s, enroll_date = %s,	Assignment statement: stores the expression result in a variable/field.
583	contact_no = %s, gender = %s, address = %s	Assignment statement: stores the expression result in a variable/field.
584	WHERE user_id = %s AND role = 'student'	General Python statement that participates in program control flow.
585	""", (	General Python statement that participates in program control flow.
586	roll_no, full_name, email, dept_id, semester,	General Python statement that participates in program control flow.
587	enroll_date if enroll_date else date.today(),	General Python statement that participates in program control flow.
588	contact_no, gender, address, edit_id	General Python statement that participates in program control flow.
589	))	General Python statement that participates in program control flow.
590	conn.commit()	Commits the active database transaction.
591	flash("Student updated successfully.", "success")	Function/method call that performs an action at runtime.
592	else:	Fallback `else` branch for the current conditional.
593	password_hash = generate_password_hash(request.form['password'])	Assignment statement: stores the expression result in a variable/field.
594	cur.execute("""INSERT INTO users	Executes a SQL query/command using the active database cursor.
595	(roll_no, full_name, email, password, role, dept_id, semester,	General Python statement that participates in program control flow.
596	enroll_date, contact_no, gender, address)	General Python statement that participates in program control flow.
597	VALUES (%s, %s, %s, %s, 'student', %s, %s, %s, %s, %s, %s)""",	General Python statement that participates in program control flow.
598	(roll_no,	General Python statement that participates in program control flow.
599	full_name,	General Python statement that participates in program control flow.
600	email,	General Python statement that participates in program control flow.
601	password_hash,	General Python statement that participates in program control flow.

Line	Code	Explanation
602	dept_id,	General Python statement that participates in program control flow.
603	semester,	General Python statement that participates in program control flow.
604	enroll_date if enroll_date else date.today(),	General Python statement that participates in program control flow.
605	contact_no,	General Python statement that participates in program control flow.
606	gender,	General Python statement that participates in program control flow.
607	address))	General Python statement that participates in program control flow.
608	conn.commit()	Commits the active database transaction.
609	flash("Student added successfully.", "success")	Function/method call that performs an action at runtime.
610	except mysql.connector.Error as e:	Exception handler for errors raised in the matching `try` block.
611	conn.rollback()	General Python statement that participates in program control flow.
612	if e.errno == 1062: # Duplicate entry	General Python statement that participates in program control flow.
613	if 'roll_no' in str(e):	Starts an `if` block that runs only when its condition is true.
614	flash("Error: This Roll No already exists.", "danger")	Function/method call that performs an action at runtime.
615	elif 'email' in str(e):	Starts an `elif` branch checked when previous conditions are false.
616	flash("Error: This Email already exists.", "danger")	Function/method call that performs an action at runtime.
617	else:	Fallback `else` branch for the current conditional.
618	flash("Error: Duplicate entry (Roll No or Email).", "danger")	Function/method call that performs an action at runtime.
619	else:	Fallback `else` branch for the current conditional.
620	flash(f"Error adding student: {e}", "danger")	Function/method call that performs an action at runtime.
621	return redirect(url_for('manage_students'))	Returns a value (or nothing) from the current function.
622		Blank line used to separate logical blocks of Python code.
623	# --- 2. HANDLE GLOBAL SEARCH & LISTING (GET) ---	Comment line used to document intent or create section dividers.
624	search_query = request.args.get('search_query')	Assignment statement: stores the expression result in a variable/field.
625		Blank line used to separate logical blocks of Python code.
626	# Base query: Join users with departments	Comment line used to document intent or create section dividers.
627	base_query = ""	Assignment statement: stores the expression result in a variable/field.
628	SELECT u.user_id, u.roll_no, u.full_name, u.email, u.semester,	General Python statement that participates in program control flow.
629	d.dept_name, u.enroll_date, u.contact_no, u.gender, u.address	General Python statement that participates in program control flow.
630	FROM users u	General Python statement that participates in program control flow.
631	LEFT JOIN departments d ON u.dept_id = d.dept_id	General Python statement that participates in program control flow.
632	WHERE u.role = 'student'	General Python statement that participates in program control flow.
633	""	General Python statement that participates in program control flow.
634		Blank line used to separate logical blocks of Python code.
635	if search_query:	Starts an `if` block that runs only when its condition is true.
636	# Search across ALL meaningful columns	Comment line used to document intent or create section dividers.
637	# We use CAST(u.enroll_date AS CHAR) to allow string searching on dates	Comment line used to document intent or create section dividers.
638	sql = base_query + ""	Assignment statement: stores the expression result in a variable/field.
639	AND (	General Python statement that participates in program control flow.
640	u.full_name LIKE %s OR	General Python statement that participates in program control flow.
641	u.roll_no LIKE %s OR	General Python statement that participates in program control flow.
642	d.dept_name LIKE %s OR	General Python statement that participates in program control flow.

Line	Code	Explanation
643	u.email LIKE %s OR	General Python statement that participates in program control flow.
644	u.contact_no LIKE %s OR	General Python statement that participates in program control flow.
645	u.gender LIKE %s OR	General Python statement that participates in program control flow.
646	u.address LIKE %s OR	General Python statement that participates in program control flow.
647	u.semester LIKE %s OR	General Python statement that participates in program control flow.
648	CAST(u.enroll_date AS CHAR) LIKE %s	General Python statement that participates in program control flow.
649	)	General Python statement that participates in program control flow.
650	ORDER BY u.roll_no ASC	General Python statement that participates in program control flow.
651	"""	General Python statement that participates in program control flow.
652	wildcard = f"%{search_query}%"	Assignment statement: stores the expression result in a variable/field.
653	# We must provide the wildcard variable once for every %s placeholder above (9 times)	Comment line used to document intent or create section dividers.
654	cur.execute(sql, (wildcard, wildcard, wildcard, wildcard, wildcard, wildcard, wildcard, wildcard, wildcard, wildcard))	Executes a SQL query/command using the active database cursor.
655	else:	Fallback `else` branch for the current conditional.
656	# Default: Show all students ordered by Roll No	Comment line used to document intent or create section dividers.
657	cur.execute(base_query + " ORDER BY u.roll_no ASC")	Executes a SQL query/command using the active database cursor.
658		Blank line used to separate logical blocks of Python code.
659	students = cur.fetchall()	Reads query results from the cursor.
660		Blank line used to separate logical blocks of Python code.
661	# --- 3. FETCH DEPARTMENTS (For Dropdown) ---	Comment line used to document intent or create section dividers.
662	cur.execute("SELECT dept_id, dept_name FROM departments ORDER BY dept_name")	Executes a SQL query/command using the active database cursor.
663	departments = cur.fetchall()	Reads query results from the cursor.
664		Blank line used to separate logical blocks of Python code.
665	edit_id = request.args.get('edit_id', type=int)	Assignment statement: stores the expression result in a variable/field.
666	if edit_id:	Starts an `if` block that runs only when its condition is true.
667	cur.execute("""	Executes a SQL query/command using the active database cursor.
668	SELECT user_id, roll_no, full_name, email, dept_id, semester,	General Python statement that participates in program control flow.
669	enroll_date, contact_no, gender, address	General Python statement that participates in program control flow.
670	FROM users	General Python statement that participates in program control flow.
671	WHERE user_id = %s AND role = 'student'	General Python statement that participates in program control flow.
672	""", (edit_id,))	General Python statement that participates in program control flow.
673	edit_student = cur.fetchone()	Reads query results from the cursor.
674	if not edit_student:	Starts an `if` block that runs only when its condition is true.
675	flash("Student not found for editing.", "danger")	Function/method call that performs an action at runtime.
676		Blank line used to separate logical blocks of Python code.
677	today = date.today().isoformat()	Assignment statement: stores the expression result in a variable/field.
678		Blank line used to separate logical blocks of Python code.
679	conn.close()	Closes a resource such as a cursor, connection, or file.
680		Blank line used to separate logical blocks of Python code.
681	return render_template(	Returns a value (or nothing) from the current function.

Line	Code	Explanation
682	'admin_students.html',	General Python statement that participates in program control flow.
683	students=students,	Assignment statement: stores the expression result in a variable/field.
684	departments=departments,	Assignment statement: stores the expression result in a variable/field.
685	today=today,	Assignment statement: stores the expression result in a variable/field.
686	edit_student=edit_student	Assignment statement: stores the expression result in a variable/field.
687	)	General Python statement that participates in program control flow.
688		Blank line used to separate logical blocks of Python code.
689	@app.route('/admin/delete_student/<int:id>', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
690	def delete_student(id):	Function definition `delete_student` with id.
691	if not check_auth('admin'):	Starts an `if` block that runs only when its condition is true.
692	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
693		Blank line used to separate logical blocks of Python code.
694	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
695	# Separate connection for POST (delete operation)	Comment line used to document intent or create section dividers.
696	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
697	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
698		Blank line used to separate logical blocks of Python code.
699	try:	Starts a `try` block for exception-safe execution.
700	# Fetch student name for success message	Comment line used to document intent or create section dividers.
701	cur.execute("""	Executes a SQL query/command using the active database cursor.
702	SELECT u.full_name	General Python statement that participates in program control flow.
703	FROM users u	General Python statement that participates in program control flow.
704	WHERE u.user_id = %s AND u.role = 'student'	General Python statement that participates in program control flow.
705	""", (id,))	General Python statement that participates in program control flow.
706	student = cur.fetchone()	Reads query results from the cursor.
707		Blank line used to separate logical blocks of Python code.
708	if not student:	Starts an `if` block that runs only when its condition is true.
709	flash("Student not found.", "danger")	Function/method call that performs an action at runtime.
710	conn.close()	Closes a resource such as a cursor, connection, or file.
711	return redirect(url_for('manage_students'))	Returns a value (or nothing) from the current function.
712		Blank line used to separate logical blocks of Python code.
713	# Start manual transaction	Comment line used to document intent or create section dividers.
714	conn.autocommit = False	Assignment statement: stores the expression result in a variable/field.
715		Blank line used to separate logical blocks of Python code.
716	# Delete related records (child tables first)	Comment line used to document intent or create section dividers.
717	related_tables = [	Assignment statement: stores the expression result in a variable/field.
718	'borrows',	General Python statement that participates in program control flow.
719	'fees',	General Python statement that participates in program control flow.
720	'student_results',	General Python statement that participates in program control flow.
721	'attendance',	General Python statement that participates in program control flow.
722	'student_enrollments'	General Python statement that participates in program control flow.

Line	Code	Explanation
723	]	General Python statement that participates in program control flow.
724		Blank line used to separate logical blocks of Python code.
725	for table in related_tables:	Starts a `for` loop over an iterable sequence.
726	cur.execute(f"DELETE FROM {table} WHERE student_id = %s", (id,))	Executes a SQL query/command using the active database cursor.
727		Blank line used to separate logical blocks of Python code.
728	# Delete the student record	Comment line used to document intent or create section dividers.
729	cur.execute("DELETE FROM users WHERE user_id = %s AND role = 'student'", (id,))	Executes a SQL query/command using the active database cursor.
730		Blank line used to separate logical blocks of Python code.
731	conn.commit()	Commits the active database transaction.
732	flash(f"Student '{student['full_name']}' and all related records deleted successfully.", "success")	Function/method call that performs an action at runtime.
733		Blank line used to separate logical blocks of Python code.
734	except Exception as e:	Exception handler for errors raised in the matching `try` block.
735	conn.rollback()	General Python statement that participates in program control flow.
736	flash(f"Error force deleting student: {str(e)}", "danger")	Function/method call that performs an action at runtime.
737	finally:	`finally` block: cleanup code that runs whether or not errors occur.
738	conn.autocommit = True	Assignment statement: stores the expression result in a variable/field.
739	cur.close()	Closes a resource such as a cursor, connection, or file.
740	conn.close()	Closes a resource such as a cursor, connection, or file.
741		Blank line used to separate logical blocks of Python code.
742	return redirect(url_for('manage_students'))	Returns a value (or nothing) from the current function.
743		Blank line used to separate logical blocks of Python code.
744	# GET request: Confirmation page (separate connection)	Comment line used to document intent or create section dividers.
745	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
746	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
747		Blank line used to separate logical blocks of Python code.
748	# Fetch student details	Comment line used to document intent or create section dividers.
749	cur.execute("""	Executes a SQL query/command using the active database cursor.
750	SELECT u.user_id, u.full_name, u.email, u.semester, d.dept_name	General Python statement that participates in program control flow.
751	FROM users u	General Python statement that participates in program control flow.
752	LEFT JOIN departments d ON u.dept_id = d.dept_id	General Python statement that participates in program control flow.
753	WHERE u.user_id = %s AND u.role = 'student'	General Python statement that participates in program control flow.
754	""", (id,))	General Python statement that participates in program control flow.
755	student = cur.fetchone()	Reads query results from the cursor.
756		Blank line used to separate logical blocks of Python code.
757	if not student:	Starts an `if` block that runs only when its condition is true.
758	flash("Student not found.", "danger")	Function/method call that performs an action at runtime.
759	conn.close()	Closes a resource such as a cursor, connection, or file.
760	return redirect(url_for('manage_students'))	Returns a value (or nothing) from the current function.
761		Blank line used to separate logical blocks of Python code.

Line	Code	Explanation
762	# Count related records	Comment line used to document intent or create section dividers.
763	counts = {}	Assignment statement: stores the expression result in a variable/field.
764	for table, label in [	General Python statement that participates in program control flow.
765	('borrows', 'borrows'),	General Python statement that participates in program control flow.
766	('fees', 'fees'),	General Python statement that participates in program control flow.
767	('student_results', 'results'),	General Python statement that participates in program control flow.
768	('attendance', 'attendance'),	General Python statement that participates in program control flow.
769	('student_enrollments', 'enrollments')	General Python statement that participates in program control flow.
770	]:	General Python statement that participates in program control flow.
771	cur.execute(f"SELECT COUNT(*) AS cnt FROM {table} WHERE student_id = %s", (id,))	Executes a SQL query/command using the active database cursor.
772	counts[label] = cur.fetchone()['cnt']	Reads query results from the cursor.
773		Blank line used to separate logical blocks of Python code.
774	total_related = sum(counts.values())	Assignment statement: stores the expression result in a variable/field.
775		Blank line used to separate logical blocks of Python code.
776	conn.close()	Closes a resource such as a cursor, connection, or file.
777		Blank line used to separate logical blocks of Python code.
778	return render_template(	Returns a value (or nothing) from the current function.
779	'admin_delete_student_confirm.html',	General Python statement that participates in program control flow.
780	student=student,	Assignment statement: stores the expression result in a variable/field.
781	counts=counts,	Assignment statement: stores the expression result in a variable/field.
782	total_related=total_related	Assignment statement: stores the expression result in a variable/field.
783	)	General Python statement that participates in program control flow.
784		Blank line used to separate logical blocks of Python code.
785	@app.route('/admin/teachers', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
786	def manage_teachers():	Function definition `manage_teachers` with no parameters.
787	if not check_auth('admin'):	Starts an `if` block that runs only when its condition is true.
788	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
789		Blank line used to separate logical blocks of Python code.
790	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
791	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
792	edit_teacher = None	Assignment statement: stores the expression result in a variable/field.
793		Blank line used to separate logical blocks of Python code.
794	# --- 1. HANDLE ADDING NEW TEACHER (POST) ---	Comment line used to document intent or create section dividers.
795	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
796	edit_id = request.form.get('edit_id')	Assignment statement: stores the expression result in a variable/field.
797	try:	Starts a `try` block for exception-safe execution.
798	contact_no = request.form['contact_no'].strip()	Assignment statement: stores the expression result in a variable/field.
799	gender = request.form['gender']	Assignment statement: stores the expression result in a variable/field.
800	address = request.form['address'].strip()	Assignment statement: stores the expression result in a variable/field.
801	full_name = request.form['full_name'].strip()	Assignment statement: stores the expression result in a variable/field.

Line	Code	Explanation
802	email = request.form['email'].strip()	Assignment statement: stores the expression result in a variable/field.
803	dept_id = request.form['dept_id']	Assignment statement: stores the expression result in a variable/field.
804		Blank line used to separate logical blocks of Python code.
805	if edit_id:	Starts an `if` block that runs only when its condition is true.
806	cur.execute(	Executes a SQL query/command using the active database cursor.
807	"SELECT user_id FROM users WHERE user_id = %s AND role = 'teacher'",	General Python statement that participates in program control flow.
808	(edit_id,)	General Python statement that participates in program control flow.
809	)	General Python statement that participates in program control flow.
810	if not cur.fetchone():	Starts an `if` block that runs only when its condition is true.
811	flash("Teacher not found.", "danger")	Function/method call that performs an action at runtime.
812	else:	Fallback `else` branch for the current conditional.
813	new_password = request.form.get('password', '').strip()	Assignment statement: stores the expression result in a variable/field.
814	if new_password:	Starts an `if` block that runs only when its condition is true.
815	password_hash = generate_password_hash(new_password)	Assignment statement: stores the expression result in a variable/field.
816	cur.execute("""	Executes a SQL query/command using the active database cursor.
817	UPDATE users	General Python statement that participates in program control flow.
818	SET full_name = %s, email = %s, password = %s, dept_id = %s,	General Python statement that participates in program control flow.
819	contact_no = %s, gender = %s, address = %s	Assignment statement: stores the expression result in a variable/field.
820	WHERE user_id = %s AND role = 'teacher'	General Python statement that participates in program control flow.
821	""", (full_name, email, password_hash, dept_id, contact_no, gender, address, edit_id))	General Python statement that participates in program control flow.
822	else:	Fallback `else` branch for the current conditional.
823	cur.execute("""	Executes a SQL query/command using the active database cursor.
824	UPDATE users	General Python statement that participates in program control flow.
825	SET full_name = %s, email = %s, dept_id = %s,	General Python statement that participates in program control flow.
826	contact_no = %s, gender = %s, address = %s	Assignment statement: stores the expression result in a variable/field.
827	WHERE user_id = %s AND role = 'teacher'	General Python statement that participates in program control flow.
828	""", (full_name, email, dept_id, contact_no, gender, address, edit_id))	General Python statement that participates in program control flow.
829	conn.commit()	Commits the active database transaction.
830	flash("Teacher updated successfully.", "success")	Function/method call that performs an action at runtime.
831	else:	Fallback `else` branch for the current conditional.
832	password_hash = generate_password_hash(request.form['password'])	Assignment statement: stores the expression result in a variable/field.
833	cur.execute("""INSERT INTO users	Executes a SQL query/command using the active database cursor.
834	(full_name, email, password, role, dept_id,	General Python statement that participates in program control flow.
835	contact_no, gender, address)	General Python statement that participates in program control flow.
836	VALUES (%s, %s, %s, 'teacher', %s, %s, %s, %s)""",	General Python statement that participates in program control flow.
837	(full_name, email, password_hash, dept_id, contact_no, gender, address))	General Python statement that participates in program control flow.
838	conn.commit()	Commits the active database transaction.
839	flash("Teacher added successfully.", "success")	Function/method call that performs an action at runtime.
840	except mysql.connector.Error as e:	Exception handler for errors raised in the matching `try` block.
841	conn.rollback()	General Python statement that participates in program control flow.

Line	Code	Explanation
842	if e.errno == 1062: # Duplicate entry	General Python statement that participates in program control flow.
843	if 'email' in str(e):	Starts an `if` block that runs only when its condition is true.
844	flash("Error: This Email already exists.", "danger")	Function/method call that performs an action at runtime.
845	else:	Fallback `else` branch for the current conditional.
846	flash("Error: Duplicate entry (Email).", "danger")	Function/method call that performs an action at runtime.
847	else:	Fallback `else` branch for the current conditional.
848	flash(f"Error adding teacher: {e}", "danger")	Function/method call that performs an action at runtime.
849	return redirect(url_for('manage_teachers'))	Returns a value (or nothing) from the current function.
850		Blank line used to separate logical blocks of Python code.
851	# --- 2. HANDLE SEARCH & LISTING (GET) ---	Comment line used to document intent or create section dividers.
852	search_query = request.args.get('search_query')	Assignment statement: stores the expression result in a variable/field.
853		Blank line used to separate logical blocks of Python code.
854	# Base query: Join users with departments	Comment line used to document intent or create section dividers.
855	base_query = ""	Assignment statement: stores the expression result in a variable/field.
856	SELECT u.user_id, u.full_name, u.email,	General Python statement that participates in program control flow.
857	d.dept_name, u.contact_no, u.gender, u.address	General Python statement that participates in program control flow.
858	FROM users u	General Python statement that participates in program control flow.
859	LEFT JOIN departments d ON u.dept_id = d.dept_id	General Python statement that participates in program control flow.
860	WHERE u.role = 'teacher'	General Python statement that participates in program control flow.
861	""	General Python statement that participates in program control flow.
862		Blank line used to separate logical blocks of Python code.
863	if search_query:	Starts an `if` block that runs only when its condition is true.
864	# Search across Name, Dept, Email, Contact, Gender, Address	Comment line used to document intent or create section dividers.
865	sql = base_query + ""	Assignment statement: stores the expression result in a variable/field.
866	AND (	General Python statement that participates in program control flow.
867	u.full_name LIKE %s OR	General Python statement that participates in program control flow.
868	d.dept_name LIKE %s OR	General Python statement that participates in program control flow.
869	u.email LIKE %s OR	General Python statement that participates in program control flow.
870	u.contact_no LIKE %s OR	General Python statement that participates in program control flow.
871	u.gender LIKE %s OR	General Python statement that participates in program control flow.
872	u.address LIKE %s	General Python statement that participates in program control flow.
873	)	General Python statement that participates in program control flow.
874	ORDER BY u.full_name ASC	General Python statement that participates in program control flow.
875	""	General Python statement that participates in program control flow.
876	wildcard = f"%{search_query}%"	Assignment statement: stores the expression result in a variable/field.
877	# Provide wildcard for each placeholder (6 times)	Comment line used to document intent or create section dividers.
878	cur.execute(sql, (wildcard, wildcard, wildcard, wildcard, wildcard, wildcard))	Executes a SQL query/command using the active database cursor.
879	else:	Fallback `else` branch for the current conditional.
880	# Default: Show all teachers	Comment line used to document intent or create section dividers.
881	cur.execute(base_query + " ORDER BY u.full_name ASC")	Executes a SQL query/command using the active database cursor.
882		Blank line used to separate logical blocks of Python code.

Line	Code	Explanation
883	teachers = cur.fetchall()	Reads query results from the cursor.
884		Blank line used to separate logical blocks of Python code.
885	# --- 3. FETCH DEPARTMENTS (For Dropdown) ---	Comment line used to document intent or create section dividers.
886	cur.execute("SELECT dept_id, dept_name FROM departments ORDER BY dept_name")	Executes a SQL query/command using the active database cursor.
887	departments = cur.fetchall()	Reads query results from the cursor.
888		Blank line used to separate logical blocks of Python code.
889	edit_id = request.args.get('edit_id', type=int)	Assignment statement: stores the expression result in a variable/field.
890	if edit_id:	Starts an `if` block that runs only when its condition is true.
891	cur.execute("""	Executes a SQL query/command using the active database cursor.
892	SELECT user_id, full_name, email, dept_id, contact_no, gender, address	General Python statement that participates in program control flow.
893	FROM users	General Python statement that participates in program control flow.
894	WHERE user_id = %s AND role = 'teacher'	General Python statement that participates in program control flow.
895	""", (edit_id,))	General Python statement that participates in program control flow.
896	edit_teacher = cur.fetchone()	Reads query results from the cursor.
897	if not edit_teacher:	Starts an `if` block that runs only when its condition is true.
898	flash("Teacher not found for editing.", "danger")	Function/method call that performs an action at runtime.
899		Blank line used to separate logical blocks of Python code.
900	conn.close()	Closes a resource such as a cursor, connection, or file.
901		Blank line used to separate logical blocks of Python code.
902	return render_template(	Returns a value (or nothing) from the current function.
903	'admin_teachers.html',	General Python statement that participates in program control flow.
904	teachers=teachers,	Assignment statement: stores the expression result in a variable/field.
905	departments=departments,	Assignment statement: stores the expression result in a variable/field.
906	edit_teacher=edit_teacher	Assignment statement: stores the expression result in a variable/field.
907	)	General Python statement that participates in program control flow.
908		Blank line used to separate logical blocks of Python code.
909	@app.route('/admin/delete_teacher/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
910	def delete_teacher(id):	Function definition `delete_teacher` with id.
911	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
912	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
913	cur.execute("DELETE FROM users WHERE user_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
914	conn.commit(); flash("Teacher Deleted")	Commits the active database transaction.
915	conn.close()	Closes a resource such as a cursor, connection, or file.
916	return redirect(url_for('manage_teachers'))	Returns a value (or nothing) from the current function.
917		Blank line used to separate logical blocks of Python code.
918	@app.route('/admin/courses', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
919	def manage_courses():	Function definition `manage_courses` with no parameters.
920	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
921	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
922	edit_course = None	Assignment statement: stores the expression result in a variable/field.
923		Blank line used to separate logical blocks of Python code.

Line	Code	Explanation
924	<code>if request.method == 'POST':</code>	Starts an `if` block that runs only when its condition is true.
925	<code>edit_id = request.form.get('edit_id')</code>	Assignment statement: stores the expression result in a variable/field.
926	<code>try:</code>	Starts a `try` block for exception-safe execution.
927	<code>if edit_id:</code>	Starts an `if` block that runs only when its condition is true.
928	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
929	<code>UPDATE courses</code>	General Python statement that participates in program control flow.
930	<code>SET course_name = %s, course_code = %s, dept_id = %s</code>	General Python statement that participates in program control flow.
931	<code>WHERE course_id = %s</code>	General Python statement that participates in program control flow.
932	<code>""", (request.form['course_name'], request.form['course_code'],</code> <code>request.form['dept_id'], edit_id))</code>	General Python statement that participates in program control flow.
933	<code>flash("Course Updated")</code>	Function/method call that performs an action at runtime.
934	<code>else:</code>	Fallback `else` branch for the current conditional.
935	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
936	<code>INSERT INTO courses (course_name, course_code, dept_id)</code>	General Python statement that participates in program control flow.
937	<code>VALUES (%s, %s, %s)</code>	General Python statement that participates in program control flow.
938	<code>""", (request.form['course_name'], request.form['course_code'],</code> <code>request.form['dept_id']))</code>	General Python statement that participates in program control flow.
939	<code>flash("Course Added")</code>	Function/method call that performs an action at runtime.
940	<code>conn.commit()</code>	Commits the active database transaction.
941	<code>except mysql.connector.Error as e:</code>	Exception handler for errors raised in the matching `try` block.
942	<code>conn.rollback()</code>	General Python statement that participates in program control flow.
943	<code>if e.errno == 1062:</code>	Starts an `if` block that runs only when its condition is true.
944	<code>flash("Error: Course code already exists.", "danger")</code>	Function/method call that performs an action at runtime.
945	<code>else:</code>	Fallback `else` branch for the current conditional.
946	<code>flash(f"Error saving course: {e}", "danger")</code>	Function/method call that performs an action at runtime.
947	<code>return redirect(url_for('manage_courses'))</code>	Returns a value (or nothing) from the current function.
948	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
949	<code>SELECT c.course_id, c.course_name, c.course_code, c.dept_id, d.dept_name</code>	General Python statement that participates in program control flow.
950	<code>FROM courses c</code>	General Python statement that participates in program control flow.
951	<code>LEFT JOIN departments d ON c.dept_id = d.dept_id</code>	General Python statement that participates in program control flow.
952	<code>""")</code>	General Python statement that participates in program control flow.
953	<code>courses = cur.fetchall()</code>	Reads query results from the cursor.
954	<code>cur.execute("SELECT * FROM departments")</code>	Executes a SQL query/command using the active database cursor.
955	<code>depts = cur.fetchall()</code>	Reads query results from the cursor.
956		Blank line used to separate logical blocks of Python code.
957	<code>edit_id = request.args.get('edit_id', type=int)</code>	Assignment statement: stores the expression result in a variable/field.
958	<code>if edit_id:</code>	Starts an `if` block that runs only when its condition is true.
959	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
960	<code>SELECT course_id, course_name, course_code, dept_id</code>	General Python statement that participates in program control flow.
961	<code>FROM courses</code>	General Python statement that participates in program control flow.
962	<code>WHERE course_id = %s</code>	General Python statement that participates in program control flow.

Line	Code	Explanation
963	""", (edit_id,))	General Python statement that participates in program control flow.
964	edit_course = cur.fetchone()	Reads query results from the cursor.
965	if not edit_course:	Starts an `if` block that runs only when its condition is true.
966	flash("Course not found for editing.", "danger")	Function/method call that performs an action at runtime.
967		Blank line used to separate logical blocks of Python code.
968	conn.close()	Closes a resource such as a cursor, connection, or file.
969	return render_template('admin_courses.html', courses=courses, departments=depts, edit_course=edit_course)	Returns a value (or nothing) from the current function.
970		Blank line used to separate logical blocks of Python code.
971	@app.route('/admin/delete_course/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
972	def delete_course(id):	Function definition `delete_course` with id.
973	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
974	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
975	cur.execute("DELETE FROM courses WHERE course_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
976	conn.commit(); flash("Course Deleted")	Commits the active database transaction.
977	conn.close()	Closes a resource such as a cursor, connection, or file.
978	return redirect(url_for('manage_courses'))	Returns a value (or nothing) from the current function.
979		Blank line used to separate logical blocks of Python code.
980	@app.route('/admin/assign', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
981	def assign_course():	Function definition `assign_course` with no parameters.
982	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
983	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
984	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
985	cur.execute("INSERT INTO teacher_courses (teacher_id, course_id) VALUES (%s, %s)",	Executes a SQL query/command using the active database cursor.
986	(request.form['teacher_id'], request.form['course_id']))	General Python statement that participates in program control flow.
987	conn.commit(); flash("Assigned Successfully")	Commits the active database transaction.
988	return redirect(url_for('assign_course'))	Returns a value (or nothing) from the current function.
989	cur.execute("SELECT tc.assign_id, u.full_name, c.course_name FROM teacher_courses tc JOIN users u ON tc.teacher_id=u.user_id JOIN courses c ON tc.course_id=c.course_id")	Executes a SQL query/command using the active database cursor.
990	assignments = cur.fetchall()	Reads query results from the cursor.
991	cur.execute("SELECT user_id, full_name FROM users WHERE role='teacher'")	Executes a SQL query/command using the active database cursor.
992	teachers = cur.fetchall()	Reads query results from the cursor.
993	cur.execute("SELECT course_id, course_name FROM courses")	Executes a SQL query/command using the active database cursor.
994	courses = cur.fetchall()	Reads query results from the cursor.
995	conn.close()	Closes a resource such as a cursor, connection, or file.
996	return render_template('admin_assign.html', assignments=assignments, teachers=teachers, courses=courses)	Returns a value (or nothing) from the current function.
997		Blank line used to separate logical blocks of Python code.
998	@app.route('/admin/delete_assign/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
999	def delete_assignment(id):	Function definition `delete_assignment` with id.

Line	Code	Explanation
1000	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1001	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
1002	cur.execute("DELETE FROM teacher_courses WHERE assign_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
1003	conn.commit(); flash("Assignment Removed")	Commits the active database transaction.
1004	conn.close()	Closes a resource such as a cursor, connection, or file.
1005	return redirect(url_for('assign_course'))	Returns a value (or nothing) from the current function.
1006		Blank line used to separate logical blocks of Python code.
1007	@app.route('/admin/exams', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1008	def manage_exams():	Function definition `manage_exams` with no parameters.
1009	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1010	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1011	edit_exam = None	Assignment statement: stores the expression result in a variable/field.
1012		Blank line used to separate logical blocks of Python code.
1013	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
1014	edit_id = request.form.get('edit_id')	Assignment statement: stores the expression result in a variable/field.
1015	if edit_id:	Starts an `if` block that runs only when its condition is true.
1016	cur.execute("""	Executes a SQL query/command using the active database cursor.
1017	UPDATE exam_schedule	General Python statement that participates in program control flow.
1018	SET course_id = %s, exam_date = %s, start_time = %s, room_no = %s	General Python statement that participates in program control flow.
1019	WHERE exam_id = %s	General Python statement that participates in program control flow.
1020	""", (	General Python statement that participates in program control flow.
1021	request.form['course_id'],	General Python statement that participates in program control flow.
1022	request.form['exam_date'],	General Python statement that participates in program control flow.
1023	request.form['exam_time'],	General Python statement that participates in program control flow.
1024	request.form['room_no'],	General Python statement that participates in program control flow.
1025	edit_id	General Python statement that participates in program control flow.
1026	))	General Python statement that participates in program control flow.
1027	conn.commit(); flash("Exam Updated")	Commits the active database transaction.
1028	else:	Fallback `else` branch for the current conditional.
1029	cur.execute("""	Executes a SQL query/command using the active database cursor.
1030	INSERT INTO exam_schedule (course_id, exam_date, start_time, room_no)	General Python statement that participates in program control flow.
1031	VALUES (%s, %s, %s, %s)	General Python statement that participates in program control flow.
1032	""", (	General Python statement that participates in program control flow.
1033	request.form['course_id'],	General Python statement that participates in program control flow.
1034	request.form['exam_date'],	General Python statement that participates in program control flow.
1035	request.form['exam_time'],	General Python statement that participates in program control flow.
1036	request.form['room_no']	General Python statement that participates in program control flow.
1037	))	General Python statement that participates in program control flow.
1038	conn.commit(); flash("Exam Scheduled")	Commits the active database transaction.
1039	return redirect(url_for('manage_exams'))	Returns a value (or nothing) from the current function.
1040	cur.execute("""	Executes a SQL query/command using the active database cursor.

Line	Code	Explanation
1041	SELECT e.exam_id, e.course_id, c.course_name, e.exam_date, e.start_time, e.room_no	General Python statement that participates in program control flow.
1042	FROM exam_schedule e	General Python statement that participates in program control flow.
1043	JOIN courses c ON e.course_id = c.course_id	General Python statement that participates in program control flow.
1044	ORDER BY e.exam_date	General Python statement that participates in program control flow.
1045	""")	General Python statement that participates in program control flow.
1046	exams = cur.fetchall()	Reads query results from the cursor.
1047	cur.execute("SELECT course_id, course_name FROM courses")	Executes a SQL query/command using the active database cursor.
1048	courses = cur.fetchall()	Reads query results from the cursor.
1049		Blank line used to separate logical blocks of Python code.
1050	edit_id = request.args.get('edit_id', type=int)	Assignment statement: stores the expression result in a variable/field.
1051	if edit_id:	Starts an `if` block that runs only when its condition is true.
1052	cur.execute("""	Executes a SQL query/command using the active database cursor.
1053	SELECT exam_id, course_id, exam_date, start_time, room_no	General Python statement that participates in program control flow.
1054	FROM exam_schedule	General Python statement that participates in program control flow.
1055	WHERE exam_id = %s	General Python statement that participates in program control flow.
1056	""", (edit_id,))	General Python statement that participates in program control flow.
1057	edit_exam = cur.fetchone()	Reads query results from the cursor.
1058	if not edit_exam:	Starts an `if` block that runs only when its condition is true.
1059	flash("Exam not found for editing.", "danger")	Function/method call that performs an action at runtime.
1060		Blank line used to separate logical blocks of Python code.
1061	conn.close()	Closes a resource such as a cursor, connection, or file.
1062	return render_template('admin_exams.html', exams=exams, courses=courses, edit_exam=edit_exam)	Returns a value (or nothing) from the current function.
1063		Blank line used to separate logical blocks of Python code.
1064	@app.route('/admin/delete_exam/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
1065	def delete_exam(id):	Function definition `delete_exam` with id.
1066	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1067	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
1068	cur.execute("DELETE FROM exam_schedule WHERE exam_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
1069	conn.commit(); flash("Exam Deleted")	Commits the active database transaction.
1070	conn.close()	Closes a resource such as a cursor, connection, or file.
1071	return redirect(url_for('manage_exams'))	Returns a value (or nothing) from the current function.
1072		Blank line used to separate logical blocks of Python code.
1073	@app.route('/admin/fees', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1074	def manage_fees():	Function definition `manage_fees` with no parameters.
1075	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1076	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1077	edit_fee = None	Assignment statement: stores the expression result in a variable/field.
1078		Blank line used to separate logical blocks of Python code.
1079	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.

Line	Code	Explanation
1080	<code>edit_id = request.form.get('edit_id')</code>	Assignment statement: stores the expression result in a variable/field.
1081	<code>if edit_id:</code>	Starts an `if` block that runs only when its condition is true.
1082	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1083	<code>UPDATE fees</code>	General Python statement that participates in program control flow.
1084	<code>SET student_id = %s, amount = %s, description = %s, status = %s</code>	General Python statement that participates in program control flow.
1085	<code>WHERE fee_id = %s</code>	General Python statement that participates in program control flow.
1086	<code>""", (</code>	General Python statement that participates in program control flow.
1087	<code>request.form['student_id'],</code>	General Python statement that participates in program control flow.
1088	<code>request.form['amount'],</code>	General Python statement that participates in program control flow.
1089	<code>request.form['description'],</code>	General Python statement that participates in program control flow.
1090	<code>request.form['status'],</code>	General Python statement that participates in program control flow.
1091	<code>edit_id</code>	General Python statement that participates in program control flow.
1092	<code>))</code>	General Python statement that participates in program control flow.
1093	<code>conn.commit(); flash("Invoice Updated")</code>	Commits the active database transaction.
1094	<code>else:</code>	Fallback `else` branch for the current conditional.
1095	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1096	<code>INSERT INTO fees (student_id, amount, description, status)</code>	General Python statement that participates in program control flow.
1097	<code>VALUES (%s, %s, %s, %s)</code>	General Python statement that participates in program control flow.
1098	<code>""", (</code>	General Python statement that participates in program control flow.
1099	<code>request.form['student_id'],</code>	General Python statement that participates in program control flow.
1100	<code>request.form['amount'],</code>	General Python statement that participates in program control flow.
1101	<code>request.form['description'],</code>	General Python statement that participates in program control flow.
1102	<code>request.form['status']</code>	General Python statement that participates in program control flow.
1103	<code>))</code>	General Python statement that participates in program control flow.
1104	<code>conn.commit(); flash("Invoice Created")</code>	Commits the active database transaction.
1105	<code>return redirect(url_for('manage_fees'))</code>	Returns a value (or nothing) from the current function.
1106	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1107	<code>SELECT f.fee_id, f.student_id, u.full_name, f.amount, f.description, f.status</code>	General Python statement that participates in program control flow.
1108	<code>FROM fees f</code>	General Python statement that participates in program control flow.
1109	<code>JOIN users u ON f.student_id = u.user_id</code>	General Python statement that participates in program control flow.
1110	<code>ORDER BY f.fee_id DESC</code>	General Python statement that participates in program control flow.
1111	<code>""")</code>	General Python statement that participates in program control flow.
1112	<code>fees = cur.fetchall()</code>	Reads query results from the cursor.
1113	<code>cur.execute("SELECT user_id, full_name FROM users WHERE role='student'")</code>	Executes a SQL query/command using the active database cursor.
1114	<code>students = cur.fetchall()</code>	Reads query results from the cursor.
1115		Blank line used to separate logical blocks of Python code.
1116	<code>edit_id = request.args.get('edit_id', type=int)</code>	Assignment statement: stores the expression result in a variable/field.
1117	<code>if edit_id:</code>	Starts an `if` block that runs only when its condition is true.
1118	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1119	<code>SELECT fee_id, student_id, amount, description, status</code>	General Python statement that participates in program control flow.
1120	<code>FROM fees</code>	General Python statement that participates in program control flow.

Line	Code	Explanation
1121	WHERE fee_id = %s	General Python statement that participates in program control flow.
1122	""", (edit_id,))	General Python statement that participates in program control flow.
1123	edit_fee = cur.fetchone()	Reads query results from the cursor.
1124	if not edit_fee:	Starts an `if` block that runs only when its condition is true.
1125	flash("Invoice not found for editing.", "danger")	Function/method call that performs an action at runtime.
1126		Blank line used to separate logical blocks of Python code.
1127	conn.close()	Closes a resource such as a cursor, connection, or file.
1128	return render_template('admin_fees.html', fees=fees, students=students, edit_fee=edit_fee)	Returns a value (or nothing) from the current function.
1129		Blank line used to separate logical blocks of Python code.
1130	@app.route('/admin/delete_fee/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
1131	def delete_fee(id):	Function definition `delete_fee` with id.
1132	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1133	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
1134	cur.execute("DELETE FROM fees WHERE fee_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
1135	conn.commit(); flash("Record Deleted")	Commits the active database transaction.
1136	conn.close()	Closes a resource such as a cursor, connection, or file.
1137	return redirect(url_for('manage_fees'))	Returns a value (or nothing) from the current function.
1138		Blank line used to separate logical blocks of Python code.
1139	@app.route('/admin/notices', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1140	def manage_notices():	Function definition `manage_notices` with no parameters.
1141	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1142	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1143	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
1144	cur.execute("INSERT INTO notices (title, content, target_role) VALUES (%s, %s, %s)",	Executes a SQL query/command using the active database cursor.
1145	(request.form['title'], request.form['content'], request.form['target_role']))	General Python statement that participates in program control flow.
1146	conn.commit(); flash("Notice Published")	Commits the active database transaction.
1147	return redirect(url_for('manage_notices'))	Returns a value (or nothing) from the current function.
1148	cur.execute("SELECT * FROM notices ORDER BY date_posted DESC")	Executes a SQL query/command using the active database cursor.
1149	notices = cur.fetchall()	Reads query results from the cursor.
1150	conn.close()	Closes a resource such as a cursor, connection, or file.
1151	return render_template('admin_notices.html', notices=notices)	Returns a value (or nothing) from the current function.
1152		Blank line used to separate logical blocks of Python code.
1153	@app.route('/admin/delete_notice/<int:id>')	Flask route decorator: binds the next function to a URL endpoint.
1154	def delete_notice(id):	Function definition `delete_notice` with id.
1155	if not check_auth('admin'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1156	conn = get_db(); cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
1157	cur.execute("DELETE FROM notices WHERE notice_id=%s", (id,))	Executes a SQL query/command using the active database cursor.
1158	conn.commit(); flash("Notice Deleted")	Commits the active database transaction.
1159	conn.close()	Closes a resource such as a cursor, connection, or file.

Line	Code	Explanation
1160	<code>return redirect(url_for('manage_notices'))</code>	Returns a value (or nothing) from the current function.
1161		Blank line used to separate logical blocks of Python code.
1162	<code># =====</code>	Comment line used to document intent or create section dividers.
1163	<code># TEACHER DASHBOARD</code>	Comment line used to document intent or create section dividers.
1164	<code># =====</code>	Comment line used to document intent or create section dividers.
1165		Blank line used to separate logical blocks of Python code.
1166	<code>@app.route('/teacher')</code>	Flask route decorator: binds the next function to a URL endpoint.
1167	<code>def teacher_dashboard():</code>	Function definition `teacher_dashboard` with no parameters.
1168	<code>if not check_auth('teacher'):</code>	Starts an `if` block that runs only when its condition is true.
1169	<code>return redirect(url_for('login'))</code>	Returns a value (or nothing) from the current function.
1170		Blank line used to separate logical blocks of Python code.
1171	<code>conn = get_db()</code>	Assignment statement: stores the expression result in a variable/field.
1172	<code>cur = conn.cursor()</code>	Assignment statement: stores the expression result in a variable/field.
1173	<code>uid = session['user_id']</code>	Assignment statement: stores the expression result in a variable/field.
1174		Blank line used to separate logical blocks of Python code.
1175	<code># Count of teacher courses</code>	Comment line used to document intent or create section dividers.
1176	<code>cur.execute("SELECT COUNT(*) FROM teacher_courses WHERE teacher_id=%s", (uid,))</code>	Executes a SQL query/command using the active database cursor.
1177	<code>classes_cnt = cur.fetchone()[0]</code>	Reads query results from the cursor.
1178		Blank line used to separate logical blocks of Python code.
1179	<code># Fetch notices for teachers (or all)</code>	Comment line used to document intent or create section dividers.
1180	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1181	<code>SELECT notice_id, title, content</code>	General Python statement that participates in program control flow.
1182	<code>FROM notices</code>	General Python statement that participates in program control flow.
1183	<code>WHERE target_role IN ('all', 'teacher')</code>	General Python statement that participates in program control flow.
1184	<code>ORDER BY date_posted DESC</code>	General Python statement that participates in program control flow.
1185	<code>""")</code>	General Python statement that participates in program control flow.
1186	<code>notices = cur.fetchall() # list of tuples</code>	Reads query results from the cursor.
1187		Blank line used to separate logical blocks of Python code.
1188	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1189		Blank line used to separate logical blocks of Python code.
1190	<code>return render_template(</code>	Returns a value (or nothing) from the current function.
1191	<code>'teacher_dash.html',</code>	General Python statement that participates in program control flow.
1192	<code>classes_count=classes_cnt,</code>	Assignment statement: stores the expression result in a variable/field.
1193	<code>notices=notices</code>	Assignment statement: stores the expression result in a variable/field.
1194	<code>)</code>	General Python statement that participates in program control flow.
1195		Blank line used to separate logical blocks of Python code.
1196		Blank line used to separate logical blocks of Python code.
1197	<code>@app.route('/teacher/classes')</code>	Flask route decorator: binds the next function to a URL endpoint.
1198	<code>def teacher_classes():</code>	Function definition `teacher_classes` with no parameters.
1199	<code>if not check_auth('teacher'): return redirect(url_for('login'))</code>	General Python statement that participates in program control flow.

Line	Code	Explanation
1200	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1201	cur.execute("""	Executes a SQL query/command using the active database cursor.
1202	SELECT c.course_id, c.course_name, c.course_code, d.dept_name	General Python statement that participates in program control flow.
1203	FROM teacher_courses tc	General Python statement that participates in program control flow.
1204	JOIN courses c ON tc.course_id=c.course_id	General Python statement that participates in program control flow.
1205	JOIN departments d ON c.dept_id=d.dept_id	General Python statement that participates in program control flow.
1206	WHERE tc.teacher_id=%s	General Python statement that participates in program control flow.
1207	""", (session['user_id'],))	General Python statement that participates in program control flow.
1208	courses = cur.fetchall()	Reads query results from the cursor.
1209	conn.close()	Closes a resource such as a cursor, connection, or file.
1210	return render_template('teacher_classes.html', courses=courses)	Returns a value (or nothing) from the current function.
1211		Blank line used to separate logical blocks of Python code.
1212	@app.route('/teacher/result/<int:course_id>', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1213	def teacher_enter_result(course_id):	Function definition `teacher_enter_result` with course_id.
1214	if not check_auth('teacher'):	Starts an `if` block that runs only when its condition is true.
1215	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1216		Blank line used to separate logical blocks of Python code.
1217	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
1218	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1219	edit_result = None	Assignment statement: stores the expression result in a variable/field.
1220		Blank line used to separate logical blocks of Python code.
1221	# Fetch course details	Comment line used to document intent or create section dividers.
1222	cur.execute("SELECT course_id, course_name FROM courses WHERE course_id=%s", (course_id,))	Executes a SQL query/command using the active database cursor.
1223	course_row = cur.fetchone()	Reads query results from the cursor.
1224	if not course_row:	Starts an `if` block that runs only when its condition is true.
1225	conn.close()	Closes a resource such as a cursor, connection, or file.
1226	flash("Course not found.", "danger")	Function/method call that performs an action at runtime.
1227	return redirect(url_for('teacher_classes'))	Returns a value (or nothing) from the current function.
1228		Blank line used to separate logical blocks of Python code.
1229	course = (course_row['course_id'], course_row['course_name'])	Assignment statement: stores the expression result in a variable/field.
1230		Blank line used to separate logical blocks of Python code.
1231	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
1232	try:	Starts a `try` block for exception-safe execution.
1233	edit_result_id = request.form.get('edit_result_id')	Assignment statement: stores the expression result in a variable/field.
1234	student_id = request.form['student_id']	Assignment statement: stores the expression result in a variable/field.
1235	if not student_id:	Starts an `if` block that runs only when its condition is true.
1236	flash("Please select a student.", "warning")	Function/method call that performs an action at runtime.
1237	return redirect(url_for('teacher_enter_result', course_id=course_id))	Returns a value (or nothing) from the current function.
1238		Blank line used to separate logical blocks of Python code.
1239	marks = float(request.form['marks'])	Assignment statement: stores the expression result in a variable/field.

Line	Code	Explanation
1240	full_marks = float(request.form['full_marks'])	Assignment statement: stores the expression result in a variable/field.
1241	exam_type = request.form['exam_type']	Assignment statement: stores the expression result in a variable/field.
1242		Blank line used to separate logical blocks of Python code.
1243	if marks > full_marks or marks < 0 or full_marks <= 0:	Starts an `if` block that runs only when its condition is true.
1244	flash("Invalid marks entered.", "warning")	Function/method call that performs an action at runtime.
1245	return redirect(url_for('teacher_enter_result', course_id=course_id))	Returns a value (or nothing) from the current function.
1246		Blank line used to separate logical blocks of Python code.
1247	# Check if student is enrolled (safety check)	Comment line used to document intent or create section dividers.
1248	cur.execute("SELECT 1 FROM student_enrollments WHERE student_id=%s AND course_id=%s",	Executes a SQL query/command using the active database cursor.
1249	(student_id, course_id))	General Python statement that participates in program control flow.
1250	if not cur.fetchone():	Starts an `if` block that runs only when its condition is true.
1251	flash("Selected student is not enrolled in this course.", "danger")	Function/method call that performs an action at runtime.
1252	return redirect(url_for('teacher_enter_result', course_id=course_id))	Returns a value (or nothing) from the current function.
1253		Blank line used to separate logical blocks of Python code.
1254	grade, gpa = calculate_grade_gpa(marks, full_marks)	General Python statement that participates in program control flow.
1255		Blank line used to separate logical blocks of Python code.
1256	if edit_result_id:	Starts an `if` block that runs only when its condition is true.
1257	cur.execute("""	Executes a SQL query/command using the active database cursor.
1258	SELECT result_id	General Python statement that participates in program control flow.
1259	FROM student_results	General Python statement that participates in program control flow.
1260	WHERE result_id = %s AND course_id = %s	General Python statement that participates in program control flow.
1261	""", (edit_result_id, course_id))	General Python statement that participates in program control flow.
1262	target = cur.fetchone()	Reads query results from the cursor.
1263	if not target:	Starts an `if` block that runs only when its condition is true.
1264	flash("Result not found for editing.", "danger")	Function/method call that performs an action at runtime.
1265	return redirect(url_for('teacher_enter_result', course_id=course_id))	Returns a value (or nothing) from the current function.
1266		Blank line used to separate logical blocks of Python code.
1267	cur.execute("""	Executes a SQL query/command using the active database cursor.
1268	UPDATE student_results	General Python statement that participates in program control flow.
1269	SET student_id = %s, exam_type = %s, marks_obtained = %s,	General Python statement that participates in program control flow.
1270	full_marks = %s, grade = %s, gpa = %s	Assignment statement: stores the expression result in a variable/field.
1271	WHERE result_id = %s AND course_id = %s	General Python statement that participates in program control flow.
1272	""", (	General Python statement that participates in program control flow.
1273	student_id, exam_type, marks, full_marks, grade, gpa,	General Python statement that participates in program control flow.
1274	edit_result_id, course_id	General Python statement that participates in program control flow.
1275	))	General Python statement that participates in program control flow.
1276	flash(f"Updated {exam_type} result", "success")	Function/method call that performs an action at runtime.
1277	else:	Fallback `else` branch for the current conditional.
1278	# Check if record already exists	Comment line used to document intent or create section dividers.
1279	cur.execute("""	Executes a SQL query/command using the active database cursor.

Line	Code	Explanation
1280	SELECT result_id FROM student_results	General Python statement that participates in program control flow.
1281	WHERE student_id=%s AND course_id=%s AND exam_type=%s	General Python statement that participates in program control flow.
1282	""", (student_id, course_id, exam_type))	General Python statement that participates in program control flow.
1283	exists = cur.fetchone()	Reads query results from the cursor.
1284		Blank line used to separate logical blocks of Python code.
1285	if exists:	Starts an `if` block that runs only when its condition is true.
1286	cur.execute("""	Executes a SQL query/command using the active database cursor.
1287	UPDATE student_results	General Python statement that participates in program control flow.
1288	SET marks_obtained=%s, full_marks=%s, grade=%s, gpa=%s	General Python statement that participates in program control flow.
1289	WHERE result_id=%s	General Python statement that participates in program control flow.
1290	""", (marks, full_marks, grade, gpa, exists['result_id']))	General Python statement that participates in program control flow.
1291	flash(f"Updated {exam_type} result", "success")	Function/method call that performs an action at runtime.
1292	else:	Fallback `else` branch for the current conditional.
1293	cur.execute("""	Executes a SQL query/command using the active database cursor.
1294	INSERT INTO student_results	General Python statement that participates in program control flow.
1295	(student_id, course_id, marks_obtained, full_marks, grade, gpa, exam_type)	General Python statement that participates in program control flow.
1296	VALUES (%s, %s, %s, %s, %s, %s, %s)	General Python statement that participates in program control flow.
1297	""", (student_id, course_id, marks, full_marks, grade, gpa, exam_type))	General Python statement that participates in program control flow.
1298	flash(f"Saved {exam_type} result", "success")	Function/method call that performs an action at runtime.
1299		Blank line used to separate logical blocks of Python code.
1300	conn.commit()	Commits the active database transaction.
1301	except ValueError:	Exception handler for errors raised in the matching `try` block.
1302	flash("Marks must be valid numbers.", "warning")	Function/method call that performs an action at runtime.
1303	except Exception as e:	Exception handler for errors raised in the matching `try` block.
1304	flash("Error saving result.", "danger")	Function/method call that performs an action at runtime.
1305		Blank line used to separate logical blocks of Python code.
1306	return redirect(url_for('teacher_enter_result', course_id=course_id))	Returns a value (or nothing) from the current function.
1307		Blank line used to separate logical blocks of Python code.
1308	edit_result_id = request.args.get('edit_result_id', type=int)	Assignment statement: stores the expression result in a variable/field.
1309	if edit_result_id:	Starts an `if` block that runs only when its condition is true.
1310	cur.execute("""	Executes a SQL query/command using the active database cursor.
1311	SELECT result_id, student_id, exam_type, marks_obtained, full_marks	General Python statement that participates in program control flow.
1312	FROM student_results	General Python statement that participates in program control flow.
1313	WHERE result_id = %s AND course_id = %s	General Python statement that participates in program control flow.
1314	""", (edit_result_id, course_id))	General Python statement that participates in program control flow.
1315	edit_result = cur.fetchone()	Reads query results from the cursor.
1316	if not edit_result:	Starts an `if` block that runs only when its condition is true.
1317	flash("Result not found for editing.", "danger")	Function/method call that performs an action at runtime.
1318		Blank line used to separate logical blocks of Python code.
1319	# GET: Load enrolled students only	Comment line used to document intent or create section dividers.
1320	cur.execute("""	Executes a SQL query/command using the active database cursor.

Line	Code	Explanation
1321	SELECT u.user_id, u.full_name, COALESCE(u.semester, 'N/A') AS semester	General Python statement that participates in program control flow.
1322	FROM users u	General Python statement that participates in program control flow.
1323	JOIN student_enrollments se ON u.user_id = se.student_id	General Python statement that participates in program control flow.
1324	WHERE se.course_id = %s	General Python statement that participates in program control flow.
1325	ORDER BY u.full_name	General Python statement that participates in program control flow.
1326	""", (course_id,))	General Python statement that participates in program control flow.
1327	students_raw = cur.fetchall()	Reads query results from the cursor.
1328	students = [(s['user_id'], s['full_name'], s['semester']) for s in students_raw]	Assignment statement: stores the expression result in a variable/field.
1329		Blank line used to separate logical blocks of Python code.
1330	# Load existing results with result_id for delete	Comment line used to document intent or create section dividers.
1331	cur.execute("""	Executes a SQL query/command using the active database cursor.
1332	SELECT u.full_name,	General Python statement that participates in program control flow.
1333	r.marks_obtained, r.full_marks, r.grade, r.gpa,	General Python statement that participates in program control flow.
1334	r.exam_type, r.result_id	General Python statement that participates in program control flow.
1335	FROM student_results r	General Python statement that participates in program control flow.
1336	JOIN users u ON r.student_id = u.user_id	General Python statement that participates in program control flow.
1337	WHERE r.course_id = %s	General Python statement that participates in program control flow.
1338	ORDER BY r.exam_type, u.full_name	General Python statement that participates in program control flow.
1339	""", (course_id,))	General Python statement that participates in program control flow.
1340	results_raw = cur.fetchall()	Reads query results from the cursor.
1341	results = [	Assignment statement: stores the expression result in a variable/field.
1342	(r['full_name'], r['marks_obtained'], r['full_marks'],	General Python statement that participates in program control flow.
1343	r['grade'], r['gpa'], r['exam_type'], r['result_id'])	General Python statement that participates in program control flow.
1344	for r in results_raw	General Python statement that participates in program control flow.
1345	]	General Python statement that participates in program control flow.
1346		Blank line used to separate logical blocks of Python code.
1347	conn.close()	Closes a resource such as a cursor, connection, or file.
1348		Blank line used to separate logical blocks of Python code.
1349	return render_template(	Returns a value (or nothing) from the current function.
1350	'teacher_enter_result.html',	General Python statement that participates in program control flow.
1351	course=course,	Assignment statement: stores the expression result in a variable/field.
1352	students=students,	Assignment statement: stores the expression result in a variable/field.
1353	results=results,	Assignment statement: stores the expression result in a variable/field.
1354	course_id=course_id,	Assignment statement: stores the expression result in a variable/field.
1355	edit_result=edit_result	Assignment statement: stores the expression result in a variable/field.
1356	)	General Python statement that participates in program control flow.
1357		Blank line used to separate logical blocks of Python code.
1358	@app.route('/teacher/result/delete/<int:result_id>', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1359	def teacher_delete_result(result_id):	Function definition `teacher_delete_result` with result_id.
1360	if not check_auth('teacher'):	Starts an `if` block that runs only when its condition is true.

Line	Code	Explanation
1361	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1362		Blank line used to separate logical blocks of Python code.
1363	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
1364	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1365		Blank line used to separate logical blocks of Python code.
1366	if request.method == 'POST':	Starts an 'if' block that runs only when its condition is true.
1367	# Perform the deletion	Comment line used to document intent or create section dividers.
1368	cur.execute("SELECT course_id FROM student_results WHERE result_id=%s", (result_id,))	Executes a SQL query/command using the active database cursor.
1369	res = cur.fetchone()	Reads query results from the cursor.
1370		Blank line used to separate logical blocks of Python code.
1371	if res:	Starts an 'if' block that runs only when its condition is true.
1372	cur.execute("DELETE FROM student_results WHERE result_id=%s", (result_id,))	Executes a SQL query/command using the active database cursor.
1373	conn.commit()	Commits the active database transaction.
1374	flash("Result deleted successfully.", "success")	Function/method call that performs an action at runtime.
1375	conn.close()	Closes a resource such as a cursor, connection, or file.
1376	return redirect(url_for('teacher_enter_result', course_id=res['course_id']))	Returns a value (or nothing) from the current function.
1377		Blank line used to separate logical blocks of Python code.
1378	conn.close()	Closes a resource such as a cursor, connection, or file.
1379	flash("Result not found.", "danger")	Function/method call that performs an action at runtime.
1380	return redirect(url_for('teacher_dashboard'))	Returns a value (or nothing) from the current function.
1381		Blank line used to separate logical blocks of Python code.
1382	# GET request → Show confirmation page	Comment line used to document intent or create section dividers.
1383	cur.execute("""	Executes a SQL query/command using the active database cursor.
1384	SELECT	General Python statement that participates in program control flow.
1385	r.course_id,	General Python statement that participates in program control flow.
1386	c.course_name,	General Python statement that participates in program control flow.
1387	u.full_name AS student_name,	General Python statement that participates in program control flow.
1388	r.exam_type,	General Python statement that participates in program control flow.
1389	r.marks_obtained,	General Python statement that participates in program control flow.
1390	r.full_marks,	General Python statement that participates in program control flow.
1391	r.grade,	General Python statement that participates in program control flow.
1392	r.gpa	General Python statement that participates in program control flow.
1393	FROM student_results r	General Python statement that participates in program control flow.
1394	JOIN courses c ON r.course_id = c.course_id	General Python statement that participates in program control flow.
1395	JOIN users u ON r.student_id = u.user_id	General Python statement that participates in program control flow.
1396	WHERE r.result_id = %s	General Python statement that participates in program control flow.
1397	""", (result_id,))	General Python statement that participates in program control flow.
1398		Blank line used to separate logical blocks of Python code.
1399	result = cur.fetchone()	Reads query results from the cursor.
1400	conn.close()	Closes a resource such as a cursor, connection, or file.

Line	Code	Explanation
1401		Blank line used to separate logical blocks of Python code.
1402	if not result:	Starts an `if` block that runs only when its condition is true.
1403	flash("Result not found.", "danger")	Function/method call that performs an action at runtime.
1404	return redirect(url_for('teacher_dashboard'))	Returns a value (or nothing) from the current function.
1405		Blank line used to separate logical blocks of Python code.
1406	return render_template(	Returns a value (or nothing) from the current function.
1407	'teacher_delete_result.html',	General Python statement that participates in program control flow.
1408	course_id=result['course_id'],	Assignment statement: stores the expression result in a variable/field.
1409	course_name=result['course_name'],	Assignment statement: stores the expression result in a variable/field.
1410	student_name=result['student_name'],	Assignment statement: stores the expression result in a variable/field.
1411	exam_type=result['exam_type'],	Assignment statement: stores the expression result in a variable/field.
1412	marks_obtained=result['marks_obtained'],	Assignment statement: stores the expression result in a variable/field.
1413	full_marks=result['full_marks'],	Assignment statement: stores the expression result in a variable/field.
1414	grade=result['grade'],	Assignment statement: stores the expression result in a variable/field.
1415	gpa=result['gpa']	Assignment statement: stores the expression result in a variable/field.
1416	)	General Python statement that participates in program control flow.
1417		Blank line used to separate logical blocks of Python code.
1418	@app.route('/teacher/attendance/<int:course_id>', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1419	def teacher_attendance(course_id):	Function definition `teacher_attendance` with course_id.
1420	if not check_auth('teacher'):	Starts an `if` block that runs only when its condition is true.
1421	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1422		Blank line used to separate logical blocks of Python code.
1423	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
1424	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1425		Blank line used to separate logical blocks of Python code.
1426	# Fetch course name	Comment line used to document intent or create section dividers.
1427	cur.execute("SELECT course_name FROM courses WHERE course_id=%s", (course_id,))	Executes a SQL query/command using the active database cursor.
1428	course_row = cur.fetchone()	Reads query results from the cursor.
1429	if not course_row:	Starts an `if` block that runs only when its condition is true.
1430	conn.close()	Closes a resource such as a cursor, connection, or file.
1431	flash("Course not found.", "danger")	Function/method call that performs an action at runtime.
1432	return redirect(url_for('teacher_classes'))	Returns a value (or nothing) from the current function.
1433	course_name = course_row['course_name']	Assignment statement: stores the expression result in a variable/field.
1434		Blank line used to separate logical blocks of Python code.
1435	# Get enrolled students	Comment line used to document intent or create section dividers.
1436	cur.execute("""	Executes a SQL query/command using the active database cursor.
1437	SELECT u.user_id, u.full_name	General Python statement that participates in program control flow.
1438	FROM users u	General Python statement that participates in program control flow.
1439	JOIN student_enrollments se ON u.user_id = se.student_id	General Python statement that participates in program control flow.
1440	WHERE se.course_id = %s	General Python statement that participates in program control flow.

Line	Code	Explanation
1441	ORDER BY u.full_name	General Python statement that participates in program control flow.
1442	""", (course_id,))	General Python statement that participates in program control flow.
1443	students = cur.fetchall()	Reads query results from the cursor.
1444		Blank line used to separate logical blocks of Python code.
1445	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
1446	dt = request.form['date']	Assignment statement: stores the expression result in a variable/field.
1447		Blank line used to separate logical blocks of Python code.
1448	# Get list of enrolled student IDs for saving (only these students get attendance records)	Comment line used to document intent or create section dividers.
1449	cur.execute("SELECT student_id AS user_id FROM student_enrollments WHERE course_id=%s", (course_id,))	Executes a SQL query/command using the active database cursor.
1450	enrolled_ids = cur.fetchall()	Reads query results from the cursor.
1451		Blank line used to separate logical blocks of Python code.
1452	for s in enrolled_ids:	Starts a `for` loop over an iterable sequence.
1453	status = 'Present' if f"status_{s['user_id']}" in request.form else 'Absent'	Assignment statement: stores the expression result in a variable/field.
1454	# Delete any existing record for this date (to allow overwrite)	Comment line used to document intent or create section dividers.
1455	cur.execute("""	Executes a SQL query/command using the active database cursor.
1456	DELETE FROM attendance	General Python statement that participates in program control flow.
1457	WHERE course_id=%s AND student_id=%s AND attendance_date=%s	General Python statement that participates in program control flow.
1458	""", (course_id, s['user_id'], dt))	General Python statement that participates in program control flow.
1459	# Insert the new record	Comment line used to document intent or create section dividers.
1460	cur.execute("""	Executes a SQL query/command using the active database cursor.
1461	INSERT INTO attendance (course_id, student_id, attendance_date, status)	General Python statement that participates in program control flow.
1462	VALUES (%s, %s, %s, %s)	General Python statement that participates in program control flow.
1463	""", (course_id, s['user_id'], dt, status))	General Python statement that participates in program control flow.
1464		Blank line used to separate logical blocks of Python code.
1465	conn.commit()	Commits the active database transaction.
1466	conn.close()	Closes a resource such as a cursor, connection, or file.
1467	flash("Attendance Saved", "success")	Function/method call that performs an action at runtime.
1468	return redirect(url_for('teacher_classes'))	Returns a value (or nothing) from the current function.
1469		Blank line used to separate logical blocks of Python code.
1470	# GET request - prepare data for template	Comment line used to document intent or create section dividers.
1471	today = date.today().isoformat()	Assignment statement: stores the expression result in a variable/field.
1472		Blank line used to separate logical blocks of Python code.
1473	# Pre-fill today's attendance if it exists	Comment line used to document intent or create section dividers.
1474	att_dict = {}	Assignment statement: stores the expression result in a variable/field.
1475	cur.execute("""	Executes a SQL query/command using the active database cursor.
1476	SELECT student_id, status	General Python statement that participates in program control flow.
1477	FROM attendance	General Python statement that participates in program control flow.
1478	WHERE course_id=%s AND attendance_date=%s	General Python statement that participates in program control flow.
1479	""", (course_id, today))	General Python statement that participates in program control flow.

Line	Code	Explanation
1480	for row in cur.fetchall():	Starts a `for` loop over an iterable sequence.
1481	att_dict[row['student_id']] = row['status']	Assignment statement: stores the expression result in a variable/field.
1482		Blank line used to separate logical blocks of Python code.
1483	conn.close()	Closes a resource such as a cursor, connection, or file.
1484		Blank line used to separate logical blocks of Python code.
1485	return render_template(	Returns a value (or nothing) from the current function.
1486	'teacher_attendance.html',	General Python statement that participates in program control flow.
1487	course={'course_name': course_name},	Assignment statement: stores the expression result in a variable/field.
1488	students=students,	Assignment statement: stores the expression result in a variable/field.
1489	today=today,	Assignment statement: stores the expression result in a variable/field.
1490	att_dict=att_dict	Assignment statement: stores the expression result in a variable/field.
1491	)	General Python statement that participates in program control flow.
1492		Blank line used to separate logical blocks of Python code.
1493	# =====	Comment line used to document intent or create section dividers.
1494	# STUDENT DASHBOARD	Comment line used to document intent or create section dividers.
1495	# =====	Comment line used to document intent or create section dividers.
1496	@app.route('/student')	Flask route decorator: binds the next function to a URL endpoint.
1497	def student_dashboard():	Function definition `student_dashboard` with no parameters.
1498	if not check_auth('student'):	Starts an `if` block that runs only when its condition is true.
1499	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1500		Blank line used to separate logical blocks of Python code.
1501	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
1502	cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
1503	sid = session['user_id']	Assignment statement: stores the expression result in a variable/field.
1504		Blank line used to separate logical blocks of Python code.
1505	# Dashboard stats	Comment line used to document intent or create section dividers.
1506	cur.execute("SELECT COUNT(*) FROM student_enrollments WHERE student_id=%s", (sid,))	Executes a SQL query/command using the active database cursor.
1507	course_cnt = cur.fetchone()[0]	Reads query results from the cursor.
1508		Blank line used to separate logical blocks of Python code.
1509	cur.execute("SELECT COUNT(*) FROM student_results WHERE student_id=%s", (sid,))	Executes a SQL query/command using the active database cursor.
1510	res_count = cur.fetchone()[0]	Reads query results from the cursor.
1511		Blank line used to separate logical blocks of Python code.
1512	cur.execute("SELECT SUM(amount) FROM fees WHERE student_id=%s AND status='Pending'", (sid,))	Executes a SQL query/command using the active database cursor.
1513	fee_row = cur.fetchone()	Reads query results from the cursor.
1514	fee_due = fee_row[0] if fee_row[0] else 0	Assignment statement: stores the expression result in a variable/field.
1515		Blank line used to separate logical blocks of Python code.
1516	# Fetch notices for students	Comment line used to document intent or create section dividers.
1517	cur.execute("""	Executes a SQL query/command using the active database cursor.
1518	SELECT notice_id, title, content	General Python statement that participates in program control flow.

Line	Code	Explanation
1519	FROM notices	General Python statement that participates in program control flow.
1520	WHERE target_role IN ('all', 'student')	General Python statement that participates in program control flow.
1521	ORDER BY date_posted DESC	General Python statement that participates in program control flow.
1522	""")	General Python statement that participates in program control flow.
1523	notices = cur.fetchall() # returns list of tuples	Reads query results from the cursor.
1524		Blank line used to separate logical blocks of Python code.
1525	conn.close()	Closes a resource such as a cursor, connection, or file.
1526		Blank line used to separate logical blocks of Python code.
1527	return render_template(	Returns a value (or nothing) from the current function.
1528	'student_dash.html',	General Python statement that participates in program control flow.
1529	course_cnt=course_cnt,	Assignment statement: stores the expression result in a variable/field.
1530	res_count=res_count,	Assignment statement: stores the expression result in a variable/field.
1531	fee_due=fee_due,	Assignment statement: stores the expression result in a variable/field.
1532	notices=notices	Assignment statement: stores the expression result in a variable/field.
1533	)	General Python statement that participates in program control flow.
1534		Blank line used to separate logical blocks of Python code.
1535		Blank line used to separate logical blocks of Python code.
1536		Blank line used to separate logical blocks of Python code.
1537	@app.route('/student/enroll', methods=['GET', 'POST'])	Flask route decorator: binds the next function to a URL endpoint.
1538	def student_enroll():	Function definition `student_enroll` with no parameters.
1539	if not check_auth('student'): return redirect(url_for('login'))	General Python statement that participates in program control flow.
1540	conn = get_db(); cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1541	sid = session['user_id']	Assignment statement: stores the expression result in a variable/field.
1542	if request.method == 'POST':	Starts an `if` block that runs only when its condition is true.
1543	cid = request.form['course_id']	Assignment statement: stores the expression result in a variable/field.
1544	cur.execute("SELECT * FROM student_enrollments WHERE student_id=%s AND course_id=%s", (sid, cid))	Executes a SQL query/command using the active database cursor.
1545	if cur.fetchone(): flash("Already Enrolled")	Reads query results from the cursor.
1546	else:	Fallback `else` branch for the current conditional.
1547	cur.execute("INSERT INTO student_enrollments (student_id, course_id, date_enrolled) VALUES (%s, %s, %s)", (sid, cid, date.today()))	Executes a SQL query/command using the active database cursor.
1548	conn.commit(); flash("Enrolled Successfully")	Commits the active database transaction.
1549	return redirect(url_for('student_enroll'))	Returns a value (or nothing) from the current function.
1550	cur.execute("SELECT dept_id FROM users WHERE user_id=%s", (sid,)); row = cur.fetchone()	Executes a SQL query/command using the active database cursor.
1551	dept_id = row['dept_id'] if row else None	Assignment statement: stores the expression result in a variable/field.
1552	if dept_id:	Starts an `if` block that runs only when its condition is true.
1553	cur.execute("SELECT c.course_id, c.course_code, c.course_name FROM courses c WHERE c.dept_id=%s AND c.course_id NOT IN (SELECT course_id FROM student_enrollments WHERE student_id=%s)", (dept_id, sid))	Executes a SQL query/command using the active database cursor.
1554	available = cur.fetchall()	Reads query results from the cursor.
1555	else: available = []	General Python statement that participates in program control flow.

Line	Code	Explanation
1556	cur.execute("SELECT c.course_code, c.course_name, e.date_enrolled FROM student_enrollments e JOIN courses c ON e.course_id=c.course_id WHERE e.student_id=%s", (sid,))	Executes a SQL query/command using the active database cursor.
1557	enrolled = cur.fetchall()	Reads query results from the cursor.
1558	conn.close()	Closes a resource such as a cursor, connection, or file.
1559	return render_template('student_enroll.html', available=available, enrolled=enrolled)	Returns a value (or nothing) from the current function.
1560		Blank line used to separate logical blocks of Python code.
1561	@app.route('/student/results')	Flask route decorator: binds the next function to a URL endpoint.
1562	def student_results():	Function definition `student_results` with no parameters.
1563	if not check_auth('student'):	Starts an `if` block that runs only when its condition is true.
1564	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1565		Blank line used to separate logical blocks of Python code.
1566	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
1567	cur = conn.cursor()	Assignment statement: stores the expression result in a variable/field.
1568		Blank line used to separate logical blocks of Python code.
1569	cur.execute("""	Executes a SQL query/command using the active database cursor.
1570	SELECT	General Python statement that participates in program control flow.
1571	c.course_code,	General Python statement that participates in program control flow.
1572	c.course_name,	General Python statement that participates in program control flow.
1573	r.marks_obtained,	General Python statement that participates in program control flow.
1574	r.full_marks,	General Python statement that participates in program control flow.
1575	r.grade,	General Python statement that participates in program control flow.
1576	r.gpa,	General Python statement that participates in program control flow.
1577	r.exam_type	General Python statement that participates in program control flow.
1578	FROM student_results r	General Python statement that participates in program control flow.
1579	JOIN courses c ON r.course_id = c.course_id	General Python statement that participates in program control flow.
1580	WHERE r.student_id = %s	General Python statement that participates in program control flow.
1581	ORDER BY r.exam_type, c.course_name	General Python statement that participates in program control flow.
1582	""", (session['user_id'],))	General Python statement that participates in program control flow.
1583		Blank line used to separate logical blocks of Python code.
1584	results = cur.fetchall()	Reads query results from the cursor.
1585	conn.close()	Closes a resource such as a cursor, connection, or file.
1586		Blank line used to separate logical blocks of Python code.
1587	return render_template('student_results.html', results=results)	Returns a value (or nothing) from the current function.
1588		Blank line used to separate logical blocks of Python code.
1589	@app.route('/student/attendance')	Flask route decorator: binds the next function to a URL endpoint.
1590	def student_attendance():	Function definition `student_attendance` with no parameters.
1591	if not check_auth('student'):	Starts an `if` block that runs only when its condition is true.
1592	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1593		Blank line used to separate logical blocks of Python code.
1594	conn = get_db()	Assignment statement: stores the expression result in a variable/field.

Line	Code	Explanation
1595	<code>cur = conn.cursor()</code>	Assignment statement: stores the expression result in a variable/field.
1596		Blank line used to separate logical blocks of Python code.
1597	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1598	<code>SELECT</code>	General Python statement that participates in program control flow.
1599	<code>c.course_name,</code>	General Python statement that participates in program control flow.
1600	<code>COUNT(*) AS total_classes,</code>	General Python statement that participates in program control flow.
1601	<code>SUM(CASE WHEN a.status='Present' THEN 1 ELSE 0 END) AS present_classes</code>	General Python statement that participates in program control flow.
1602	<code>FROM attendance a</code>	General Python statement that participates in program control flow.
1603	<code>JOIN courses c ON a.course_id = c.course_id</code>	General Python statement that participates in program control flow.
1604	<code>WHERE a.student_id = %s</code>	General Python statement that participates in program control flow.
1605	<code>GROUP BY c.course_id</code>	General Python statement that participates in program control flow.
1606	<code>""", (session['user_id'],))</code>	General Python statement that participates in program control flow.
1607		Blank line used to separate logical blocks of Python code.
1608	<code>records = cur.fetchall()</code>	Reads query results from the cursor.
1609	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1610		Blank line used to separate logical blocks of Python code.
1611	<code>return render_template('student_attendance.html', records=records)</code>	Returns a value (or nothing) from the current function.
1612		Blank line used to separate logical blocks of Python code.
1613		Blank line used to separate logical blocks of Python code.
1614	<code>@app.route('/student/fees')</code>	Flask route decorator: binds the next function to a URL endpoint.
1615	<code>def student_fees():</code>	Function definition `student_fees` with no parameters.
1616	<code>if not check_auth('student'):</code>	Starts an `if` block that runs only when its condition is true.
1617	<code>return redirect(url_for('login'))</code>	Returns a value (or nothing) from the current function.
1618		Blank line used to separate logical blocks of Python code.
1619	<code>conn = get_db()</code>	Assignment statement: stores the expression result in a variable/field.
1620	<code>cur = conn.cursor(dictionary=True) # <- IMPORTANT</code>	Assignment statement: stores the expression result in a variable/field.
1621	<code>cur.execute("SELECT * FROM fees WHERE student_id=%s", (session['user_id'],))</code>	Executes a SQL query/command using the active database cursor.
1622	<code>fees = cur.fetchall()</code>	Reads query results from the cursor.
1623	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1624		Blank line used to separate logical blocks of Python code.
1625	<code>return render_template('student_fees.html', fees=fees)</code>	Returns a value (or nothing) from the current function.
1626		Blank line used to separate logical blocks of Python code.
1627		Blank line used to separate logical blocks of Python code.
1628	<code>@app.route('/student/exams')</code>	Flask route decorator: binds the next function to a URL endpoint.
1629	<code>def student_exams():</code>	Function definition `student_exams` with no parameters.
1630	<code>if not check_auth('student'):</code>	Starts an `if` block that runs only when its condition is true.
1631	<code>return redirect(url_for('login'))</code>	Returns a value (or nothing) from the current function.
1632		Blank line used to separate logical blocks of Python code.
1633	<code>conn = get_db()</code>	Assignment statement: stores the expression result in a variable/field.
1634	<code>cur = conn.cursor() # ■ NO dictionary=True</code>	Assignment statement: stores the expression result in a variable/field.
1635		Blank line used to separate logical blocks of Python code.

Line	Code	Explanation
1636	cur.execute("""	Executes a SQL query/command using the active database cursor.
1637	SELECT	General Python statement that participates in program control flow.
1638	c.course_name,	General Python statement that participates in program control flow.
1639	e.exam_date,	General Python statement that participates in program control flow.
1640	e.start_time,	General Python statement that participates in program control flow.
1641	e.room_no	General Python statement that participates in program control flow.
1642	FROM exam_schedule e	General Python statement that participates in program control flow.
1643	JOIN courses c ON e.course_id = c.course_id	General Python statement that participates in program control flow.
1644	ORDER BY e.exam_date, e.start_time	General Python statement that participates in program control flow.
1645	""")	General Python statement that participates in program control flow.
1646		Blank line used to separate logical blocks of Python code.
1647	exams = cur.fetchall()	Reads query results from the cursor.
1648	conn.close()	Closes a resource such as a cursor, connection, or file.
1649		Blank line used to separate logical blocks of Python code.
1650	return render_template('student_exams.html', exams=exams)	Returns a value (or nothing) from the current function.
1651		Blank line used to separate logical blocks of Python code.
1652	# =====	Comment line used to document intent or create section dividers.
1653	# STUDENT LIBRARY ROUTES (ADDED)	Comment line used to document intent or create section dividers.
1654	# =====	Comment line used to document intent or create section dividers.
1655		Blank line used to separate logical blocks of Python code.
1656	@app.route('/student/library')	Flask route decorator: binds the next function to a URL endpoint.
1657	def library_view():	Function definition `library_view` with no parameters.
1658	if not check_auth('student'):	Starts an `if` block that runs only when its condition is true.
1659	return redirect(url_for('login'))	Returns a value (or nothing) from the current function.
1660		Blank line used to separate logical blocks of Python code.
1661	conn = get_db()	Assignment statement: stores the expression result in a variable/field.
1662	cur = conn.cursor(dictionary=True)	Assignment statement: stores the expression result in a variable/field.
1663		Blank line used to separate logical blocks of Python code.
1664	# Fetch books with calculated availability (Total - Unreturned Borrows)	Comment line used to document intent or create section dividers.
1665	query = """	Assignment statement: stores the expression result in a variable/field.
1666	SELECT b.book_id, b.title, b.author, b.isbn, b.copies_total,	General Python statement that participates in program control flow.
1667	c.name as category_name,	General Python statement that participates in program control flow.
1668	(b.copies_total - (	General Python statement that participates in program control flow.
1669	SELECT COUNT(*)	General Python statement that participates in program control flow.
1670	FROM borrows	General Python statement that participates in program control flow.
1671	WHERE book_id = b.book_id AND return_date IS NULL	General Python statement that participates in program control flow.
1672	)) as copies_available	General Python statement that participates in program control flow.
1673	FROM library_books b	General Python statement that participates in program control flow.
1674	LEFT JOIN book_categories c ON b.category_id = c.category_id	General Python statement that participates in program control flow.
1675	ORDER BY b.title	General Python statement that participates in program control flow.
1676	"""	General Python statement that participates in program control flow.

Line	Code	Explanation
1677	<code>cur.execute(query)</code>	Executes a SQL query/command using the active database cursor.
1678	<code>books = cur.fetchall()</code>	Reads query results from the cursor.
1679		Blank line used to separate logical blocks of Python code.
1680	<code># Sanitize None values in case database is fresh</code>	Comment line used to document intent or create section dividers.
1681	<code>for book in books:</code>	Starts a `for` loop over an iterable sequence.
1682	<code>if book['copies_available'] is None:</code>	Starts an `if` block that runs only when its condition is true.
1683	<code>book['copies_available'] = book['copies_total']</code>	Assignment statement: stores the expression result in a variable/field.
1684		Blank line used to separate logical blocks of Python code.
1685	<code># ■ Fetch this student's borrowed details for "My Borrows" section</code>	Comment line used to document intent or create section dividers.
1686	<code>student_id = session['user_id']</code>	Assignment statement: stores the expression result in a variable/field.
1687	<code>my_borrows_query = """</code>	Assignment statement: stores the expression result in a variable/field.
1688	<code>SELECT</code>	General Python statement that participates in program control flow.
1689	<code>br.borrow_id,</code>	General Python statement that participates in program control flow.
1690	<code>b.title AS book_title,</code>	General Python statement that participates in program control flow.
1691	<code>br.borrow_date,</code>	General Python statement that participates in program control flow.
1692	<code>br.due_date,</code>	General Python statement that participates in program control flow.
1693	<code>br.return_date,</code>	General Python statement that participates in program control flow.
1694	<code>br.fine</code>	General Python statement that participates in program control flow.
1695	<code>FROM borrows br</code>	General Python statement that participates in program control flow.
1696	<code>JOIN library_books b ON br.book_id = b.book_id</code>	General Python statement that participates in program control flow.
1697	<code>WHERE br.student_id = %s AND br.return_date IS NULL</code>	General Python statement that participates in program control flow.
1698	<code>ORDER BY br.borrow_date DESC</code>	General Python statement that participates in program control flow.
1699	<code>"""</code>	General Python statement that participates in program control flow.
1700	<code>cur.execute(my_borrows_query, (student_id,))</code>	Executes a SQL query/command using the active database cursor.
1701	<code>my_borrows = cur.fetchall()</code>	Reads query results from the cursor.
1702		Blank line used to separate logical blocks of Python code.
1703	<code># ■ Compute is_overdue exactly how your template expects</code>	Comment line used to document intent or create section dividers.
1704	<code>today = date.today()</code>	Assignment statement: stores the expression result in a variable/field.
1705	<code>for br in my_borrows:</code>	Starts a `for` loop over an iterable sequence.
1706	<code>br['fine'] = br['fine'] if br['fine'] is not None else 0</code>	Assignment statement: stores the expression result in a variable/field.
1707	<code>br['is_overdue'] = (br['return_date'] is None and br['due_date'] is not None and br['due_date'] < today)</code>	Assignment statement: stores the expression result in a variable/field.
1708		Blank line used to separate logical blocks of Python code.
1709	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1710	<code>return render_template('library_view.html', books=books, my_borrows=my_borrows)</code>	Returns a value (or nothing) from the current function.
1711		Blank line used to separate logical blocks of Python code.
1712		Blank line used to separate logical blocks of Python code.
1713	<code>@app.route('/student/borrow/<int:book_id>')</code>	Flask route decorator: binds the next function to a URL endpoint.
1714	<code>def student_borrow_book(book_id):</code>	Function definition `student_borrow_book` with book_id.
1715	<code>if not check_auth('student'):</code>	Starts an `if` block that runs only when its condition is true.

Line	Code	Explanation
1716	<code>return redirect(url_for('login'))</code>	Returns a value (or nothing) from the current function.
1717		Blank line used to separate logical blocks of Python code.
1718	<code>student_id = session['user_id']</code>	Assignment statement: stores the expression result in a variable/field.
1719	<code>today = date.today()</code>	Assignment statement: stores the expression result in a variable/field.
1720	<code>due_date = today + timedelta(days=14) # 2 weeks due date</code>	Assignment statement: stores the expression result in a variable/field.
1721		Blank line used to separate logical blocks of Python code.
1722	<code>conn = get_db()</code>	Assignment statement: stores the expression result in a variable/field.
1723	<code>cur = conn.cursor(dictionary=True)</code>	Assignment statement: stores the expression result in a variable/field.
1724		Blank line used to separate logical blocks of Python code.
1725	<code># 1. Check if student already has THIS book borrowed and not returned</code>	Comment line used to document intent or create section dividers.
1726	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1727	<code>SELECT * FROM borrows</code>	General Python statement that participates in program control flow.
1728	<code>WHERE student_id = %s AND book_id = %s AND return_date IS NULL</code>	General Python statement that participates in program control flow.
1729	<code>""", (student_id, book_id))</code>	General Python statement that participates in program control flow.
1730	<code>existing_borrow = cur.fetchone()</code>	Reads query results from the cursor.
1731		Blank line used to separate logical blocks of Python code.
1732	<code>if existing_borrow:</code>	Starts an `if` block that runs only when its condition is true.
1733	<code>flash("You have already borrowed this book and haven't returned it yet.",</code> <code>"warning")</code>	Function/method call that performs an action at runtime.
1734	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1735	<code>return redirect(url_for('library_view'))</code>	Returns a value (or nothing) from the current function.
1736		Blank line used to separate logical blocks of Python code.
1737	<code># 2. Check stock availability</code>	Comment line used to document intent or create section dividers.
1738	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1739	<code>SELECT (copies_total - (</code>	General Python statement that participates in program control flow.
1740	<code>SELECT COUNT(*) FROM borrows WHERE book_id = %s AND return_date IS NULL</code>	General Python statement that participates in program control flow.
1741	<code>)) as available</code>	General Python statement that participates in program control flow.
1742	<code>FROM library_books WHERE book_id = %s</code>	General Python statement that participates in program control flow.
1743	<code>""", (book_id, book_id))</code>	General Python statement that participates in program control flow.
1744	<code>result = cur.fetchone()</code>	Reads query results from the cursor.
1745		Blank line used to separate logical blocks of Python code.
1746	<code>if result and result['available'] > 0:</code>	Starts an `if` block that runs only when its condition is true.
1747	<code># 3. Issue the book</code>	Comment line used to document intent or create section dividers.
1748	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1749	<code>INSERT INTO borrows (book_id, student_id, borrow_date, due_date)</code>	General Python statement that participates in program control flow.
1750	<code>VALUES (%s, %s, %s, %s)</code>	General Python statement that participates in program control flow.
1751	<code>""", (book_id, student_id, today, due_date))</code>	General Python statement that participates in program control flow.
1752	<code>conn.commit()</code>	Commits the active database transaction.
1753	<code>flash(f"Book borrowed successfully! Please return by {due_date}", "success")</code>	Function/method call that performs an action at runtime.
1754	<code>else:</code>	Fallback `else` branch for the current conditional.
1755	<code>flash("Sorry, this book is currently out of stock.", "danger")</code>	Function/method call that performs an action at runtime.

Line	Code	Explanation
1756		Blank line used to separate logical blocks of Python code.
1757	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1758	<code>return redirect(url_for('library_view'))</code>	Returns a value (or nothing) from the current function.
1759		Blank line used to separate logical blocks of Python code.
1760		Blank line used to separate logical blocks of Python code.
1761	<code># ■ This route is required because your library_view.html calls url_for('student_return_book', borrow_id=...)</code>	Comment line used to document intent or create section dividers.
1762	<code>@app.route('/student/return/<int:borrow_id>')</code>	Flask route decorator: binds the next function to a URL endpoint.
1763	<code>def student_return_book(borrow_id):</code>	Function definition `student_return_book` with <code>borrow_id</code> .
1764	<code>if not check_auth('student'):</code>	Starts an `if` block that runs only when its condition is true.
1765	<code>return redirect(url_for('login'))</code>	Returns a value (or nothing) from the current function.
1766		Blank line used to separate logical blocks of Python code.
1767	<code>student_id = session['user_id']</code>	Assignment statement: stores the expression result in a variable/field.
1768	<code>today = date.today()</code>	Assignment statement: stores the expression result in a variable/field.
1769		Blank line used to separate logical blocks of Python code.
1770	<code>conn = get_db()</code>	Assignment statement: stores the expression result in a variable/field.
1771	<code>cur = conn.cursor(dictionary=True)</code>	Assignment statement: stores the expression result in a variable/field.
1772		Blank line used to separate logical blocks of Python code.
1773	<code># Only allow returning your own active borrow record</code>	Comment line used to document intent or create section dividers.
1774	<code>cur.execute("""</code>	Executes a SQL query/command using the active database cursor.
1775	<code>UPDATE borrows</code>	General Python statement that participates in program control flow.
1776	<code>SET return_date = %s</code>	General Python statement that participates in program control flow.
1777	<code>WHERE borrow_id = %s AND student_id = %s AND return_date IS NULL</code>	General Python statement that participates in program control flow.
1778	<code>""", (today, borrow_id, student_id))</code>	General Python statement that participates in program control flow.
1779		Blank line used to separate logical blocks of Python code.
1780	<code>conn.commit()</code>	Commits the active database transaction.
1781	<code>conn.close()</code>	Closes a resource such as a cursor, connection, or file.
1782		Blank line used to separate logical blocks of Python code.
1783	<code>flash("Book returned successfully.", "success")</code>	Function/method call that performs an action at runtime.
1784	<code>return redirect(url_for('library_view'))</code>	Returns a value (or nothing) from the current function.
1785		Blank line used to separate logical blocks of Python code.
1786		Blank line used to separate logical blocks of Python code.
1787	<code>if __name__ == '__main__':</code>	Starts an `if` block that runs only when its condition is true.
1788	<code>app.run(debug=True)</code>	General Python statement that participates in program control flow.

File: static/style.css (Lines: 1734)

Line	Code	Explanation
1	/* =====	CSS comment used to document the stylesheet.
2	NCIT_SIS - Modern Visual Theme	CSS selector/value syntax line used by nearby declarations.
3	===== */	CSS comment used to document the stylesheet.
4	@import url("https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@400;500;600;700&family=Fraunces:opsz,wght@9..144,600;700&display=swap");	Imports external stylesheet content (for example web fonts).
5		Blank line used to group related style declarations.
6	:root {	Opens a CSS rule block for the selector on this line.
7	--bg-0: #f6f9fc;	Defines CSS custom property `--bg-0` with value `#f6f9fc`.
8	--bg-1: #eef3f8;	Defines CSS custom property `--bg-1` with value `#eef3f8`.
9	--ink: #0b1220;	Defines CSS custom property `--ink` with value `#0b1220`.
10	--ink-2: #1f2937;	Defines CSS custom property `--ink-2` with value `#1f2937`.
11	--muted: #6b7280;	Defines CSS custom property `--muted` with value `#6b7280`.
12	--card: rgba(255, 255, 255, 0.84);	Defines CSS custom property `--card` with value `rgba(255, 255, 255, 0.84)`.
13	--card-solid: #ffffff;	Defines CSS custom property `--card-solid` with value `#ffffff`.
14	--line: rgba(15, 23, 42, 0.09);	Defines CSS custom property `--line` with value `rgba(15, 23, 42, 0.09)`.
15	--accent: #00b7ff;	Defines CSS custom property `--accent` with value `#00b7ff`.
16	--accent-2: #2dd4bf;	Defines CSS custom property `--accent-2` with value `#2dd4bf`.
17	--accent-3: #3b82f6;	Defines CSS custom property `--accent-3` with value `#3b82f6`.
18	--success: #22c55e;	Defines CSS custom property `--success` with value `#22c55e`.
19	--warning: #f59e0b;	Defines CSS custom property `--warning` with value `#f59e0b`.
20	--danger: #ef4444;	Defines CSS custom property `--danger` with value `#ef4444`.
21	--sidebar-w: 260px;	Defines CSS custom property `--sidebar-w` with value `260px`.
22	--header-h: 72px;	Defines CSS custom property `--header-h` with value `72px`.
23	--radius: 16px;	Defines CSS custom property `--radius` with value `16px`.
24	--radius-sm: 12px;	Defines CSS custom property `--radius-sm` with value `12px`.
25	--shadow-sm: 0 6px 14px rgba(11, 18, 32, 0.08);	Defines CSS custom property `--shadow-sm` with value `0 6px 14px rgba(11, 18, 32, 0.08)`.
26	--shadow-md: 0 16px 36px rgba(11, 18, 32, 0.12);	Defines CSS custom property `--shadow-md` with value `0 16px 36px rgba(11, 18, 32, 0.12)`.
27	--shadow-lg: 0 28px 70px rgba(11, 18, 32, 0.18);	Defines CSS custom property `--shadow-lg` with value `0 28px 70px rgba(11, 18, 32, 0.18)`.
28	--glow: 0 0 0 6px rgba(0, 183, 255, 0.16);	Defines CSS custom property `--glow` with value `0 0 0 6px rgba(0, 183, 255, 0.16)`.
29	--transition: 180ms ease;	Defines CSS custom property `--transition` with value `180ms ease`.
30	--gradient-primary: linear-gradient(135deg, #00b7ff, #2dd4bf);	Defines CSS custom property `--gradient-primary` with value `linear-gradient(135deg, #00b7ff, #2dd4bf)`.
31	--gradient-deep: linear-gradient(135deg, #0f172a, #0b1220);	Defines CSS custom property `--gradient-deep` with value `linear-gradient(135deg, #0f172a, #0b1220)`.
32		Blank line used to group related style declarations.
33	/* Bootstrap theme overrides */	CSS comment used to document the stylesheet.
34	--bs-body-font-family: "Space Grotesk", sans-serif;	Defines CSS custom property `--bs-body-font-family` with value `"Space Grotesk", sans-serif`.
35	--bs-body-bg: var(--bg-0);	Defines CSS custom property `--bs-body-bg` with value `var(--bg-0)`.
36	--bs-body-color: var(--ink);	Defines CSS custom property `--bs-body-color` with value `var(--ink)`.
37	--bs-primary: var(--accent);	Defines CSS custom property `--bs-primary` with value `var(--accent)`.
38	--bs-secondary: #64748b;	Defines CSS custom property `--bs-secondary` with value `#64748b`.

Line	Code	Explanation
39	--bs-success: var(--success);	Defines CSS custom property `--bs-success` with value `var(--success)`.
40	--bs-info: #38bdf8;	Defines CSS custom property `--bs-info` with value `#38bdf8`.
41	--bs-warning: var(--warning);	Defines CSS custom property `--bs-warning` with value `var(--warning)`.
42	--bs-danger: var(--danger);	Defines CSS custom property `--bs-danger` with value `var(--danger)`.
43	--bs-dark: #0b1220;	Defines CSS custom property `--bs-dark` with value `#0b1220`.
44	--bs-border-color: var(--line);	Defines CSS custom property `--bs-border-color` with value `var(--line)`.
45	--bs-border-radius: 0.85rem;	Defines CSS custom property `--bs-border-radius` with value `0.85rem`.
46	}	Closes the current CSS rule block.
47		Blank line used to group related style declarations.
48	* {	CSS comment used to document the stylesheet.
49	box-sizing: border-box;	Sets CSS property `box-sizing` to `border-box` for the active selector.
50	}	Closes the current CSS rule block.
51		Blank line used to group related style declarations.
52	html {	Opens a CSS rule block for the selector on this line.
53	scroll-behavior: smooth;	Sets CSS property `scroll-behavior` to `smooth` for the active selector.
54	}	Closes the current CSS rule block.
55		Blank line used to group related style declarations.
56	body {	Opens a CSS rule block for the selector on this line.
57	margin: 0;	Sets CSS property `margin` to `0` for the active selector.
58	min-height: 100vh;	Sets CSS property `min-height` to `100vh` for the active selector.
59	display: flex;	Sets CSS property `display` to `flex` for the active selector.
60	font-family: "Space Grotesk", sans-serif;	Sets CSS property `font-family` to `"Space Grotesk", sans-serif` for the active selector.
61	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
62	line-height: 1.6;	Sets CSS property `line-height` to `1.6` for the active selector.
63	background-image:	Starts a multi-line CSS property value that continues on next lines.
64	radial-gradient(1200px 600px at 10% -10%, rgba(0, 183, 255, 0.18), transparent 60%),	CSS selector/value syntax line used by nearby declarations.
65	radial-gradient(900px 600px at 90% 10%, rgba(45, 212, 191, 0.18), transparent 60%),	CSS selector/value syntax line used by nearby declarations.
66	radial-gradient(900px 600px at 60% 110%, rgba(59, 130, 246, 0.12), transparent 60%),	CSS selector/value syntax line used by nearby declarations.
67	linear-gradient(180deg, var(--bg-0) 0%, var(--bg-1) 100%);	CSS selector/value syntax line used by nearby declarations.
68	background-attachment: fixed;	Sets CSS property `background-attachment` to `fixed` for the active selector.
69	position: relative;	Sets CSS property `position` to `relative` for the active selector.
70	}	Closes the current CSS rule block.
71		Blank line used to group related style declarations.
72	body::before {	Opens a CSS rule block for the selector on this line.
73	content: "";	Sets CSS property `content` to `""` for the active selector.
74	position: fixed;	Sets CSS property `position` to `fixed` for the active selector.
75	inset: 0;	Sets CSS property `inset` to `0` for the active selector.
76	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
77	z-index: 0;	Sets CSS property `z-index` to `0` for the active selector.

Line	Code	Explanation
78	opacity: 0.15;	Sets CSS property `opacity` to `0.15` for the active selector.
79	background-image:	Starts a multi-line CSS property value that continues on next lines.
80	linear-gradient(transparent 0 31px, rgba(15, 23, 42, 0.03) 32px),	CSS selector/value syntax line used by nearby declarations.
81	linear-gradient(90deg, transparent 0 31px, rgba(15, 23, 42, 0.03) 32px);	CSS selector/value syntax line used by nearby declarations.
82	background-size: 32px 32px;	Sets CSS property `background-size` to `32px 32px` for the active selector.
83	}	Closes the current CSS rule block.
84		Blank line used to group related style declarations.
85	body::after {	Opens a CSS rule block for the selector on this line.
86	content: "";	Sets CSS property `content` to `""` for the active selector.
87	position: fixed;	Sets CSS property `position` to `fixed` for the active selector.
88	inset: 0;	Sets CSS property `inset` to `0` for the active selector.
89	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
90	z-index: 0;	Sets CSS property `z-index` to `0` for the active selector.
91	opacity: 0.12;	Sets CSS property `opacity` to `0.12` for the active selector.
92	background-image:	Starts a multi-line CSS property value that continues on next lines.
93	radial-gradient(240px 240px at 20% 30%, rgba(0, 183, 255, 0.35), transparent 65%),	CSS selector/value syntax line used by nearby declarations.
94	radial-gradient(260px 260px at 85% 18%, rgba(45, 212, 191, 0.35), transparent 65%),	CSS selector/value syntax line used by nearby declarations.
95	radial-gradient(280px 280px at 70% 80%, rgba(59, 130, 246, 0.35), transparent 65%);	CSS selector/value syntax line used by nearby declarations.
96	}	Closes the current CSS rule block.
97		Blank line used to group related style declarations.
98	img {	Opens a CSS rule block for the selector on this line.
99	max-width: 100%;	Sets CSS property `max-width` to `100%` for the active selector.
100	display: block;	Sets CSS property `display` to `block` for the active selector.
101	}	Closes the current CSS rule block.
102		Blank line used to group related style declarations.
103	a {	Opens a CSS rule block for the selector on this line.
104	color: inherit;	Sets CSS property `color` to `inherit` for the active selector.
105	}	Closes the current CSS rule block.
106		Blank line used to group related style declarations.
107	h1, h2, h3, h4 {	Opens a CSS rule block for the selector on this line.
108	font-family: "Fraunces", serif;	Sets CSS property `font-family` to `"Fraunces", serif` for the active selector.
109	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
110	letter-spacing: 0.2px;	Sets CSS property `letter-spacing` to `0.2px` for the active selector.
111	margin: 0 0 0.6rem 0;	Sets CSS property `margin` to `0 0 0.6rem 0` for the active selector.
112	}	Closes the current CSS rule block.
113		Blank line used to group related style declarations.
114	/* =====	CSS comment used to document the stylesheet.
115	LAYOUT: SIDEBAR & MAIN CONTENT	Starts a multi-line CSS property value that continues on next lines.
116	===== */	CSS comment used to document the stylesheet.

Line	Code	Explanation
117	.sidebar {	Opens a CSS rule block for the selector on this line.
118	width: var(--sidebar-w);	Sets CSS property `width` to `var(--sidebar-w)` for the active selector.
119	background: linear-gradient(180deg, #0b1220 0%, #111827 60%, #0b1220 100%);	Sets CSS property `background` to `linear-gradient(180deg, #0b1220 0%, #111827 60%, #0b1220 100%)` for the active selector.
120	color: #e2e8f0;	Sets CSS property `color` to `#e2e8f0` for the active selector.
121	position: fixed;	Sets CSS property `position` to `fixed` for the active selector.
122	top: 0;	Sets CSS property `top` to `0` for the active selector.
123	left: 0;	Sets CSS property `left` to `0` for the active selector.
124	height: 100vh;	Sets CSS property `height` to `100vh` for the active selector.
125	display: flex;	Sets CSS property `display` to `flex` for the active selector.
126	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
127	box-shadow: 2px 0 20px rgba(2, 6, 23, 0.25);	Sets CSS property `box-shadow` to `2px 0 20px rgba(2, 6, 23, 0.25)` for the active selector.
128	border-right: 1px solid rgba(255, 255, 255, 0.08);	Sets CSS property `border-right` to `1px solid rgba(255, 255, 255, 0.08)` for the active selector.
129	z-index: 2;	Sets CSS property `z-index` to `2` for the active selector.
130	}	Closes the current CSS rule block.
131		Blank line used to group related style declarations.
132	.brand {	Opens a CSS rule block for the selector on this line.
133	padding: 24px 22px;	Sets CSS property `padding` to `24px 22px` for the active selector.
134	font-size: 1.4rem;	Sets CSS property `font-size` to `1.4rem` for the active selector.
135	font-weight: 700;	Sets CSS property `font-weight` to `700` for the active selector.
136	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
137	background: rgba(255, 255, 255, 0.05);	Sets CSS property `background` to `rgba(255, 255, 255, 0.05)` for the active selector.
138	border-bottom: 1px solid rgba(255, 255, 255, 0.08);	Sets CSS property `border-bottom` to `1px solid rgba(255, 255, 255, 0.08)` for the active selector.
139	letter-spacing: 0.6px;	Sets CSS property `letter-spacing` to `0.6px` for the active selector.
140	}	Closes the current CSS rule block.
141		Blank line used to group related style declarations.
142	.menu {	Opens a CSS rule block for the selector on this line.
143	list-style: none;	Sets CSS property `list-style` to `none` for the active selector.
144	padding: 0;	Sets CSS property `padding` to `0` for the active selector.
145	margin: 18px 0;	Sets CSS property `margin` to `18px 0` for the active selector.
146	flex-grow: 1;	Sets CSS property `flex-grow` to `1` for the active selector.
147	overflow-y: auto;	Sets CSS property `overflow-y` to `auto` for the active selector.
148	}	Closes the current CSS rule block.
149		Blank line used to group related style declarations.
150	.menu li a {	Opens a CSS rule block for the selector on this line.
151	display: flex;	Sets CSS property `display` to `flex` for the active selector.
152	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
153	gap: 10px;	Sets CSS property `gap` to `10px` for the active selector.
154	padding: 12px 16px;	Sets CSS property `padding` to `12px 16px` for the active selector.
155	margin: 6px 12px;	Sets CSS property `margin` to `6px 12px` for the active selector.
156	border-radius: 10px;	Sets CSS property `border-radius` to `10px` for the active selector.
157	color: #cbd5f5;	Sets CSS property `color` to `#cbd5f5` for the active selector.

Line	Code	Explanation
158	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
159	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
160	position: relative;	Sets CSS property `position` to `relative` for the active selector.
161	border: 1px solid transparent;	Sets CSS property `border` to `1px solid transparent` for the active selector.
162	}	Closes the current CSS rule block.
163		Blank line used to group related style declarations.
164	.menu li a::before {	Opens a CSS rule block for the selector on this line.
165	content: "";	Sets CSS property `content` to `""` for the active selector.
166	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
167	left: -6px;	Sets CSS property `left` to `-6px` for the active selector.
168	top: 8px;	Sets CSS property `top` to `8px` for the active selector.
169	bottom: 8px;	Sets CSS property `bottom` to `8px` for the active selector.
170	width: 3px;	Sets CSS property `width` to `3px` for the active selector.
171	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
172	background: var(--accent);	Sets CSS property `background` to `var(--accent)` for the active selector.
173	opacity: 0;	Sets CSS property `opacity` to `0` for the active selector.
174	transform: scaleY(0.6);	Sets CSS property `transform` to `scaleY(0.6)` for the active selector.
175	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
176	}	Closes the current CSS rule block.
177		Blank line used to group related style declarations.
178	.menu li a:hover,	CSS selector/value syntax line used by nearby declarations.
179	.menu li a.active {	Opens a CSS rule block for the selector on this line.
180	background: rgba(0, 183, 255, 0.18);	Sets CSS property `background` to `rgba(0, 183, 255, 0.18)` for the active selector.
181	color: #ffffff;	Sets CSS property `color` to `#ffffff` for the active selector.
182	border-color: rgba(0, 183, 255, 0.35);	Sets CSS property `border-color` to `rgba(0, 183, 255, 0.35)` for the active selector.
183	}	Closes the current CSS rule block.
184		Blank line used to group related style declarations.
185	.menu li a:hover::before,	CSS selector/value syntax line used by nearby declarations.
186	.menu li a.active::before {	Opens a CSS rule block for the selector on this line.
187	opacity: 1;	Sets CSS property `opacity` to `1` for the active selector.
188	transform: scaleY(1);	Sets CSS property `transform` to `scaleY(1)` for the active selector.
189	}	Closes the current CSS rule block.
190		Blank line used to group related style declarations.
191	.menu i {	Opens a CSS rule block for the selector on this line.
192	width: 22px;	Sets CSS property `width` to `22px` for the active selector.
193	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
194	opacity: 0.9;	Sets CSS property `opacity` to `0.9` for the active selector.
195	}	Closes the current CSS rule block.
196		Blank line used to group related style declarations.
197	.logout {	Opens a CSS rule block for the selector on this line.
198	padding: 18px;	Sets CSS property `padding` to `18px` for the active selector.

Line	Code	Explanation
199	border-top: 1px solid rgba(255, 255, 255, 0.08);	Sets CSS property `border-top` to `1px solid rgba(255, 255, 255, 0.08)` for the active selector.
200	}	Closes the current CSS rule block.
201		Blank line used to group related style declarations.
202	.logout a {	Opens a CSS rule block for the selector on this line.
203	color: #fecaca !important;	Sets CSS property `color` to `#fecaca !important` for the active selector.
204	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
205	display: flex;	Sets CSS property `display` to `flex` for the active selector.
206	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
207	gap: 10px;	Sets CSS property `gap` to `10px` for the active selector.
208	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
209	}	Closes the current CSS rule block.
210		Blank line used to group related style declarations.
211	/* Main Content Area */	CSS comment used to document the stylesheet.
212	.content {	Opens a CSS rule block for the selector on this line.
213	margin-left: var(--sidebar-w);	Sets CSS property `margin-left` to `var(--sidebar-w)` for the active selector.
214	width: calc(100% - var(--sidebar-w));	Sets CSS property `width` to `calc(100% - var(--sidebar-w))` for the active selector.
215	padding: 32px 36px 60px;	Sets CSS property `padding` to `32px 36px 60px` for the active selector.
216	position: relative;	Sets CSS property `position` to `relative` for the active selector.
217	z-index: 1;	Sets CSS property `z-index` to `1` for the active selector.
218	min-height: 100vh;	Sets CSS property `min-height` to `100vh` for the active selector.
219	}	Closes the current CSS rule block.
220		Blank line used to group related style declarations.
221	.overlay {	Opens a CSS rule block for the selector on this line.
222	position: fixed;	Sets CSS property `position` to `fixed` for the active selector.
223	inset: 0;	Sets CSS property `inset` to `0` for the active selector.
224	background: rgba(15, 23, 42, 0.45);	Sets CSS property `background` to `rgba(15, 23, 42, 0.45)` for the active selector.
225	backdrop-filter: blur(2px);	Sets CSS property `backdrop-filter` to `blur(2px)` for the active selector.
226	opacity: 0;	Sets CSS property `opacity` to `0` for the active selector.
227	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
228	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
229	z-index: 1;	Sets CSS property `z-index` to `1` for the active selector.
230	}	Closes the current CSS rule block.
231		Blank line used to group related style declarations.
232	.overlay.active {	Opens a CSS rule block for the selector on this line.
233	opacity: 1;	Sets CSS property `opacity` to `1` for the active selector.
234	pointer-events: auto;	Sets CSS property `pointer-events` to `auto` for the active selector.
235	}	Closes the current CSS rule block.
236		Blank line used to group related style declarations.
237	/* Header & Cards */	CSS comment used to document the stylesheet.
238	.header {	Opens a CSS rule block for the selector on this line.
239	display: flex;	Sets CSS property `display` to `flex` for the active selector.

Line	Code	Explanation
240	<code>justify-content: space-between;</code>	Sets CSS property `justify-content` to `space-between` for the active selector.
241	<code>align-items: center;</code>	Sets CSS property `align-items` to `center` for the active selector.
242	<code>gap: 18px;</code>	Sets CSS property `gap` to `18px` for the active selector.
243	<code>margin-bottom: 26px;</code>	Sets CSS property `margin-bottom` to `26px` for the active selector.
244	<code>background: linear-gradient(135deg, rgba(255, 255, 255, 0.96), rgba(255, 255, 255, 0.82));</code>	Sets CSS property `background` to `linear-gradient(135deg, rgba(255, 255, 255, 0.96), rgba(255, 255, 255, 0.82))` for the active selector.
245	<code>padding: 22px 24px;</code>	Sets CSS property `padding` to `22px 24px` for the active selector.
246	<code>border-radius: var(--radius);</code>	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
247	<code>border: 1px solid var(--line);</code>	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
248	<code>box-shadow: var(--shadow-sm);</code>	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
249	<code>backdrop-filter: blur(12px);</code>	Sets CSS property `backdrop-filter` to `blur(12px)` for the active selector.
250	<code>position: sticky;</code>	Sets CSS property `position` to `sticky` for the active selector.
251	<code>top: 0;</code>	Sets CSS property `top` to `0` for the active selector.
252	<code>z-index: 5;</code>	Sets CSS property `z-index` to `5` for the active selector.
253	<code>min-height: var(--header-h);</code>	Sets CSS property `min-height` to `var(--header-h)` for the active selector.
254	<code>overflow: visible;</code>	Sets CSS property `overflow` to `visible` for the active selector.
255	<code>}</code>	Closes the current CSS rule block.
256		Blank line used to group related style declarations.
257	<code>.header::before {</code>	Opens a CSS rule block for the selector on this line.
258	<code>content: "";</code>	Sets CSS property `content` to `""` for the active selector.
259	<code>position: absolute;</code>	Sets CSS property `position` to `absolute` for the active selector.
260	<code>left: 0;</code>	Sets CSS property `left` to `0` for the active selector.
261	<code>top: 0;</code>	Sets CSS property `top` to `0` for the active selector.
262	<code>width: 100%;</code>	Sets CSS property `width` to `100%` for the active selector.
263	<code>height: 3px;</code>	Sets CSS property `height` to `3px` for the active selector.
264	<code>background: var(--gradient-primary);</code>	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
265	<code>}</code>	Closes the current CSS rule block.
266		Blank line used to group related style declarations.
267	<code>.card {</code>	Opens a CSS rule block for the selector on this line.
268	<code>background: linear-gradient(180deg, rgba(255, 255, 255, 0.98), rgba(255, 255, 255, 0.86));</code>	Sets CSS property `background` to `linear-gradient(180deg, rgba(255, 255, 255, 0.98), rgba(255, 255, 255, 0.86))` for the active selector.
269	<code>padding: 24px;</code>	Sets CSS property `padding` to `24px` for the active selector.
270	<code>border-radius: var(--radius);</code>	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
271	<code>border: 1px solid var(--line);</code>	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
272	<code>box-shadow: var(--shadow-sm);</code>	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
273	<code>margin-bottom: 24px;</code>	Sets CSS property `margin-bottom` to `24px` for the active selector.
274	<code>backdrop-filter: blur(10px);</code>	Sets CSS property `backdrop-filter` to `blur(10px)` for the active selector.
275	<code>transition: var(--transition);</code>	Sets CSS property `transition` to `var(--transition)` for the active selector.
276	<code>position: relative;</code>	Sets CSS property `position` to `relative` for the active selector.
277	<code>overflow: hidden;</code>	Sets CSS property `overflow` to `hidden` for the active selector.
278	<code>}</code>	Closes the current CSS rule block.

Line	Code	Explanation
279		Blank line used to group related style declarations.
280	.card::before {	Opens a CSS rule block for the selector on this line.
281	content: "";	Sets CSS property `content` to `""` for the active selector.
282	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
283	left: -40%;	Sets CSS property `left` to `-40%` for the active selector.
284	top: -60%;	Sets CSS property `top` to `-60%` for the active selector.
285	width: 160%;	Sets CSS property `width` to `160%` for the active selector.
286	height: 160%;	Sets CSS property `height` to `160%` for the active selector.
287	background: radial-gradient(circle, rgba(0, 183, 255, 0.10), transparent 60%);	Sets CSS property `background` to `radial-gradient(circle, rgba(0, 183, 255, 0.10), transparent 60%)` for the active selector.
288	opacity: 0.6;	Sets CSS property `opacity` to `0.6` for the active selector.
289	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
290	}	Closes the current CSS rule block.
291		Blank line used to group related style declarations.
292	.card:hover {	Opens a CSS rule block for the selector on this line.
293	transform: translateY(-2px);	Sets CSS property `transform` to `translateY(-2px)` for the active selector.
294	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
295	}	Closes the current CSS rule block.
296		Blank line used to group related style declarations.
297	/* =====	CSS comment used to document the stylesheet.
298	FORMS & BUTTONS	CSS selector/value syntax line used by nearby declarations.
299	===== */	CSS comment used to document the stylesheet.
300	form {	Opens a CSS rule block for the selector on this line.
301	display: grid;	Sets CSS property `display` to `grid` for the active selector.
302	gap: 14px;	Sets CSS property `gap` to `14px` for the active selector.
303	}	Closes the current CSS rule block.
304		Blank line used to group related style declarations.
305	input, select, textarea {	Opens a CSS rule block for the selector on this line.
306	padding: 12px 14px;	Sets CSS property `padding` to `12px 14px` for the active selector.
307	border: 1px solid rgba(148, 163, 184, 0.35);	Sets CSS property `border` to `1px solid rgba(148, 163, 184, 0.35)` for the active selector.
308	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
309	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
310	box-sizing: border-box;	Sets CSS property `box-sizing` to `border-box` for the active selector.
311	font-family: inherit;	Sets CSS property `font-family` to `inherit` for the active selector.
312	background: rgba(255, 255, 255, 0.85);	Sets CSS property `background` to `rgba(255, 255, 255, 0.85)` for the active selector.
313	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
314	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
315	}	Closes the current CSS rule block.
316		Blank line used to group related style declarations.
317	textarea {	Opens a CSS rule block for the selector on this line.
318	min-height: 120px;	Sets CSS property `min-height` to `120px` for the active selector.
319	resize: vertical;	Sets CSS property `resize` to `vertical` for the active selector.

Line	Code	Explanation
320	}	Closes the current CSS rule block.
321		Blank line used to group related style declarations.
322	input:focus, select:focus, textarea:focus {	Opens a CSS rule block for the selector on this line.
323	outline: none;	Sets CSS property `outline` to `none` for the active selector.
324	border-color: var(--accent);	Sets CSS property `border-color` to `var(--accent)` for the active selector.
325	box-shadow: var(--glow);	Sets CSS property `box-shadow` to `var(--glow)` for the active selector.
326	}	Closes the current CSS rule block.
327		Blank line used to group related style declarations.
328	button {	Opens a CSS rule block for the selector on this line.
329	padding: 12px 20px;	Sets CSS property `padding` to `12px 20px` for the active selector.
330	background: linear-gradient(135deg, var(--accent), var(--accent-2));	Sets CSS property `background` to `linear-gradient(135deg, var(--accent), var(--accent-2))` for the active selector.
331	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
332	border: none;	Sets CSS property `border` to `none` for the active selector.
333	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
334	cursor: pointer;	Sets CSS property `cursor` to `pointer` for the active selector.
335	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
336	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
337	box-shadow: 0 12px 24px rgba(0, 183, 255, 0.22);	Sets CSS property `box-shadow` to `0 12px 24px rgba(0, 183, 255, 0.22)` for the active selector.
338	}	Closes the current CSS rule block.
339		Blank line used to group related style declarations.
340	button:hover {	Opens a CSS rule block for the selector on this line.
341	transform: translateY(-1px);	Sets CSS property `transform` to `translateY(-1px)` for the active selector.
342	filter: brightness(1.03);	Sets CSS property `filter` to `brightness(1.03)` for the active selector.
343	}	Closes the current CSS rule block.
344		Blank line used to group related style declarations.
345	button:active {	Opens a CSS rule block for the selector on this line.
346	transform: translateY(0);	Sets CSS property `transform` to `translateY(0)` for the active selector.
347	}	Closes the current CSS rule block.
348		Blank line used to group related style declarations.
349	/* Action Buttons */	CSS comment used to document the stylesheet.
350	.btn-edit, .btn-delete, .btn-add {	Opens a CSS rule block for the selector on this line.
351	display: inline-block;	Sets CSS property `display` to `inline-block` for the active selector.
352	padding: 8px 16px;	Sets CSS property `padding` to `8px 16px` for the active selector.
353	border-radius: 10px;	Sets CSS property `border-radius` to `10px` for the active selector.
354	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
355	font-size: 0.9rem;	Sets CSS property `font-size` to `0.9rem` for the active selector.
356	cursor: pointer;	Sets CSS property `cursor` to `pointer` for the active selector.
357	border: none;	Sets CSS property `border` to `none` for the active selector.
358	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
359	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
360	}	Closes the current CSS rule block.

Line	Code	Explanation
361		Blank line used to group related style declarations.
362	.btn-add {	Opens a CSS rule block for the selector on this line.
363	background: var(--gradient-deep);	Sets CSS property `background` to `var(--gradient-deep)` for the active selector.
364	padding: 10px 18px;	Sets CSS property `padding` to `10px 18px` for the active selector.
365	margin-bottom: 16px;	Sets CSS property `margin-bottom` to `16px` for the active selector.
366	box-shadow: 0 8px 20px rgba(15, 23, 42, 0.2);	Sets CSS property `box-shadow` to `0 8px 20px rgba(15, 23, 42, 0.2)` for the active selector.
367	}	Closes the current CSS rule block.
368		Blank line used to group related style declarations.
369	.btn-edit {	Opens a CSS rule block for the selector on this line.
370	background: linear-gradient(135deg, #00b7ff, #3b82f6);	Sets CSS property `background` to `linear-gradient(135deg, #00b7ff, #3b82f6)` for the active selector.
371	margin-right: 6px;	Sets CSS property `margin-right` to `6px` for the active selector.
372	}	Closes the current CSS rule block.
373		Blank line used to group related style declarations.
374	.btn-delete {	Opens a CSS rule block for the selector on this line.
375	background: linear-gradient(135deg, #ef4444, #fb7185);	Sets CSS property `background` to `linear-gradient(135deg, #ef4444, #fb7185)` for the active selector.
376	}	Closes the current CSS rule block.
377		Blank line used to group related style declarations.
378	/* Return Dashboard Button */	CSS comment used to document the stylesheet.
379	.btn-dashboard-return {	Opens a CSS rule block for the selector on this line.
380	background-color: transparent;	Sets CSS property `background-color` to `transparent` for the active selector.
381	color: var(--accent);	Sets CSS property `color` to `var(--accent)` for the active selector.
382	border: 2px solid var(--accent);	Sets CSS property `border` to `2px solid var(--accent)` for the active selector.
383	padding: 8px 16px;	Sets CSS property `padding` to `8px 16px` for the active selector.
384	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
385	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
386	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
387	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
388	}	Closes the current CSS rule block.
389		Blank line used to group related style declarations.
390	.btn-dashboard-return:hover {	Opens a CSS rule block for the selector on this line.
391	background-color: var(--accent);	Sets CSS property `background-color` to `var(--accent)` for the active selector.
392	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
393	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
394	}	Closes the current CSS rule block.
395		Blank line used to group related style declarations.
396	/* Borrow Book Button */	CSS comment used to document the stylesheet.
397	.btn-borrow {	Opens a CSS rule block for the selector on this line.
398	background-color: var(--success);	Sets CSS property `background-color` to `var(--success)` for the active selector.
399	color: #fff !important;	Sets CSS property `color` to `#fff !important` for the active selector.
400	border: none;	Sets CSS property `border` to `none` for the active selector.
401	padding: 6px 14px;	Sets CSS property `padding` to `6px 14px` for the active selector.

Line	Code	Explanation
402	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
403	font-size: 0.9rem;	Sets CSS property `font-size` to `0.9rem` for the active selector.
404	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
405	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
406	display: inline-block;	Sets CSS property `display` to `inline-block` for the active selector.
407	}	Closes the current CSS rule block.
408		Blank line used to group related style declarations.
409	.btn-borrow:hover {	Opens a CSS rule block for the selector on this line.
410	filter: brightness(0.96);	Sets CSS property `filter` to `brightness(0.96)` for the active selector.
411	}	Closes the current CSS rule block.
412		Blank line used to group related style declarations.
413	/* Return Book Button */	CSS comment used to document the stylesheet.
414	.btn-return {	Opens a CSS rule block for the selector on this line.
415	background-color: var(--warning);	Sets CSS property `background-color` to `var(--warning)` for the active selector.
416	color: #fff !important;	Sets CSS property `color` to `#fff !important` for the active selector.
417	border: none;	Sets CSS property `border` to `none` for the active selector.
418	padding: 6px 14px;	Sets CSS property `padding` to `6px 14px` for the active selector.
419	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
420	font-size: 0.9rem;	Sets CSS property `font-size` to `0.9rem` for the active selector.
421	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
422	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
423	display: inline-block;	Sets CSS property `display` to `inline-block` for the active selector.
424	}	Closes the current CSS rule block.
425		Blank line used to group related style declarations.
426	.btn-return:hover {	Opens a CSS rule block for the selector on this line.
427	filter: brightness(0.95);	Sets CSS property `filter` to `brightness(0.95)` for the active selector.
428	}	Closes the current CSS rule block.
429		Blank line used to group related style declarations.
430	/* =====	CSS comment used to document the stylesheet.
431	BOOTSTRAP OVERRIDES	CSS selector/value syntax line used by nearby declarations.
432	===== */	CSS comment used to document the stylesheet.
433	.btn {	Opens a CSS rule block for the selector on this line.
434	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
435	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
436	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
437	}	Closes the current CSS rule block.
438		Blank line used to group related style declarations.
439	.btn-primary {	Opens a CSS rule block for the selector on this line.
440	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
441	border: none;	Sets CSS property `border` to `none` for the active selector.
442	box-shadow: 0 10px 22px rgba(0, 183, 255, 0.25);	Sets CSS property `box-shadow` to `0 10px 22px rgba(0, 183, 255, 0.25)` for the active selector.

Line	Code	Explanation
443	}	Closes the current CSS rule block.
444		Blank line used to group related style declarations.
445	.btn-primary:hover {	Opens a CSS rule block for the selector on this line.
446	filter: brightness(1.03);	Sets CSS property `filter` to `brightness(1.03)` for the active selector.
447	transform: translateY(-1px);	Sets CSS property `transform` to `translateY(-1px)` for the active selector.
448	}	Closes the current CSS rule block.
449		Blank line used to group related style declarations.
450	.btn-outline-primary {	Opens a CSS rule block for the selector on this line.
451	border-color: var(--accent);	Sets CSS property `border-color` to `var(--accent)` for the active selector.
452	color: var(--accent);	Sets CSS property `color` to `var(--accent)` for the active selector.
453	}	Closes the current CSS rule block.
454		Blank line used to group related style declarations.
455	.btn-outline-primary:hover {	Opens a CSS rule block for the selector on this line.
456	background: var(--accent);	Sets CSS property `background` to `var(--accent)` for the active selector.
457	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
458	}	Closes the current CSS rule block.
459		Blank line used to group related style declarations.
460	.btn-secondary {	Opens a CSS rule block for the selector on this line.
461	background: #64748b;	Sets CSS property `background` to `#64748b` for the active selector.
462	border: none;	Sets CSS property `border` to `none` for the active selector.
463	}	Closes the current CSS rule block.
464		Blank line used to group related style declarations.
465	.btn-dark {	Opens a CSS rule block for the selector on this line.
466	background: #0f172a;	Sets CSS property `background` to `#0f172a` for the active selector.
467	border: none;	Sets CSS property `border` to `none` for the active selector.
468	}	Closes the current CSS rule block.
469		Blank line used to group related style declarations.
470	.btn-success {	Opens a CSS rule block for the selector on this line.
471	background: linear-gradient(135deg, #22c55e, #16a34a);	Sets CSS property `background` to `linear-gradient(135deg, #22c55e, #16a34a)` for the active selector.
472	border: none;	Sets CSS property `border` to `none` for the active selector.
473	}	Closes the current CSS rule block.
474		Blank line used to group related style declarations.
475	.btn-warning {	Opens a CSS rule block for the selector on this line.
476	background: linear-gradient(135deg, #f59e0b, #f97316);	Sets CSS property `background` to `linear-gradient(135deg, #f59e0b, #f97316)` for the active selector.
477	border: none;	Sets CSS property `border` to `none` for the active selector.
478	color: #111827;	Sets CSS property `color` to `#111827` for the active selector.
479	}	Closes the current CSS rule block.
480		Blank line used to group related style declarations.
481	.btn-danger {	Opens a CSS rule block for the selector on this line.
482	background: linear-gradient(135deg, #fb7185, #ef4444);	Sets CSS property `background` to `linear-gradient(135deg, #fb7185, #ef4444)` for the active selector.
483	border: none;	Sets CSS property `border` to `none` for the active selector.

Line	Code	Explanation
484	}	Closes the current CSS rule block.
485		Blank line used to group related style declarations.
486	.btn-info {	Opens a CSS rule block for the selector on this line.
487	background: linear-gradient(135deg, #38bdf8, #0ea5e9);	Sets CSS property `background` to `linear-gradient(135deg, #38bdf8, #0ea5e9)` for the active selector.
488	border: none;	Sets CSS property `border` to `none` for the active selector.
489	}	Closes the current CSS rule block.
490		Blank line used to group related style declarations.
491	.btn-light {	Opens a CSS rule block for the selector on this line.
492	background: #f8fafc;	Sets CSS property `background` to `#f8fafc` for the active selector.
493	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
494	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
495	}	Closes the current CSS rule block.
496		Blank line used to group related style declarations.
497	.btn-link {	Opens a CSS rule block for the selector on this line.
498	color: var(--accent);	Sets CSS property `color` to `var(--accent)` for the active selector.
499	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
500	}	Closes the current CSS rule block.
501		Blank line used to group related style declarations.
502	.btn-link:hover {	Opens a CSS rule block for the selector on this line.
503	color: var(--accent-2);	Sets CSS property `color` to `var(--accent-2)` for the active selector.
504	}	Closes the current CSS rule block.
505		Blank line used to group related style declarations.
506	.btn-sm {	Opens a CSS rule block for the selector on this line.
507	border-radius: 10px;	Sets CSS property `border-radius` to `10px` for the active selector.
508	}	Closes the current CSS rule block.
509		Blank line used to group related style declarations.
510	.badge {	Opens a CSS rule block for the selector on this line.
511	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
512	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
513	padding: 0.35em 0.65em;	Sets CSS property `padding` to `0.35em 0.65em` for the active selector.
514	}	Closes the current CSS rule block.
515		Blank line used to group related style declarations.
516	.alert {	Opens a CSS rule block for the selector on this line.
517	border-radius: var(--radius-sm);	Sets CSS property `border-radius` to `var(--radius-sm)` for the active selector.
518	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
519	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
520	}	Closes the current CSS rule block.
521		Blank line used to group related style declarations.
522	.alert-success {	Opens a CSS rule block for the selector on this line.
523	background: rgba(34, 197, 94, 0.12);	Sets CSS property `background` to `rgba(34, 197, 94, 0.12)` for the active selector.
524	color: #166534;	Sets CSS property `color` to `#166534` for the active selector.

Line	Code	Explanation
525	}	Closes the current CSS rule block.
526		Blank line used to group related style declarations.
527	.alert-warning {	Opens a CSS rule block for the selector on this line.
528	background: rgba(245, 158, 11, 0.15);	Sets CSS property `background` to `rgba(245, 158, 11, 0.15)` for the active selector.
529	color: #92400e;	Sets CSS property `color` to `#92400e` for the active selector.
530	}	Closes the current CSS rule block.
531		Blank line used to group related style declarations.
532	.alert-danger {	Opens a CSS rule block for the selector on this line.
533	background: rgba(239, 68, 68, 0.12);	Sets CSS property `background` to `rgba(239, 68, 68, 0.12)` for the active selector.
534	color: #b91c1c;	Sets CSS property `color` to `#b91c1c` for the active selector.
535	}	Closes the current CSS rule block.
536		Blank line used to group related style declarations.
537	.alert-info {	Opens a CSS rule block for the selector on this line.
538	background: rgba(56, 189, 248, 0.14);	Sets CSS property `background` to `rgba(56, 189, 248, 0.14)` for the active selector.
539	color: #0c4a6e;	Sets CSS property `color` to `#0c4a6e` for the active selector.
540	}	Closes the current CSS rule block.
541		Blank line used to group related style declarations.
542	.alert-primary {	Opens a CSS rule block for the selector on this line.
543	background: rgba(14, 165, 163, 0.14);	Sets CSS property `background` to `rgba(14, 165, 163, 0.14)` for the active selector.
544	color: #0f766e;	Sets CSS property `color` to `#0f766e` for the active selector.
545	}	Closes the current CSS rule block.
546		Blank line used to group related style declarations.
547	.dropdown-menu {	Opens a CSS rule block for the selector on this line.
548	border-radius: var(--radius-sm);	Sets CSS property `border-radius` to `var(--radius-sm)` for the active selector.
549	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
550	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
551	padding: 8px;	Sets CSS property `padding` to `8px` for the active selector.
552	}	Closes the current CSS rule block.
553		Blank line used to group related style declarations.
554	.dropdown-item {	Opens a CSS rule block for the selector on this line.
555	border-radius: 8px;	Sets CSS property `border-radius` to `8px` for the active selector.
556	padding: 8px 12px;	Sets CSS property `padding` to `8px 12px` for the active selector.
557	font-weight: 500;	Sets CSS property `font-weight` to `500` for the active selector.
558	}	Closes the current CSS rule block.
559		Blank line used to group related style declarations.
560	.dropdown-item:hover {	Opens a CSS rule block for the selector on this line.
561	background: rgba(14, 165, 163, 0.12);	Sets CSS property `background` to `rgba(14, 165, 163, 0.12)` for the active selector.
562	}	Closes the current CSS rule block.
563		Blank line used to group related style declarations.
564	.form-control,	CSS selector/value syntax line used by nearby declarations.
565	.form-select {	Opens a CSS rule block for the selector on this line.

Line	Code	Explanation
566	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
567	border-color: rgba(148, 163, 184, 0.4);	Sets CSS property `border-color` to `rgba(148, 163, 184, 0.4)` for the active selector.
568	background: rgba(255, 255, 255, 0.9);	Sets CSS property `background` to `rgba(255, 255, 255, 0.9)` for the active selector.
569	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
570	}	Closes the current CSS rule block.
571		Blank line used to group related style declarations.
572	.form-control:focus,	CSS selector/value syntax line used by nearby declarations.
573	.form-select:focus {	Opens a CSS rule block for the selector on this line.
574	border-color: var(--accent);	Sets CSS property `border-color` to `var(--accent)` for the active selector.
575	box-shadow: var(--glow);	Sets CSS property `box-shadow` to `var(--glow)` for the active selector.
576	}	Closes the current CSS rule block.
577		Blank line used to group related style declarations.
578	/* =====	CSS comment used to document the stylesheet.
579	TABLES	CSS selector/value syntax line used by nearby declarations.
580	===== */	CSS comment used to document the stylesheet.
581	table,	CSS selector/value syntax line used by nearby declarations.
582	.table {	Opens a CSS rule block for the selector on this line.
583	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
584	border-collapse: separate;	Sets CSS property `border-collapse` to `separate` for the active selector.
585	border-spacing: 0 10px;	Sets CSS property `border-spacing` to `0 10px` for the active selector.
586	margin-top: 15px;	Sets CSS property `margin-top` to `15px` for the active selector.
587	}	Closes the current CSS rule block.
588		Blank line used to group related style declarations.
589	.table-responsive {	Opens a CSS rule block for the selector on this line.
590	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
591	}	Closes the current CSS rule block.
592		Blank line used to group related style declarations.
593	table thead th,	CSS selector/value syntax line used by nearby declarations.
594	.table thead th {	Opens a CSS rule block for the selector on this line.
595	text-align: left;	Sets CSS property `text-align` to `left` for the active selector.
596	padding: 10px 14px;	Sets CSS property `padding` to `10px 14px` for the active selector.
597	text-transform: uppercase;	Sets CSS property `text-transform` to `uppercase` for the active selector.
598	letter-spacing: 0.08em;	Sets CSS property `letter-spacing` to `0.08em` for the active selector.
599	font-size: 0.74rem;	Sets CSS property `font-size` to `0.74rem` for the active selector.
600	color: var(--muted);	Sets CSS property `color` to `var(--muted)` for the active selector.
601	border-bottom: none;	Sets CSS property `border-bottom` to `none` for the active selector.
602	}	Closes the current CSS rule block.
603		Blank line used to group related style declarations.
604	thead.table-dark th,	CSS selector/value syntax line used by nearby declarations.
605	.table thead.table-dark th,	CSS selector/value syntax line used by nearby declarations.
606	.table-dark th {	Opens a CSS rule block for the selector on this line.

Line	Code	Explanation
607	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
608	background: #0f172a;	Sets CSS property `background` to `#0f172a` for the active selector.
609	}	Closes the current CSS rule block.
610		Blank line used to group related style declarations.
611	thead.table-light th,	CSS selector/value syntax line used by nearby declarations.
612	.table thead.table-light th,	CSS selector/value syntax line used by nearby declarations.
613	.table-light th {	Opens a CSS rule block for the selector on this line.
614	color: #1f2937;	Sets CSS property `color` to `#1f2937` for the active selector.
615	background: #f8fafc;	Sets CSS property `background` to `#f8fafc` for the active selector.
616	}	Closes the current CSS rule block.
617		Blank line used to group related style declarations.
618	table tbody tr,	CSS selector/value syntax line used by nearby declarations.
619	.table tbody tr {	Opens a CSS rule block for the selector on this line.
620	background: var(--card-solid);	Sets CSS property `background` to `var(--card-solid)` for the active selector.
621	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
622	}	Closes the current CSS rule block.
623		Blank line used to group related style declarations.
624	table tbody td,	CSS selector/value syntax line used by nearby declarations.
625	.table tbody td {	Opens a CSS rule block for the selector on this line.
626	padding: 14px;	Sets CSS property `padding` to `14px` for the active selector.
627	border-top: 1px solid rgba(15, 23, 42, 0.04);	Sets CSS property `border-top` to `1px solid rgba(15, 23, 42, 0.04)` for the active selector.
628	border-bottom: 1px solid rgba(15, 23, 42, 0.04);	Sets CSS property `border-bottom` to `1px solid rgba(15, 23, 42, 0.04)` for the active selector.
629	}	Closes the current CSS rule block.
630		Blank line used to group related style declarations.
631	table tbody td:first-child,	CSS selector/value syntax line used by nearby declarations.
632	.table tbody td:first-child {	Opens a CSS rule block for the selector on this line.
633	border-radius: 12px 0 0 12px;	Sets CSS property `border-radius` to `12px 0 0 12px` for the active selector.
634	}	Closes the current CSS rule block.
635		Blank line used to group related style declarations.
636	table tbody td:last-child,	CSS selector/value syntax line used by nearby declarations.
637	.table tbody td:last-child {	Opens a CSS rule block for the selector on this line.
638	border-radius: 0 12px 12px 0;	Sets CSS property `border-radius` to `0 12px 12px 0` for the active selector.
639	}	Closes the current CSS rule block.
640		Blank line used to group related style declarations.
641	.table.table-striped > tbody > tr:nth-of-type(odd) {	Opens a CSS rule block for the selector on this line.
642	background: rgba(15, 23, 42, 0.02);	Sets CSS property `background` to `rgba(15, 23, 42, 0.02)` for the active selector.
643	}	Closes the current CSS rule block.
644		Blank line used to group related style declarations.
645	.table.table-hover > tbody > tr:hover {	Opens a CSS rule block for the selector on this line.
646	background: rgba(14, 165, 163, 0.08);	Sets CSS property `background` to `rgba(14, 165, 163, 0.08)` for the active selector.
647	}	Closes the current CSS rule block.

Line	Code	Explanation
648		Blank line used to group related style declarations.
649	table tbody tr:hover,	CSS selector/value syntax line used by nearby declarations.
650	.table tbody tr:hover {	Opens a CSS rule block for the selector on this line.
651	transform: translateY(-1px);	Sets CSS property `transform` to `translateY(-1px)` for the active selector.
652	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
653	}	Closes the current CSS rule block.
654		Blank line used to group related style declarations.
655	/* --- Admin Library Table --- */	CSS comment used to document the stylesheet.
656	.library-table {	Opens a CSS rule block for the selector on this line.
657	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
658	border-collapse: separate;	Sets CSS property `border-collapse` to `separate` for the active selector.
659	border-spacing: 0;	Sets CSS property `border-spacing` to `0` for the active selector.
660	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
661	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
662	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
663	margin-top: 0;	Sets CSS property `margin-top` to `0` for the active selector.
664	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
665	}	Closes the current CSS rule block.
666		Blank line used to group related style declarations.
667	.library-table thead th {	Opens a CSS rule block for the selector on this line.
668	background: linear-gradient(135deg, #0f172a, #1f2937);	Sets CSS property `background` to `linear-gradient(135deg, #0f172a, #1f2937)` for the active selector.
669	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
670	padding: 14px 16px;	Sets CSS property `padding` to `14px 16px` for the active selector.
671	text-transform: none;	Sets CSS property `text-transform` to `none` for the active selector.
672	letter-spacing: 0;	Sets CSS property `letter-spacing` to `0` for the active selector.
673	font-size: 0.9rem;	Sets CSS property `font-size` to `0.9rem` for the active selector.
674	}	Closes the current CSS rule block.
675		Blank line used to group related style declarations.
676	.library-table td {	Opens a CSS rule block for the selector on this line.
677	padding: 14px;	Sets CSS property `padding` to `14px` for the active selector.
678	border-bottom: 1px solid rgba(15, 23, 42, 0.06);	Sets CSS property `border-bottom` to `1px solid rgba(15, 23, 42, 0.06)` for the active selector.
679	vertical-align: middle;	Sets CSS property `vertical-align` to `middle` for the active selector.
680	}	Closes the current CSS rule block.
681		Blank line used to group related style declarations.
682	.library-table tr:last-child td {	Opens a CSS rule block for the selector on this line.
683	border-bottom: none;	Sets CSS property `border-bottom` to `none` for the active selector.
684	}	Closes the current CSS rule block.
685		Blank line used to group related style declarations.
686	/* =====	CSS comment used to document the stylesheet.
687	DASHBOARD WIDGETS	CSS selector/value syntax line used by nearby declarations.
688	===== */	CSS comment used to document the stylesheet.

Line	Code	Explanation
689	.stats-grid {	Opens a CSS rule block for the selector on this line.
690	display: grid;	Sets CSS property `display` to `grid` for the active selector.
691	grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));	Sets CSS property `grid-template-columns` to `repeat(auto-fit, minmax(220px, 1fr))` for the active selector.
692	gap: 18px;	Sets CSS property `gap` to `18px` for the active selector.
693	}	Closes the current CSS rule block.
694		Blank line used to group related style declarations.
695	.stat-box {	Opens a CSS rule block for the selector on this line.
696	padding: 22px;	Sets CSS property `padding` to `22px` for the active selector.
697	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
698	background: var(--card);	Sets CSS property `background` to `var(--card)` for the active selector.
699	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
700	text-align: left;	Sets CSS property `text-align` to `left` for the active selector.
701	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
702	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
703	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
704	position: relative;	Sets CSS property `position` to `relative` for the active selector.
705	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
706	}	Closes the current CSS rule block.
707		Blank line used to group related style declarations.
708	.stat-box::after {	Opens a CSS rule block for the selector on this line.
709	content: "";	Sets CSS property `content` to `""` for the active selector.
710	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
711	right: -30px;	Sets CSS property `right` to `-30px` for the active selector.
712	top: -30px;	Sets CSS property `top` to `-30px` for the active selector.
713	width: 120px;	Sets CSS property `width` to `120px` for the active selector.
714	height: 120px;	Sets CSS property `height` to `120px` for the active selector.
715	background: rgba(0, 183, 255, 0.18);	Sets CSS property `background` to `rgba(0, 183, 255, 0.18)` for the active selector.
716	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
717	}	Closes the current CSS rule block.
718		Blank line used to group related style declarations.
719	.stat-box:hover {	Opens a CSS rule block for the selector on this line.
720	transform: translateY(-3px);	Sets CSS property `transform` to `translateY(-3px)` for the active selector.
721	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
722	}	Closes the current CSS rule block.
723		Blank line used to group related style declarations.
724	.stat-box.bg-blue {	Opens a CSS rule block for the selector on this line.
725	background: linear-gradient(135deg, #00b7ff, #3b82f6);	Sets CSS property `background` to `linear-gradient(135deg, #00b7ff, #3b82f6)` for the active selector.
726	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
727	}	Closes the current CSS rule block.
728		Blank line used to group related style declarations.
729	.stat-box.bg-green {	Opens a CSS rule block for the selector on this line.

Line	Code	Explanation
730	background: linear-gradient(135deg, #22c55e, #10b981);	Sets CSS property `background` to `linear-gradient(135deg, #22c55e, #10b981)` for the active selector.
731	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
732	}	Closes the current CSS rule block.
733		Blank line used to group related style declarations.
734	.stat-box.bg-orange {	Opens a CSS rule block for the selector on this line.
735	background: linear-gradient(135deg, #f59e0b, #f97316);	Sets CSS property `background` to `linear-gradient(135deg, #f59e0b, #f97316)` for the active selector.
736	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
737	}	Closes the current CSS rule block.
738		Blank line used to group related style declarations.
739	.stat-box.bg-red,	CSS selector/value syntax line used by nearby declarations.
740	.stat-box.bg-danger {	Opens a CSS rule block for the selector on this line.
741	background: linear-gradient(135deg, #ef4444, #fb7185);	Sets CSS property `background` to `linear-gradient(135deg, #ef4444, #fb7185)` for the active selector.
742	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
743	}	Closes the current CSS rule block.
744		Blank line used to group related style declarations.
745	.stat-box.bg-blue h3,	CSS selector/value syntax line used by nearby declarations.
746	.stat-box.bg-green h3,	CSS selector/value syntax line used by nearby declarations.
747	.stat-box.bg-orange h3,	CSS selector/value syntax line used by nearby declarations.
748	.stat-box.bg-red h3,	CSS selector/value syntax line used by nearby declarations.
749	.stat-box.bg-danger h3 {	Opens a CSS rule block for the selector on this line.
750	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
751	}	Closes the current CSS rule block.
752		Blank line used to group related style declarations.
753	/* =====	CSS comment used to document the stylesheet.
754	LIBRARY SPECIFIC MODULES	CSS selector/value syntax line used by nearby declarations.
755	===== */	CSS comment used to document the stylesheet.
756	.library-split-container {	Opens a CSS rule block for the selector on this line.
757	display: flex;	Sets CSS property `display` to `flex` for the active selector.
758	gap: 30px;	Sets CSS property `gap` to `30px` for the active selector.
759	align-items: flex-start;	Sets CSS property `align-items` to `flex-start` for the active selector.
760	}	Closes the current CSS rule block.
761		Blank line used to group related style declarations.
762	.library-book-section {	Opens a CSS rule block for the selector on this line.
763	flex: 2;	Sets CSS property `flex` to `2` for the active selector.
764	}	Closes the current CSS rule block.
765		Blank line used to group related style declarations.
766	.library-history-section {	Opens a CSS rule block for the selector on this line.
767	flex: 1;	Sets CSS property `flex` to `1` for the active selector.
768	position: sticky;	Sets CSS property `position` to `sticky` for the active selector.
769	top: 30px;	Sets CSS property `top` to `30px` for the active selector.
770	}	Closes the current CSS rule block.

Line	Code	Explanation
771		Blank line used to group related style declarations.
772	@media (max-width: 1100px) {	Starts a media query block for responsive behavior.
773	.library-split-container {	Opens a CSS rule block for the selector on this line.
774	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
775	}	Closes the current CSS rule block.
776	.library-history-section {	Opens a CSS rule block for the selector on this line.
777	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
778	position: static;	Sets CSS property `position` to `static` for the active selector.
779	margin-top: 30px;	Sets CSS property `margin-top` to `30px` for the active selector.
780	}	Closes the current CSS rule block.
781	}	Closes the current CSS rule block.
782		Blank line used to group related style declarations.
783	.library-grid {	Opens a CSS rule block for the selector on this line.
784	display: grid;	Sets CSS property `display` to `grid` for the active selector.
785	grid-template-columns: repeat(auto-fill, minmax(240px, 1fr));	Sets CSS property `grid-template-columns` to `repeat(auto-fill, minmax(240px, 1fr))` for the active selector.
786	gap: 18px;	Sets CSS property `gap` to `18px` for the active selector.
787	padding: 10px 0;	Sets CSS property `padding` to `10px 0` for the active selector.
788	}	Closes the current CSS rule block.
789		Blank line used to group related style declarations.
790	.book-card {	Opens a CSS rule block for the selector on this line.
791	background: var(--card-solid);	Sets CSS property `background` to `var(--card-solid)` for the active selector.
792	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
793	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
794	padding: 20px;	Sets CSS property `padding` to `20px` for the active selector.
795	display: flex;	Sets CSS property `display` to `flex` for the active selector.
796	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
797	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
798	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
799	position: relative;	Sets CSS property `position` to `relative` for the active selector.
800	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
801	}	Closes the current CSS rule block.
802		Blank line used to group related style declarations.
803	.book-card::before {	Opens a CSS rule block for the selector on this line.
804	content: "";	Sets CSS property `content` to `""` for the active selector.
805	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
806	top: 0;	Sets CSS property `top` to `0` for the active selector.
807	left: 0;	Sets CSS property `left` to `0` for the active selector.
808	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
809	height: 4px;	Sets CSS property `height` to `4px` for the active selector.
810	background: linear-gradient(90deg, var(--accent), var(--accent-2));	Sets CSS property `background` to `linear-gradient(90deg, var(--accent), var(--accent-2))` for the active selector.
811	}	Closes the current CSS rule block.

Line	Code	Explanation
812		Blank line used to group related style declarations.
813	.book-card:hover {	Opens a CSS rule block for the selector on this line.
814	transform: translateY(-3px);	Sets CSS property `transform` to `translateY(-3px)` for the active selector.
815	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
816	}	Closes the current CSS rule block.
817		Blank line used to group related style declarations.
818	.book-icon {	Opens a CSS rule block for the selector on this line.
819	font-size: 34px;	Sets CSS property `font-size` to `34px` for the active selector.
820	color: var(--ink-2);	Sets CSS property `color` to `var(--ink-2)` for the active selector.
821	margin-bottom: 12px;	Sets CSS property `margin-bottom` to `12px` for the active selector.
822	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
823	}	Closes the current CSS rule block.
824		Blank line used to group related style declarations.
825	.book-title {	Opens a CSS rule block for the selector on this line.
826	font-size: 1.08rem;	Sets CSS property `font-size` to `1.08rem` for the active selector.
827	font-weight: 700;	Sets CSS property `font-weight` to `700` for the active selector.
828	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
829	margin-bottom: 8px;	Sets CSS property `margin-bottom` to `8px` for the active selector.
830	line-height: 1.3;	Sets CSS property `line-height` to `1.3` for the active selector.
831	}	Closes the current CSS rule block.
832		Blank line used to group related style declarations.
833	.book-detail {	Opens a CSS rule block for the selector on this line.
834	color: var(--muted);	Sets CSS property `color` to `var(--muted)` for the active selector.
835	font-size: 0.92rem;	Sets CSS property `font-size` to `0.92rem` for the active selector.
836	margin-bottom: 4px;	Sets CSS property `margin-bottom` to `4px` for the active selector.
837	}	Closes the current CSS rule block.
838		Blank line used to group related style declarations.
839	.book-detail i {	Opens a CSS rule block for the selector on this line.
840	margin-right: 8px;	Sets CSS property `margin-right` to `8px` for the active selector.
841	color: #94a3b8;	Sets CSS property `color` to `#94a3b8` for the active selector.
842	width: 16px;	Sets CSS property `width` to `16px` for the active selector.
843	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
844	}	Closes the current CSS rule block.
845		Blank line used to group related style declarations.
846	.status-text {	Opens a CSS rule block for the selector on this line.
847	font-weight: 700;	Sets CSS property `font-weight` to `700` for the active selector.
848	font-size: 0.85rem;	Sets CSS property `font-size` to `0.85rem` for the active selector.
849	margin-top: 10px;	Sets CSS property `margin-top` to `10px` for the active selector.
850	display: block;	Sets CSS property `display` to `block` for the active selector.
851	}	Closes the current CSS rule block.
852		Blank line used to group related style declarations.

Line	Code	Explanation
853	.text-success {	Opens a CSS rule block for the selector on this line.
854	color: #16a34a;	Sets CSS property `color` to `#16a34a` for the active selector.
855	}	Closes the current CSS rule block.
856		Blank line used to group related style declarations.
857	.text-danger {	Opens a CSS rule block for the selector on this line.
858	color: #dc2626;	Sets CSS property `color` to `#dc2626` for the active selector.
859	}	Closes the current CSS rule block.
860		Blank line used to group related style declarations.
861	.book-footer {	Opens a CSS rule block for the selector on this line.
862	margin-top: auto;	Sets CSS property `margin-top` to `auto` for the active selector.
863	padding-top: 15px;	Sets CSS property `padding-top` to `15px` for the active selector.
864	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
865	}	Closes the current CSS rule block.
866		Blank line used to group related style declarations.
867	.compact-table {	Opens a CSS rule block for the selector on this line.
868	font-size: 0.9rem;	Sets CSS property `font-size` to `0.9rem` for the active selector.
869	}	Closes the current CSS rule block.
870		Blank line used to group related style declarations.
871	.compact-table th,	CSS selector/value syntax line used by nearby declarations.
872	.compact-table td {	Opens a CSS rule block for the selector on this line.
873	padding: 10px;	Sets CSS property `padding` to `10px` for the active selector.
874	}	Closes the current CSS rule block.
875		Blank line used to group related style declarations.
876	.row-overdue {	Opens a CSS rule block for the selector on this line.
877	background-color: #fff5f5;	Sets CSS property `background-color` to `#fff5f5` for the active selector.
878	}	Closes the current CSS rule block.
879		Blank line used to group related style declarations.
880	.row-overdue td {	Opens a CSS rule block for the selector on this line.
881	color: #b91c1c;	Sets CSS property `color` to `#b91c1c` for the active selector.
882	}	Closes the current CSS rule block.
883		Blank line used to group related style declarations.
884	.fine-badge {	Opens a CSS rule block for the selector on this line.
885	background: #dc2626;	Sets CSS property `background` to `#dc2626` for the active selector.
886	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
887	padding: 2px 6px;	Sets CSS property `padding` to `2px 6px` for the active selector.
888	border-radius: 6px;	Sets CSS property `border-radius` to `6px` for the active selector.
889	font-size: 0.75rem;	Sets CSS property `font-size` to `0.75rem` for the active selector.
890	margin-left: 5px;	Sets CSS property `margin-left` to `5px` for the active selector.
891	}	Closes the current CSS rule block.
892		Blank line used to group related style declarations.
893	/* =====	CSS comment used to document the stylesheet.

Line	Code	Explanation
894	UTILITIES	CSS selector/value syntax line used by nearby declarations.
895	===== */	CSS comment used to document the stylesheet.
896	.bg-blue { background: var(--accent); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
897	.bg-green { background: var(--success); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
898	.bg-orange { background: var(--warning); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
899	.bg-red { background: var(--danger); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
900	.bg-dark { background: #111827; color: #fff; }	CSS selector/value syntax line used by nearby declarations.
901		Blank line used to group related style declarations.
902	.bg-success { background: var(--success); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
903	.bg-warning { background: var(--warning); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
904	.bg-danger { background: var(--danger); color: #fff; }	CSS selector/value syntax line used by nearby declarations.
905		Blank line used to group related style declarations.
906	/* Gradient helpers used in dashboards */	CSS comment used to document the stylesheet.
907	.bg-gradient-primary {	Opens a CSS rule block for the selector on this line.
908	background: linear-gradient(135deg, #00b7ff, #3b82f6);	Sets CSS property `background` to `linear-gradient(135deg, #00b7ff, #3b82f6)` for the active selector.
909	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
910	}	Closes the current CSS rule block.
911		Blank line used to group related style declarations.
912	.bg-gradient-success {	Opens a CSS rule block for the selector on this line.
913	background: linear-gradient(135deg, #22c55e, #10b981);	Sets CSS property `background` to `linear-gradient(135deg, #22c55e, #10b981)` for the active selector.
914	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
915	}	Closes the current CSS rule block.
916		Blank line used to group related style declarations.
917	.bg-gradient-info {	Opens a CSS rule block for the selector on this line.
918	background: linear-gradient(135deg, #38bdf8, #0ea5e9);	Sets CSS property `background` to `linear-gradient(135deg, #38bdf8, #0ea5e9)` for the active selector.
919	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
920	}	Closes the current CSS rule block.
921		Blank line used to group related style declarations.
922	.bg-gradient-warning {	Opens a CSS rule block for the selector on this line.
923	background: linear-gradient(135deg, #f59e0b, #f97316);	Sets CSS property `background` to `linear-gradient(135deg, #f59e0b, #f97316)` for the active selector.
924	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
925	}	Closes the current CSS rule block.
926		Blank line used to group related style declarations.
927	.bg-gradient-blue {	Opens a CSS rule block for the selector on this line.
928	background: linear-gradient(135deg, #00b7ff, #3b82f6);	Sets CSS property `background` to `linear-gradient(135deg, #00b7ff, #3b82f6)` for the active selector.
929	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
930	}	Closes the current CSS rule block.
931		Blank line used to group related style declarations.
932	.bg-gradient-green {	Opens a CSS rule block for the selector on this line.
933	background: linear-gradient(135deg, #22c55e, #16a34a);	Sets CSS property `background` to `linear-gradient(135deg, #22c55e, #16a34a)` for the active selector.
934	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.

Line	Code	Explanation
935	}	Closes the current CSS rule block.
936		Blank line used to group related style declarations.
937	.bg-gradient-red {	Opens a CSS rule block for the selector on this line.
938	background: linear-gradient(135deg, #ef4444, #fb7185);	Sets CSS property `background` to `linear-gradient(135deg, #ef4444, #fb7185)` for the active selector.
939	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
940	}	Closes the current CSS rule block.
941		Blank line used to group related style declarations.
942	/* =====	CSS comment used to document the stylesheet.
943	MOTION	CSS selector/value syntax line used by nearby declarations.
944	===== */	CSS comment used to document the stylesheet.
945	@keyframes fadeUp {	Starts a keyframes animation definition.
946	from {	Opens a CSS rule block for the selector on this line.
947	opacity: 0;	Sets CSS property `opacity` to `0` for the active selector.
948	transform: translateY(12px);	Sets CSS property `transform` to `translateY(12px)` for the active selector.
949	}	Closes the current CSS rule block.
950	to {	Opens a CSS rule block for the selector on this line.
951	opacity: 1;	Sets CSS property `opacity` to `1` for the active selector.
952	transform: translateY(0);	Sets CSS property `transform` to `translateY(0)` for the active selector.
953	}	Closes the current CSS rule block.
954	}	Closes the current CSS rule block.
955		Blank line used to group related style declarations.
956	.header,	CSS selector/value syntax line used by nearby declarations.
957	.card,	CSS selector/value syntax line used by nearby declarations.
958	.stat-box,	CSS selector/value syntax line used by nearby declarations.
959	.library-table,	CSS selector/value syntax line used by nearby declarations.
960	.book-card {	Opens a CSS rule block for the selector on this line.
961	animation: fadeUp 0.45s ease both;	Sets CSS property `animation` to `fadeUp 0.45s ease both` for the active selector.
962	}	Closes the current CSS rule block.
963		Blank line used to group related style declarations.
964	.stats-grid .stat-box:nth-child(2),	CSS selector/value syntax line used by nearby declarations.
965	.content > .card:nth-child(2) {	Opens a CSS rule block for the selector on this line.
966	animation-delay: 0.06s;	Sets CSS property `animation-delay` to `0.06s` for the active selector.
967	}	Closes the current CSS rule block.
968		Blank line used to group related style declarations.
969	.stats-grid .stat-box:nth-child(3),	CSS selector/value syntax line used by nearby declarations.
970	.content > .card:nth-child(3) {	Opens a CSS rule block for the selector on this line.
971	animation-delay: 0.12s;	Sets CSS property `animation-delay` to `0.12s` for the active selector.
972	}	Closes the current CSS rule block.
973		Blank line used to group related style declarations.
974	.stats-grid .stat-box:nth-child(4),	CSS selector/value syntax line used by nearby declarations.
975	.content > .card:nth-child(4) {	Opens a CSS rule block for the selector on this line.

Line	Code	Explanation
976	animation-delay: 0.18s;	Sets CSS property `animation-delay` to `0.18s` for the active selector.
977	}	Closes the current CSS rule block.
978		Blank line used to group related style declarations.
979	@media (prefers-reduced-motion: reduce) {	Starts a media query block for responsive behavior.
980	* {	CSS comment used to document the stylesheet.
981	animation: none !important;	Sets CSS property `animation` to `none !important` for the active selector.
982	transition: none !important;	Sets CSS property `transition` to `none !important` for the active selector.
983	}	Closes the current CSS rule block.
984	}	Closes the current CSS rule block.
985		Blank line used to group related style declarations.
986	/* =====	CSS comment used to document the stylesheet.
987	RESPONSIVE	CSS selector/value syntax line used by nearby declarations.
988	===== */	CSS comment used to document the stylesheet.
989	@media (max-width: 980px) {	Starts a media query block for responsive behavior.
990	body {	Opens a CSS rule block for the selector on this line.
991	display: block;	Sets CSS property `display` to `block` for the active selector.
992	}	Closes the current CSS rule block.
993	.sidebar {	Opens a CSS rule block for the selector on this line.
994	position: fixed;	Sets CSS property `position` to `fixed` for the active selector.
995	width: var(--sidebar-w);	Sets CSS property `width` to `var(--sidebar-w)` for the active selector.
996	height: 100vh;	Sets CSS property `height` to `100vh` for the active selector.
997	transform: translateX(-100%);	Sets CSS property `transform` to `translateX(-100%)` for the active selector.
998	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
999	}	Closes the current CSS rule block.
1000	.sidebar.active {	Opens a CSS rule block for the selector on this line.
1001	transform: translateX(0);	Sets CSS property `transform` to `translateX(0)` for the active selector.
1002	}	Closes the current CSS rule block.
1003	.content {	Opens a CSS rule block for the selector on this line.
1004	margin-left: 0;	Sets CSS property `margin-left` to `0` for the active selector.
1005	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
1006	padding: 24px;	Sets CSS property `padding` to `24px` for the active selector.
1007	}	Closes the current CSS rule block.
1008	}	Closes the current CSS rule block.
1009		Blank line used to group related style declarations.
1010	@media (max-width: 640px) {	Starts a media query block for responsive behavior.
1011	.header {	Opens a CSS rule block for the selector on this line.
1012	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
1013	align-items: flex-start;	Sets CSS property `align-items` to `flex-start` for the active selector.
1014	}	Closes the current CSS rule block.
1015	table {	Opens a CSS rule block for the selector on this line.
1016	display: block;	Sets CSS property `display` to `block` for the active selector.

Line	Code	Explanation
1017	overflow-x: auto;	Sets CSS property `overflow-x` to `auto` for the active selector.
1018	white-space: nowrap;	Sets CSS property `white-space` to `nowrap` for the active selector.
1019	}	Closes the current CSS rule block.
1020	}	Closes the current CSS rule block.
1021		Blank line used to group related style declarations.
1022	/* =====	CSS comment used to document the stylesheet.
1023	2026 UI POLISH: LUMINOUS GLASS OVERRIDES	CSS selector/value syntax line used by nearby declarations.
1024	===== */	CSS comment used to document the stylesheet.
1025	:root {	Opens a CSS rule block for the selector on this line.
1026	--radius: 18px;	Defines CSS custom property `--radius` with value `18px`.
1027	--radius-lg: 24px;	Defines CSS custom property `--radius-lg` with value `24px`.
1028	--shadow-sm: 0 8px 18px rgba(8, 14, 29, 0.08);	Defines CSS custom property `--shadow-sm` with value `0 8px 18px rgba(8, 14, 29, 0.08)`.
1029	--shadow-md: 0 18px 42px rgba(8, 14, 29, 0.14);	Defines CSS custom property `--shadow-md` with value `0 18px 42px rgba(8, 14, 29, 0.14)`.
1030	--shadow-lg: 0 30px 80px rgba(8, 14, 29, 0.2);	Defines CSS custom property `--shadow-lg` with value `0 30px 80px rgba(8, 14, 29, 0.2)`.
1031	--shadow-xl: 0 40px 90px rgba(8, 14, 29, 0.28);	Defines CSS custom property `--shadow-xl` with value `0 40px 90px rgba(8, 14, 29, 0.28)`.
1032	--line: rgba(15, 23, 42, 0.08);	Defines CSS custom property `--line` with value `rgba(15, 23, 42, 0.08)`.
1033	--card: rgba(255, 255, 255, 0.78);	Defines CSS custom property `--card` with value `rgba(255, 255, 255, 0.78)`.
1034	--glass: linear-gradient(180deg, rgba(255, 255, 255, 0.96), rgba(255, 255, 255, 0.84));	Defines CSS custom property `--glass` with value `linear-gradient(180deg, rgba(255, 255, 255, 0.96), rgba(255, 255, 255, 0.84))`.
1035	--glass-strong: linear-gradient(180deg, rgba(255, 255, 255, 0.98), rgba(255, 255, 255, 0.9));	Defines CSS custom property `--glass-strong` with value `linear-gradient(180deg, rgba(255, 255, 255, 0.98), rgba(255, 255, 255, 0.9))`.
1036	--accent-4: #f97316;	Defines CSS custom property `--accent-4` with value `#f97316`.
1037	--gradient-primary: linear-gradient(135deg, #00b7ff 0%, #2dd4bf 45%, #3b82f6 100%);	Defines CSS custom property `--gradient-primary` with value `linear-gradient(135deg, #00b7ff 0%, #2dd4bf 45%, #3b82f6 100%)`.
1038	--gradient-warm: linear-gradient(135deg, #f97316 0%, #f59e0b 55%, #f43f5e 100%);	Defines CSS custom property `--gradient-warm` with value `linear-gradient(135deg, #f97316 0%, #f59e0b 55%, #f43f5e 100%)`.
1039	--gradient-cool: linear-gradient(135deg, #38bdf8 0%, #0ea5e9 50%, #6366f1 100%);	Defines CSS custom property `--gradient-cool` with value `linear-gradient(135deg, #38bdf8 0%, #0ea5e9 50%, #6366f1 100%)`.
1040	--glow: 0 0 0 7px rgba(0, 183, 255, 0.18);	Defines CSS custom property `--glow` with value `0 0 0 7px rgba(0, 183, 255, 0.18)`.
1041	}	Closes the current CSS rule block.
1042		Blank line used to group related style declarations.
1043	body {	Opens a CSS rule block for the selector on this line.
1044	overflow-x: hidden;	Sets CSS property `overflow-x` to `hidden` for the active selector.
1045	-webkit-font-smoothing: antialiased;	Sets CSS property `webkit-font-smoothing` to `antialiased` for the active selector.
1046	text-rendering: optimizeLegibility;	Sets CSS property `text-rendering` to `optimizeLegibility` for the active selector.
1047	background-image:	Starts a multi-line CSS property value that continues on next lines.
1048	radial-gradient(1200px 800px at 0% -10%, rgba(0, 183, 255, 0.24), transparent 60%),	CSS selector/value syntax line used by nearby declarations.
1049	radial-gradient(1000px 700px at 100% 10%, rgba(45, 212, 191, 0.22), transparent 60%),	CSS selector/value syntax line used by nearby declarations.
1050	radial-gradient(900px 700px at 60% 120%, rgba(59, 130, 246, 0.18), transparent 60%),	CSS selector/value syntax line used by nearby declarations.
1051	linear-gradient(180deg, var(--bg-0) 0%, var(--bg-1) 100%);	CSS selector/value syntax line used by nearby declarations.

Line	Code	Explanation
1052	}	Closes the current CSS rule block.
1053		Blank line used to group related style declarations.
1054	body::before {	Opens a CSS rule block for the selector on this line.
1055	opacity: 0.12;	Sets CSS property `opacity` to `0.12` for the active selector.
1056	}	Closes the current CSS rule block.
1057		Blank line used to group related style declarations.
1058	body::after {	Opens a CSS rule block for the selector on this line.
1059	opacity: 0.18;	Sets CSS property `opacity` to `0.18` for the active selector.
1060	mix-blend-mode: screen;	Sets CSS property `mix-blend-mode` to `screen` for the active selector.
1061	}	Closes the current CSS rule block.
1062		Blank line used to group related style declarations.
1063	.brand {	Opens a CSS rule block for the selector on this line.
1064	display: flex;	Sets CSS property `display` to `flex` for the active selector.
1065	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
1066	justify-content: center;	Sets CSS property `justify-content` to `center` for the active selector.
1067	gap: 10px;	Sets CSS property `gap` to `10px` for the active selector.
1068	font-family: "Fraunces", serif;	Sets CSS property `font-family` to `"Fraunces", serif` for the active selector.
1069	}	Closes the current CSS rule block.
1070		Blank line used to group related style declarations.
1071	.brand-logo {	Opens a CSS rule block for the selector on this line.
1072	width: 34px;	Sets CSS property `width` to `34px` for the active selector.
1073	height: 34px;	Sets CSS property `height` to `34px` for the active selector.
1074	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
1075	padding: 6px;	Sets CSS property `padding` to `6px` for the active selector.
1076	background: rgba(255, 255, 255, 0.08);	Sets CSS property `background` to `rgba(255, 255, 255, 0.08)` for the active selector.
1077	border: 1px solid rgba(255, 255, 255, 0.2);	Sets CSS property `border` to `1px solid rgba(255, 255, 255, 0.2)` for the active selector.
1078	box-shadow: 0 8px 20px rgba(2, 6, 23, 0.45);	Sets CSS property `box-shadow` to `0 8px 20px rgba(2, 6, 23, 0.45)` for the active selector.
1079	}	Closes the current CSS rule block.
1080		Blank line used to group related style declarations.
1081	.brand-text {	Opens a CSS rule block for the selector on this line.
1082	letter-spacing: 0.6px;	Sets CSS property `letter-spacing` to `0.6px` for the active selector.
1083	}	Closes the current CSS rule block.
1084		Blank line used to group related style declarations.
1085	.sidebar {	Opens a CSS rule block for the selector on this line.
1086	background: linear-gradient(180deg, rgba(9, 15, 32, 0.98) 0%, rgba(12, 18, 38, 0.96) 55%, rgba(9, 15, 32, 0.98) 100%);	Sets CSS property `background` to `linear-gradient(180deg, rgba(9, 15, 32, 0.98) 0%, rgba(12, 18, 38, 0.96) 55%, rgba(9, 15, 32, 0.98) 100%)` for the active selector.
1087	}	Closes the current CSS rule block.
1088		Blank line used to group related style declarations.
1089	.sidebar::after {	Opens a CSS rule block for the selector on this line.
1090	content: "";	Sets CSS property `content` to `""` for the active selector.
1091	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.

Line	Code	Explanation
1092	inset: 0;	Sets CSS property `inset` to `0` for the active selector.
1093	background: radial-gradient(240px 240px at 20% 10%, rgba(0, 183, 255, 0.22), transparent 60%),	Starts a multi-line CSS property value that continues on next lines.
1094	radial-gradient(220px 220px at 80% 70%, rgba(45, 212, 191, 0.18), transparent 60%);	CSS selector/value syntax line used by nearby declarations.
1095	opacity: 0.6;	Sets CSS property `opacity` to `0.6` for the active selector.
1096	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
1097	z-index: 0;	Sets CSS property `z-index` to `0` for the active selector.
1098	}	Closes the current CSS rule block.
1099		Blank line used to group related style declarations.
1100	.sidebar > * {	Opens a CSS rule block for the selector on this line.
1101	position: relative;	Sets CSS property `position` to `relative` for the active selector.
1102	z-index: 1;	Sets CSS property `z-index` to `1` for the active selector.
1103	}	Closes the current CSS rule block.
1104		Blank line used to group related style declarations.
1105	.menu li a {	Opens a CSS rule block for the selector on this line.
1106	background: rgba(255, 255, 255, 0.02);	Sets CSS property `background` to `rgba(255, 255, 255, 0.02)` for the active selector.
1107	backdrop-filter: blur(6px);	Sets CSS property `backdrop-filter` to `blur(6px)` for the active selector.
1108	}	Closes the current CSS rule block.
1109		Blank line used to group related style declarations.
1110	.menu li a:hover,	CSS selector/value syntax line used by nearby declarations.
1111	.menu li a.active {	Opens a CSS rule block for the selector on this line.
1112	box-shadow: 0 12px 26px rgba(0, 183, 255, 0.18);	Sets CSS property `box-shadow` to `0 12px 26px rgba(0, 183, 255, 0.18)` for the active selector.
1113	}	Closes the current CSS rule block.
1114		Blank line used to group related style declarations.
1115	.content {	Opens a CSS rule block for the selector on this line.
1116	padding: 28px 36px 70px;	Sets CSS property `padding` to `28px 36px 70px` for the active selector.
1117	}	Closes the current CSS rule block.
1118		Blank line used to group related style declarations.
1119	.header {	Opens a CSS rule block for the selector on this line.
1120	background: var(--glass);	Sets CSS property `background` to `var(--glass)` for the active selector.
1121	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1122	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
1123	}	Closes the current CSS rule block.
1124		Blank line used to group related style declarations.
1125	.header::after {	Opens a CSS rule block for the selector on this line.
1126	content: "";	Sets CSS property `content` to `""` for the active selector.
1127	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1128	inset: 0;	Sets CSS property `inset` to `0` for the active selector.
1129	border-radius: inherit;	Sets CSS property `border-radius` to `inherit` for the active selector.
1130	box-shadow: inset 0 1px 0 rgba(255, 255, 255, 0.7);	Sets CSS property `box-shadow` to `inset 0 1px 0 rgba(255, 255, 255, 0.7)` for the active selector.

Line	Code	Explanation
1131	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
1132	}	Closes the current CSS rule block.
1133		Blank line used to group related style declarations.
1134	.card {	Opens a CSS rule block for the selector on this line.
1135	background: var(--glass);	Sets CSS property `background` to `var(--glass)` for the active selector.
1136	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1137	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1138	}	Closes the current CSS rule block.
1139		Blank line used to group related style declarations.
1140	.card::before {	Opens a CSS rule block for the selector on this line.
1141	opacity: 0.35;	Sets CSS property `opacity` to `0.35` for the active selector.
1142	}	Closes the current CSS rule block.
1143		Blank line used to group related style declarations.
1144	.card-header {	Opens a CSS rule block for the selector on this line.
1145	background: transparent;	Sets CSS property `background` to `transparent` for the active selector.
1146	border-bottom: 1px solid var(--line);	Sets CSS property `border-bottom` to `1px solid var(--line)` for the active selector.
1147	}	Closes the current CSS rule block.
1148		Blank line used to group related style declarations.
1149	.card-header h5,	CSS selector/value syntax line used by nearby declarations.
1150	.card-header h6 {	Opens a CSS rule block for the selector on this line.
1151	letter-spacing: 0.3px;	Sets CSS property `letter-spacing` to `0.3px` for the active selector.
1152	}	Closes the current CSS rule block.
1153		Blank line used to group related style declarations.
1154	.container-fluid > .d-flex.justify-content-between.align-items-center.mb-4,	CSS selector/value syntax line used by nearby declarations.
1155	.container-fluid > .d-sm-flex.align-items-center.justify-content-between.mb-4 {	Opens a CSS rule block for the selector on this line.
1156	position: relative;	Sets CSS property `position` to `relative` for the active selector.
1157	padding: 18px 22px;	Sets CSS property `padding` to `18px 22px` for the active selector.
1158	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1159	background: var(--glass);	Sets CSS property `background` to `var(--glass)` for the active selector.
1160	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1161	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1162	backdrop-filter: blur(12px);	Sets CSS property `backdrop-filter` to `blur(12px)` for the active selector.
1163	}	Closes the current CSS rule block.
1164		Blank line used to group related style declarations.
1165	.container-fluid > .d-flex.justify-content-between.align-items-center.mb-4::before,	CSS selector/value syntax line used by nearby declarations.
1166	.container-fluid > .d-sm-flex.align-items-center.justify-content-between.mb-4::before {	Opens a CSS rule block for the selector on this line.
1167	content: "";	Sets CSS property `content` to `""` for the active selector.
1168	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1169	left: 0;	Sets CSS property `left` to `0` for the active selector.

Line	Code	Explanation
1170	top: 0;	Sets CSS property `top` to `0` for the active selector.
1171	height: 3px;	Sets CSS property `height` to `3px` for the active selector.
1172	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
1173	border-radius: var(--radius) var(--radius) 0 0;	Sets CSS property `border-radius` to `var(--radius) var(--radius) 0 0` for the active selector.
1174	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1175	}	Closes the current CSS rule block.
1176		Blank line used to group related style declarations.
1177	.container-fluid > .d-flex.justify-content-between.align-items-center.mb-4 > div:first-child h2,	CSS selector/value syntax line used by nearby declarations.
1178	.container-fluid > .d-flex.justify-content-between.align-items-center.mb-4 > div:first-child h3,	CSS selector/value syntax line used by nearby declarations.
1179	.container-fluid > .d-sm-flex.align-items-center.justify-content-between.mb-4 > div:first-child h3 {	Opens a CSS rule block for the selector on this line.
1180	margin-bottom: 6px;	Sets CSS property `margin-bottom` to `6px` for the active selector.
1181	}	Closes the current CSS rule block.
1182		Blank line used to group related style declarations.
1183	.table thead th {	Opens a CSS rule block for the selector on this line.
1184	font-size: 0.72rem;	Sets CSS property `font-size` to `0.72rem` for the active selector.
1185	letter-spacing: 0.16em;	Sets CSS property `letter-spacing` to `0.16em` for the active selector.
1186	}	Closes the current CSS rule block.
1187		Blank line used to group related style declarations.
1188	.table tbody tr,	CSS selector/value syntax line used by nearby declarations.
1189	table tbody tr {	Opens a CSS rule block for the selector on this line.
1190	background: var(--glass-strong);	Sets CSS property `background` to `var(--glass-strong)` for the active selector.
1191	}	Closes the current CSS rule block.
1192		Blank line used to group related style declarations.
1193	.table.table-hover > tbody > tr:hover,	CSS selector/value syntax line used by nearby declarations.
1194	table tbody tr:hover {	Opens a CSS rule block for the selector on this line.
1195	background: rgba(0, 183, 255, 0.08);	Sets CSS property `background` to `rgba(0, 183, 255, 0.08)` for the active selector.
1196	}	Closes the current CSS rule block.
1197		Blank line used to group related style declarations.
1198	.badge {	Opens a CSS rule block for the selector on this line.
1199	box-shadow: 0 6px 14px rgba(15, 23, 42, 0.12);	Sets CSS property `box-shadow` to `0 6px 14px rgba(15, 23, 42, 0.12)` for the active selector.
1200	}	Closes the current CSS rule block.
1201		Blank line used to group related style declarations.
1202	.btn:not(.btn-link) {	Opens a CSS rule block for the selector on this line.
1203	box-shadow: 0 12px 24px rgba(15, 23, 42, 0.12);	Sets CSS property `box-shadow` to `0 12px 24px rgba(15, 23, 42, 0.12)` for the active selector.
1204	}	Closes the current CSS rule block.
1205		Blank line used to group related style declarations.
1206	.btn:focus-visible,	CSS selector/value syntax line used by nearby declarations.
1207	.form-control:focus-visible,	CSS selector/value syntax line used by nearby declarations.
1208	.form-select:focus-visible,	CSS selector/value syntax line used by nearby declarations.

Line	Code	Explanation
1209	textarea:focus-visible {	Opens a CSS rule block for the selector on this line.
1210	outline: none;	Sets CSS property `outline` to `none` for the active selector.
1211	box-shadow: var(--glow);	Sets CSS property `box-shadow` to `var(--glow)` for the active selector.
1212	}	Closes the current CSS rule block.
1213		Blank line used to group related style declarations.
1214	.input-group-text {	Opens a CSS rule block for the selector on this line.
1215	border: 1px solid rgba(148, 163, 184, 0.35);	Sets CSS property `border` to `1px solid rgba(148, 163, 184, 0.35)` for the active selector.
1216	background: rgba(255, 255, 255, 0.9);	Sets CSS property `background` to `rgba(255, 255, 255, 0.9)` for the active selector.
1217	}	Closes the current CSS rule block.
1218		Blank line used to group related style declarations.
1219	.list-group-item {	Opens a CSS rule block for the selector on this line.
1220	background: transparent;	Sets CSS property `background` to `transparent` for the active selector.
1221	border-color: var(--line);	Sets CSS property `border-color` to `var(--line)` for the active selector.
1222	}	Closes the current CSS rule block.
1223		Blank line used to group related style declarations.
1224	.progress {	Opens a CSS rule block for the selector on this line.
1225	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
1226	}	Closes the current CSS rule block.
1227		Blank line used to group related style declarations.
1228	.transition-base {	Opens a CSS rule block for the selector on this line.
1229	transition: var(--transition);	Sets CSS property `transition` to `var(--transition)` for the active selector.
1230	}	Closes the current CSS rule block.
1231		Blank line used to group related style declarations.
1232	/* Dashboard widgets shared */	CSS comment used to document the stylesheet.
1233	.stat-card {	Opens a CSS rule block for the selector on this line.
1234	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1235	border: 1px solid rgba(255, 255, 255, 0.35);	Sets CSS property `border` to `1px solid rgba(255, 255, 255, 0.35)` for the active selector.
1236	color: var(--ink);	Sets CSS property `color` to `var(--ink)` for the active selector.
1237	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1238	position: relative;	Sets CSS property `position` to `relative` for the active selector.
1239	transition: transform 0.3s ease, box-shadow 0.3s ease;	Sets CSS property `transition` to `transform 0.3s ease, box-shadow 0.3s ease` for the active selector.
1240	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1241	min-height: 140px;	Sets CSS property `min-height` to `140px` for the active selector.
1242	background: var(--glass-strong);	Sets CSS property `background` to `var(--glass-strong)` for the active selector.
1243	}	Closes the current CSS rule block.
1244		Blank line used to group related style declarations.
1245	.stat-card .card-body {	Opens a CSS rule block for the selector on this line.
1246	position: relative;	Sets CSS property `position` to `relative` for the active selector.
1247	z-index: 2;	Sets CSS property `z-index` to `2` for the active selector.
1248	padding: 1.5rem;	Sets CSS property `padding` to `1.5rem` for the active selector.
1249	}	Closes the current CSS rule block.

Line	Code	Explanation
1250		Blank line used to group related style declarations.
1251	.stat-card:hover {	Opens a CSS rule block for the selector on this line.
1252	transform: translateY(-4px);	Sets CSS property `transform` to `translateY(-4px)` for the active selector.
1253	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
1254	}	Closes the current CSS rule block.
1255		Blank line used to group related style declarations.
1256	.stat-card .icon-bg {	Opens a CSS rule block for the selector on this line.
1257	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1258	right: 16px;	Sets CSS property `right` to `16px` for the active selector.
1259	top: 50%;	Sets CSS property `top` to `50%` for the active selector.
1260	transform: translateY(-50%) rotate(-12deg);	Sets CSS property `transform` to `translateY(-50%) rotate(-12deg)` for the active selector.
1261	font-size: 4.4rem;	Sets CSS property `font-size` to `4.4rem` for the active selector.
1262	opacity: 0.12;	Sets CSS property `opacity` to `0.12` for the active selector.
1263	z-index: 1;	Sets CSS property `z-index` to `1` for the active selector.
1264	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
1265	}	Closes the current CSS rule block.
1266		Blank line used to group related style declarations.
1267	.stat-card.bg-gradient-primary,	CSS selector/value syntax line used by nearby declarations.
1268	.stat-card.bg-gradient-success,	CSS selector/value syntax line used by nearby declarations.
1269	.stat-card.bg-gradient-info,	CSS selector/value syntax line used by nearby declarations.
1270	.stat-card.bg-gradient-warning,	CSS selector/value syntax line used by nearby declarations.
1271	.stat-card.bg-gradient-red,	CSS selector/value syntax line used by nearby declarations.
1272	.stat-card.bg-gradient-blue,	CSS selector/value syntax line used by nearby declarations.
1273	.stat-card.bg-gradient-green {	Opens a CSS rule block for the selector on this line.
1274	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
1275	}	Closes the current CSS rule block.
1276		Blank line used to group related style declarations.
1277	.stat-card.bg-gradient-primary { background: linear-gradient(135deg, #00b7ff 0%, #3b82f6 100%); }	CSS selector/value syntax line used by nearby declarations.
1278	.stat-card.bg-gradient-success { background: linear-gradient(135deg, #22c55e 0%, #10b981 100%); }	CSS selector/value syntax line used by nearby declarations.
1279	.stat-card.bg-gradient-info { background: linear-gradient(135deg, #38bdf8 0%, #0ea5e9 100%); }	CSS selector/value syntax line used by nearby declarations.
1280	.stat-card.bg-gradient-warning { background: linear-gradient(135deg, #f59e0b 0%, #f97316 100%); }	CSS selector/value syntax line used by nearby declarations.
1281	.stat-card.bg-gradient-red { background: linear-gradient(135deg, #ef4444 0%, #fb7185 100%); }	CSS selector/value syntax line used by nearby declarations.
1282	.stat-card.bg-gradient-blue { background: linear-gradient(135deg, #00b7ff 0%, #3b82f6 100%); }	CSS selector/value syntax line used by nearby declarations.
1283	.stat-card.bg-gradient-green { background: linear-gradient(135deg, #22c55e 0%, #16a34a 100%); }	CSS selector/value syntax line used by nearby declarations.
1284		Blank line used to group related style declarations.
1285	.stat-card.bg-gradient-primary .text-uppercase,	CSS selector/value syntax line used by nearby declarations.

Line	Code	Explanation
1286	<code>.stat-card.bg-gradient-success .text-uppercase,</code>	CSS selector/value syntax line used by nearby declarations.
1287	<code>.stat-card.bg-gradient-info .text-uppercase,</code>	CSS selector/value syntax line used by nearby declarations.
1288	<code>.stat-card.bg-gradient-warning .text-uppercase,</code>	CSS selector/value syntax line used by nearby declarations.
1289	<code>.stat-card.bg-gradient-red .text-uppercase,</code>	CSS selector/value syntax line used by nearby declarations.
1290	<code>.stat-card.bg-gradient-blue .text-uppercase,</code>	CSS selector/value syntax line used by nearby declarations.
1291	<code>.stat-card.bg-gradient-green .text-uppercase,</code>	CSS selector/value syntax line used by nearby declarations.
1292	<code>.stat-card.bg-gradient-primary .small,</code>	CSS selector/value syntax line used by nearby declarations.
1293	<code>.stat-card.bg-gradient-success .small,</code>	CSS selector/value syntax line used by nearby declarations.
1294	<code>.stat-card.bg-gradient-info .small,</code>	CSS selector/value syntax line used by nearby declarations.
1295	<code>.stat-card.bg-gradient-warning .small,</code>	CSS selector/value syntax line used by nearby declarations.
1296	<code>.stat-card.bg-gradient-red .small,</code>	CSS selector/value syntax line used by nearby declarations.
1297	<code>.stat-card.bg-gradient-blue .small,</code>	CSS selector/value syntax line used by nearby declarations.
1298	<code>.stat-card.bg-gradient-green .small {</code>	Opens a CSS rule block for the selector on this line.
1299	<code>color: rgba(255, 255, 255, 0.82);</code>	Sets CSS property <code>color</code> to <code>rgba(255, 255, 255, 0.82)</code> for the active selector.
1300	<code>}</code>	Closes the current CSS rule block.
1301		Blank line used to group related style declarations.
1302	<code>.notice-item {</code>	Opens a CSS rule block for the selector on this line.
1303	<code>border-left: 4px solid transparent;</code>	Sets CSS property <code>border-left</code> to <code>4px solid transparent</code> for the active selector.
1304	<code>border-bottom: 1px solid var(--line);</code>	Sets CSS property <code>border-bottom</code> to <code>1px solid var(--line)</code> for the active selector.
1305	<code>transition: all 0.2s ease;</code>	Sets CSS property <code>transition</code> to <code>all 0.2s ease</code> for the active selector.
1306	<code>padding: 1.25rem;</code>	Sets CSS property <code>padding</code> to <code>1.25rem</code> for the active selector.
1307	<code>}</code>	Closes the current CSS rule block.
1308		Blank line used to group related style declarations.
1309	<code>.notice-item:last-child {</code>	Opens a CSS rule block for the selector on this line.
1310	<code>border-bottom: none;</code>	Sets CSS property <code>border-bottom</code> to <code>none</code> for the active selector.
1311	<code>}</code>	Closes the current CSS rule block.
1312		Blank line used to group related style declarations.
1313	<code>.notice-item:hover {</code>	Opens a CSS rule block for the selector on this line.
1314	<code>background: rgba(0, 183, 255, 0.08);</code>	Sets CSS property <code>background</code> to <code>rgba(0, 183, 255, 0.08)</code> for the active selector.
1315	<code>border-left-color: var(--accent);</code>	Sets CSS property <code>border-left-color</code> to <code>var(--accent)</code> for the active selector.
1316	<code>padding-left: 1.5rem;</code>	Sets CSS property <code>padding-left</code> to <code>1.5rem</code> for the active selector.
1317	<code>}</code>	Closes the current CSS rule block.
1318		Blank line used to group related style declarations.
1319	<code>.btn-quick,</code>	CSS selector/value syntax line used by nearby declarations.
1320	<code>.btn-action-card {</code>	Opens a CSS rule block for the selector on this line.
1321	<code>border-radius: 12px;</code>	Sets CSS property <code>border-radius</code> to <code>12px</code> for the active selector.
1322	<code>padding: 14px 16px;</code>	Sets CSS property <code>padding</code> to <code>14px 16px</code> for the active selector.
1323	<code>font-weight: 600;</code>	Sets CSS property <code>font-weight</code> to <code>600</code> for the active selector.
1324	<code>display: flex;</code>	Sets CSS property <code>display</code> to <code>flex</code> for the active selector.
1325	<code>align-items: center;</code>	Sets CSS property <code>align-items</code> to <code>center</code> for the active selector.
1326	<code>gap: 14px;</code>	Sets CSS property <code>gap</code> to <code>14px</code> for the active selector.

Line	Code	Explanation
1327	transition: all 0.25s ease;	Sets CSS property `transition` to `all 0.25s ease` for the active selector.
1328	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1329	background: var(--card-solid);	Sets CSS property `background` to `var(--card-solid)` for the active selector.
1330	color: var(--muted);	Sets CSS property `color` to `var(--muted)` for the active selector.
1331	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1332	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
1333	margin-bottom: 12px;	Sets CSS property `margin-bottom` to `12px` for the active selector.
1334	}	Closes the current CSS rule block.
1335		Blank line used to group related style declarations.
1336	.btn-quick:hover,	CSS selector/value syntax line used by nearby declarations.
1337	.btn-action-card:hover {	Opens a CSS rule block for the selector on this line.
1338	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1339	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
1340	border-color: transparent;	Sets CSS property `border-color` to `transparent` for the active selector.
1341	transform: translateX(6px);	Sets CSS property `transform` to `translateX(6px)` for the active selector.
1342	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
1343	}	Closes the current CSS rule block.
1344		Blank line used to group related style declarations.
1345	.btn-quick i,	CSS selector/value syntax line used by nearby declarations.
1346	.btn-action-card i {	Opens a CSS rule block for the selector on this line.
1347	font-size: 1.2rem;	Sets CSS property `font-size` to `1.2rem` for the active selector.
1348	width: 30px;	Sets CSS property `width` to `30px` for the active selector.
1349	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
1350	}	Closes the current CSS rule block.
1351		Blank line used to group related style declarations.
1352	.btn-quick:hover i,	CSS selector/value syntax line used by nearby declarations.
1353	.btn-action-card:hover i,	CSS selector/value syntax line used by nearby declarations.
1354	.btn-action-card:hover .text-dark,	CSS selector/value syntax line used by nearby declarations.
1355	.btn-action-card:hover .text-muted {	Opens a CSS rule block for the selector on this line.
1356	color: #fff !important;	Sets CSS property `color` to `#fff !important` for the active selector.
1357	}	Closes the current CSS rule block.
1358		Blank line used to group related style declarations.
1359	.status-indicator {	Opens a CSS rule block for the selector on this line.
1360	width: 10px;	Sets CSS property `width` to `10px` for the active selector.
1361	height: 10px;	Sets CSS property `height` to `10px` for the active selector.
1362	background-color: var(--success);	Sets CSS property `background-color` to `var(--success)` for the active selector.
1363	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
1364	display: inline-block;	Sets CSS property `display` to `inline-block` for the active selector.
1365	margin-right: 8px;	Sets CSS property `margin-right` to `8px` for the active selector.
1366	box-shadow: 0 0 0 rgba(34, 197, 94, 0.4);	Sets CSS property `box-shadow` to `0 0 0 rgba(34, 197, 94, 0.4)` for the active selector.
1367	animation: pulse 2s infinite;	Sets CSS property `animation` to `pulse 2s infinite` for the active selector.

Line	Code	Explanation
1368	}	Closes the current CSS rule block.
1369		Blank line used to group related style declarations.
1370	/* Enrollment */	CSS comment used to document the stylesheet.
1371	.enroll-card {	Opens a CSS rule block for the selector on this line.
1372	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1373	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1374	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1375	}	Closes the current CSS rule block.
1376		Blank line used to group related style declarations.
1377	.course-select-wrapper {	Opens a CSS rule block for the selector on this line.
1378	background: rgba(248, 250, 252, 0.9);	Sets CSS property `background` to `rgba(248, 250, 252, 0.9)` for the active selector.
1379	padding: 18px;	Sets CSS property `padding` to `18px` for the active selector.
1380	border-radius: 14px;	Sets CSS property `border-radius` to `14px` for the active selector.
1381	border: 1px dashed rgba(148, 163, 184, 0.4);	Sets CSS property `border` to `1px dashed rgba(148, 163, 184, 0.4)` for the active selector.
1382	}	Closes the current CSS rule block.
1383		Blank line used to group related style declarations.
1384	.table-card {	Opens a CSS rule block for the selector on this line.
1385	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1386	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1387	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1388	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1389	}	Closes the current CSS rule block.
1390		Blank line used to group related style declarations.
1391	.course-badge {	Opens a CSS rule block for the selector on this line.
1392	width: 48px;	Sets CSS property `width` to `48px` for the active selector.
1393	height: 48px;	Sets CSS property `height` to `48px` for the active selector.
1394	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
1395	background: rgba(0, 183, 255, 0.12);	Sets CSS property `background` to `rgba(0, 183, 255, 0.12)` for the active selector.
1396	color: var(--accent);	Sets CSS property `color` to `var(--accent)` for the active selector.
1397	display: flex;	Sets CSS property `display` to `flex` for the active selector.
1398	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
1399	justify-content: center;	Sets CSS property `justify-content` to `center` for the active selector.
1400	font-weight: 700;	Sets CSS property `font-weight` to `700` for the active selector.
1401	font-size: 1.05rem;	Sets CSS property `font-size` to `1.05rem` for the active selector.
1402	}	Closes the current CSS rule block.
1403		Blank line used to group related style declarations.
1404	/* Teacher classes */	CSS comment used to document the stylesheet.
1405	.course-card {	Opens a CSS rule block for the selector on this line.
1406	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1407	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1408	transition: transform 0.3s ease, box-shadow 0.3s ease;	Sets CSS property `transition` to `transform 0.3s ease, box-shadow 0.3s ease` for the active selector.

Line	Code	Explanation
1409	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1410	}	Closes the current CSS rule block.
1411		Blank line used to group related style declarations.
1412	.course-card:hover {	Opens a CSS rule block for the selector on this line.
1413	transform: translateY(-5px);	Sets CSS property `transform` to `translateY(-5px)` for the active selector.
1414	box-shadow: var(--shadow-md);	Sets CSS property `box-shadow` to `var(--shadow-md)` for the active selector.
1415	}	Closes the current CSS rule block.
1416		Blank line used to group related style declarations.
1417	.card-border-top {	Opens a CSS rule block for the selector on this line.
1418	height: 6px;	Sets CSS property `height` to `6px` for the active selector.
1419	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1420	}	Closes the current CSS rule block.
1421		Blank line used to group related style declarations.
1422	/* Attendance */	CSS comment used to document the stylesheet.
1423	.student-avatar {	Opens a CSS rule block for the selector on this line.
1424	width: 42px;	Sets CSS property `width` to `42px` for the active selector.
1425	height: 42px;	Sets CSS property `height` to `42px` for the active selector.
1426	background: var(--gradient-cool);	Sets CSS property `background` to `var(--gradient-cool)` for the active selector.
1427	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
1428	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
1429	display: flex;	Sets CSS property `display` to `flex` for the active selector.
1430	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
1431	justify-content: center;	Sets CSS property `justify-content` to `center` for the active selector.
1432	font-weight: 700;	Sets CSS property `font-weight` to `700` for the active selector.
1433	font-size: 0.9rem;	Sets CSS property `font-size` to `0.9rem` for the active selector.
1434	}	Closes the current CSS rule block.
1435		Blank line used to group related style declarations.
1436	.attendance-card {	Opens a CSS rule block for the selector on this line.
1437	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1438	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1439	box-shadow: var(--shadow-sm);	Sets CSS property `box-shadow` to `var(--shadow-sm)` for the active selector.
1440	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1441	}	Closes the current CSS rule block.
1442		Blank line used to group related style declarations.
1443	.row-hover:hover {	Opens a CSS rule block for the selector on this line.
1444	background-color: rgba(0, 183, 255, 0.06);	Sets CSS property `background-color` to `rgba(0, 183, 255, 0.06)` for the active selector.
1445	}	Closes the current CSS rule block.
1446		Blank line used to group related style declarations.
1447	.form-check-input:checked {	Opens a CSS rule block for the selector on this line.
1448	background-color: #22c55e;	Sets CSS property `background-color` to `#22c55e` for the active selector.
1449	border-color: #22c55e;	Sets CSS property `border-color` to `#22c55e` for the active selector.

Line	Code	Explanation
1450	}	Closes the current CSS rule block.
1451		Blank line used to group related style declarations.
1452	/* Settings */	CSS comment used to document the stylesheet.
1453	.settings-card {	Opens a CSS rule block for the selector on this line.
1454	border-radius: 20px;	Sets CSS property `border-radius` to `20px` for the active selector.
1455	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1456	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1457	box-shadow: var(--shadow-lg);	Sets CSS property `box-shadow` to `var(--shadow-lg)` for the active selector.
1458	animation: slideUp 0.5s ease-out;	Sets CSS property `animation` to `slideUp 0.5s ease-out` for the active selector.
1459	background: var(--glass-strong);	Sets CSS property `background` to `var(--glass-strong)` for the active selector.
1460	}	Closes the current CSS rule block.
1461		Blank line used to group related style declarations.
1462	.settings-header {	Opens a CSS rule block for the selector on this line.
1463	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1464	padding: 30px;	Sets CSS property `padding` to `30px` for the active selector.
1465	color: white;	Sets CSS property `color` to `white` for the active selector.
1466	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
1467	position: relative;	Sets CSS property `position` to `relative` for the active selector.
1468	}	Closes the current CSS rule block.
1469		Blank line used to group related style declarations.
1470	.settings-back {	Opens a CSS rule block for the selector on this line.
1471	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1472	top: 16px;	Sets CSS property `top` to `16px` for the active selector.
1473	left: 16px;	Sets CSS property `left` to `16px` for the active selector.
1474	display: inline-flex;	Sets CSS property `display` to `inline-flex` for the active selector.
1475	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
1476	gap: 8px;	Sets CSS property `gap` to `8px` for the active selector.
1477	padding: 8px 14px;	Sets CSS property `padding` to `8px 14px` for the active selector.
1478	border-radius: 999px;	Sets CSS property `border-radius` to `999px` for the active selector.
1479	background: rgba(255, 255, 255, 0.2);	Sets CSS property `background` to `rgba(255, 255, 255, 0.2)` for the active selector.
1480	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
1481	text-decoration: none;	Sets CSS property `text-decoration` to `none` for the active selector.
1482	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
1483	border: 1px solid rgba(255, 255, 255, 0.35);	Sets CSS property `border` to `1px solid rgba(255, 255, 255, 0.35)` for the active selector.
1484	transition: all 0.2s ease;	Sets CSS property `transition` to `all 0.2s ease` for the active selector.
1485	}	Closes the current CSS rule block.
1486		Blank line used to group related style declarations.
1487	.settings-back:hover {	Opens a CSS rule block for the selector on this line.
1488	background: rgba(255, 255, 255, 0.32);	Sets CSS property `background` to `rgba(255, 255, 255, 0.32)` for the active selector.
1489	color: #fff;	Sets CSS property `color` to `#fff` for the active selector.
1490	transform: translateY(-1px);	Sets CSS property `transform` to `translateY(-1px)` for the active selector.

Line	Code	Explanation
1491	}	Closes the current CSS rule block.
1492		Blank line used to group related style declarations.
1493	.header-circle {	Opens a CSS rule block for the selector on this line.
1494	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1495	width: 160px;	Sets CSS property `width` to `160px` for the active selector.
1496	height: 160px;	Sets CSS property `height` to `160px` for the active selector.
1497	background: rgba(255, 255, 255, 0.12);	Sets CSS property `background` to `rgba(255, 255, 255, 0.12)` for the active selector.
1498	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
1499	top: -55px;	Sets CSS property `top` to `-55px` for the active selector.
1500	right: -55px;	Sets CSS property `right` to `-55px` for the active selector.
1501	}	Closes the current CSS rule block.
1502		Blank line used to group related style declarations.
1503	.btn-update {	Opens a CSS rule block for the selector on this line.
1504	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1505	border: none;	Sets CSS property `border` to `none` for the active selector.
1506	padding: 12px;	Sets CSS property `padding` to `12px` for the active selector.
1507	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
1508	font-weight: 600;	Sets CSS property `font-weight` to `600` for the active selector.
1509	transition: all 0.3s ease;	Sets CSS property `transition` to `all 0.3s ease` for the active selector.
1510	}	Closes the current CSS rule block.
1511		Blank line used to group related style declarations.
1512	.btn-update:hover {	Opens a CSS rule block for the selector on this line.
1513	filter: brightness(1.03);	Sets CSS property `filter` to `brightness(1.03)` for the active selector.
1514	transform: translateY(-2px);	Sets CSS property `transform` to `translateY(-2px)` for the active selector.
1515	box-shadow: 0 12px 24px rgba(0, 183, 255, 0.25);	Sets CSS property `box-shadow` to `0 12px 24px rgba(0, 183, 255, 0.25)` for the active selector.
1516	}	Closes the current CSS rule block.
1517		Blank line used to group related style declarations.
1518	@keyframes slideUp {	Starts a keyframes animation definition.
1519	from { opacity: 0; transform: translateY(20px); }	CSS selector/value syntax line used by nearby declarations.
1520	to { opacity: 1; transform: translateY(0); }	CSS selector/value syntax line used by nearby declarations.
1521	}	Closes the current CSS rule block.
1522		Blank line used to group related style declarations.
1523	/* Auth */	CSS comment used to document the stylesheet.
1524	.auth-page {	Opens a CSS rule block for the selector on this line.
1525	display: grid;	Sets CSS property `display` to `grid` for the active selector.
1526	place-items: center;	Sets CSS property `place-items` to `center` for the active selector.
1527	min-height: 100vh;	Sets CSS property `min-height` to `100vh` for the active selector.
1528	padding: 24px;	Sets CSS property `padding` to `24px` for the active selector.
1529	background: transparent;	Sets CSS property `background` to `transparent` for the active selector.
1530	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1531	}	Closes the current CSS rule block.

Line	Code	Explanation
1532		Blank line used to group related style declarations.
1533	.login-container {	Opens a CSS rule block for the selector on this line.
1534	width: 100%;	Sets CSS property `width` to `100%` for the active selector.
1535	max-width: 1020px;	Sets CSS property `max-width` to `1020px` for the active selector.
1536	min-height: 620px;	Sets CSS property `min-height` to `620px` for the active selector.
1537	background: var(--glass);	Sets CSS property `background` to `var(--glass)` for the active selector.
1538	border-radius: 28px;	Sets CSS property `border-radius` to `28px` for the active selector.
1539	border: 1px solid var(--line);	Sets CSS property `border` to `1px solid var(--line)` for the active selector.
1540	box-shadow: var(--shadow-lg);	Sets CSS property `box-shadow` to `var(--shadow-lg)` for the active selector.
1541	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1542	display: flex;	Sets CSS property `display` to `flex` for the active selector.
1543	animation: fadeIn 0.8s ease-in-out;	Sets CSS property `animation` to `fadeIn 0.8s ease-in-out` for the active selector.
1544	}	Closes the current CSS rule block.
1545		Blank line used to group related style declarations.
1546	.login-visual {	Opens a CSS rule block for the selector on this line.
1547	flex: 1;	Sets CSS property `flex` to `1` for the active selector.
1548	background: var(--gradient-primary);	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1549	display: flex;	Sets CSS property `display` to `flex` for the active selector.
1550	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
1551	justify-content: center;	Sets CSS property `justify-content` to `center` for the active selector.
1552	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
1553	color: white;	Sets CSS property `color` to `white` for the active selector.
1554	padding: 44px;	Sets CSS property `padding` to `44px` for the active selector.
1555	position: relative;	Sets CSS property `position` to `relative` for the active selector.
1556	overflow: hidden;	Sets CSS property `overflow` to `hidden` for the active selector.
1557	}	Closes the current CSS rule block.
1558		Blank line used to group related style declarations.
1559	.visual-content {	Opens a CSS rule block for the selector on this line.
1560	z-index: 2;	Sets CSS property `z-index` to `2` for the active selector.
1561	text-align: center;	Sets CSS property `text-align` to `center` for the active selector.
1562	display: flex;	Sets CSS property `display` to `flex` for the active selector.
1563	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
1564	align-items: center;	Sets CSS property `align-items` to `center` for the active selector.
1565	}	Closes the current CSS rule block.
1566		Blank line used to group related style declarations.
1567	.visual-content h1 {	Opens a CSS rule block for the selector on this line.
1568	font-weight: 700;	Sets CSS property `font-weight` to `700` for the active selector.
1569	font-size: 2.6rem;	Sets CSS property `font-size` to `2.6rem` for the active selector.
1570	margin-bottom: 10px;	Sets CSS property `margin-bottom` to `10px` for the active selector.
1571	}	Closes the current CSS rule block.
1572		Blank line used to group related style declarations.

Line	Code	Explanation
1573	.visual-content p {	Opens a CSS rule block for the selector on this line.
1574	font-size: 1.1rem;	Sets CSS property `font-size` to `1.1rem` for the active selector.
1575	opacity: 0.9;	Sets CSS property `opacity` to `0.9` for the active selector.
1576	}	Closes the current CSS rule block.
1577		Blank line used to group related style declarations.
1578	.login-visual::after {	Opens a CSS rule block for the selector on this line.
1579	content: "";	Sets CSS property `content` to `""` for the active selector.
1580	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1581	inset: 0;	Sets CSS property `inset` to `0` for the active selector.
1582	background: radial-gradient(400px 400px at 15% 20%, rgba(255, 255, 255, 0.18), transparent 60%),	Starts a multi-line CSS property value that continues on next lines.
1583	radial-gradient(360px 360px at 85% 80%, rgba(255, 255, 255, 0.12), transparent 60%);	CSS selector/value syntax line used by nearby declarations.
1584	opacity: 0.8;	Sets CSS property `opacity` to `0.8` for the active selector.
1585	pointer-events: none;	Sets CSS property `pointer-events` to `none` for the active selector.
1586	}	Closes the current CSS rule block.
1587		Blank line used to group related style declarations.
1588	.circle {	Opens a CSS rule block for the selector on this line.
1589	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1590	background: rgba(255, 255, 255, 0.12);	Sets CSS property `background` to `rgba(255, 255, 255, 0.12)` for the active selector.
1591	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
1592	}	Closes the current CSS rule block.
1593		Blank line used to group related style declarations.
1594	.circle-1 { width: 320px; height: 320px; top: -60px; left: -60px; }	CSS selector/value syntax line used by nearby declarations.
1595	.circle-2 { width: 220px; height: 220px; bottom: 40px; right: -60px; }	CSS selector/value syntax line used by nearby declarations.
1596		Blank line used to group related style declarations.
1597	.login-logo {	Opens a CSS rule block for the selector on this line.
1598	width: 120px;	Sets CSS property `width` to `120px` for the active selector.
1599	height: auto;	Sets CSS property `height` to `auto` for the active selector.
1600	border-radius: 50%;	Sets CSS property `border-radius` to `50%` for the active selector.
1601	background: rgba(255, 255, 255, 0.22);	Sets CSS property `background` to `rgba(255, 255, 255, 0.22)` for the active selector.
1602	padding: 16px;	Sets CSS property `padding` to `16px` for the active selector.
1603	box-shadow: 0 16px 30px rgba(11, 18, 32, 0.3);	Sets CSS property `box-shadow` to `0 16px 30px rgba(11, 18, 32, 0.3)` for the active selector.
1604	backdrop-filter: blur(6px);	Sets CSS property `backdrop-filter` to `blur(6px)` for the active selector.
1605	margin-bottom: 1.5rem;	Sets CSS property `margin-bottom` to `1.5rem` for the active selector.
1606	}	Closes the current CSS rule block.
1607		Blank line used to group related style declarations.
1608	.login-form-wrapper {	Opens a CSS rule block for the selector on this line.
1609	flex: 1;	Sets CSS property `flex` to `1` for the active selector.
1610	padding: 50px;	Sets CSS property `padding` to `50px` for the active selector.
1611	display: flex;	Sets CSS property `display` to `flex` for the active selector.

Line	Code	Explanation
1612	<code>flex-direction: column;</code>	Sets CSS property `flex-direction` to `column` for the active selector.
1613	<code>justify-content: center;</code>	Sets CSS property `justify-content` to `center` for the active selector.
1614	<code>position: relative;</code>	Sets CSS property `position` to `relative` for the active selector.
1615	<code>}</code>	Closes the current CSS rule block.
1616		Blank line used to group related style declarations.
1617	<code>.login-header {</code>	Opens a CSS rule block for the selector on this line.
1618	<code>margin-bottom: 30px;</code>	Sets CSS property `margin-bottom` to `30px` for the active selector.
1619	<code>}</code>	Closes the current CSS rule block.
1620		Blank line used to group related style declarations.
1621	<code>.login-header h2 {</code>	Opens a CSS rule block for the selector on this line.
1622	<code>font-weight: 700;</code>	Sets CSS property `font-weight` to `700` for the active selector.
1623	<code>color: var(--ink);</code>	Sets CSS property `color` to `var(--ink)` for the active selector.
1624	<code>}</code>	Closes the current CSS rule block.
1625		Blank line used to group related style declarations.
1626	<code>.login-header p {</code>	Opens a CSS rule block for the selector on this line.
1627	<code>color: var(--muted);</code>	Sets CSS property `color` to `var(--muted)` for the active selector.
1628	<code>}</code>	Closes the current CSS rule block.
1629		Blank line used to group related style declarations.
1630	<code>.btn-login {</code>	Opens a CSS rule block for the selector on this line.
1631	<code>background: var(--gradient-primary);</code>	Sets CSS property `background` to `var(--gradient-primary)` for the active selector.
1632	<code>border: none;</code>	Sets CSS property `border` to `none` for the active selector.
1633	<code>height: 52px;</code>	Sets CSS property `height` to `52px` for the active selector.
1634	<code>border-radius: 12px;</code>	Sets CSS property `border-radius` to `12px` for the active selector.
1635	<code>font-size: 1rem;</code>	Sets CSS property `font-size` to `1rem` for the active selector.
1636	<code>font-weight: 600;</code>	Sets CSS property `font-weight` to `600` for the active selector.
1637	<code>color: white;</code>	Sets CSS property `color` to `white` for the active selector.
1638	<code>transition: all 0.3s ease;</code>	Sets CSS property `transition` to `all 0.3s ease` for the active selector.
1639	<code>width: 100%;</code>	Sets CSS property `width` to `100%` for the active selector.
1640	<code>margin-top: 10px;</code>	Sets CSS property `margin-top` to `10px` for the active selector.
1641	<code>box-shadow: 0 14px 26px rgba(0, 183, 255, 0.25);</code>	Sets CSS property `box-shadow` to `0 14px 26px rgba(0, 183, 255, 0.25)` for the active selector.
1642	<code>}</code>	Closes the current CSS rule block.
1643		Blank line used to group related style declarations.
1644	<code>.btn-login:hover {</code>	Opens a CSS rule block for the selector on this line.
1645	<code>filter: brightness(1.03);</code>	Sets CSS property `filter` to `brightness(1.03)` for the active selector.
1646	<code>transform: translateY(-2px);</code>	Sets CSS property `transform` to `translateY(-2px)` for the active selector.
1647	<code>box-shadow: 0 18px 34px rgba(0, 183, 255, 0.3);</code>	Sets CSS property `box-shadow` to `0 18px 34px rgba(0, 183, 255, 0.3)` for the active selector.
1648	<code>}</code>	Closes the current CSS rule block.
1649		Blank line used to group related style declarations.
1650	<code>.auth-page .form-floating > .form-control {</code>	Opens a CSS rule block for the selector on this line.
1651	<code>height: 55px;</code>	Sets CSS property `height` to `55px` for the active selector.
1652	<code>border: 1px solid rgba(148, 163, 184, 0.35);</code>	Sets CSS property `border` to `1px solid rgba(148, 163, 184, 0.35)` for the active selector.

Line	Code	Explanation
1653	border-radius: 12px;	Sets CSS property `border-radius` to `12px` for the active selector.
1654	}	Closes the current CSS rule block.
1655		Blank line used to group related style declarations.
1656	.auth-page .form-floating > .form-control:focus {	Opens a CSS rule block for the selector on this line.
1657	border-color: var(--accent);	Sets CSS property `border-color` to `var(--accent)` for the active selector.
1658	box-shadow: 0 0 0 0.25rem rgba(0, 183, 255, 0.2);	Sets CSS property `box-shadow` to `0 0 0 0.25rem rgba(0, 183, 255, 0.2)` for the active selector.
1659	}	Closes the current CSS rule block.
1660		Blank line used to group related style declarations.
1661	.auth-page .form-floating > label {	Opens a CSS rule block for the selector on this line.
1662	color: var(--muted);	Sets CSS property `color` to `var(--muted)` for the active selector.
1663	}	Closes the current CSS rule block.
1664		Blank line used to group related style declarations.
1665	.auth-page .input-group-text {	Opens a CSS rule block for the selector on this line.
1666	background: none;	Sets CSS property `background` to `none` for the active selector.
1667	border: none;	Sets CSS property `border` to `none` for the active selector.
1668	position: absolute;	Sets CSS property `position` to `absolute` for the active selector.
1669	right: 15px;	Sets CSS property `right` to `15px` for the active selector.
1670	top: 18px;	Sets CSS property `top` to `18px` for the active selector.
1671	z-index: 4;	Sets CSS property `z-index` to `4` for the active selector.
1672	color: #94a3b8;	Sets CSS property `color` to `#94a3b8` for the active selector.
1673	}	Closes the current CSS rule block.
1674		Blank line used to group related style declarations.
1675	@keyframes fadeIn {	Starts a keyframes animation definition.
1676	from { opacity: 0; transform: translateY(20px); }	CSS selector/value syntax line used by nearby declarations.
1677	to { opacity: 1; transform: translateY(0); }	CSS selector/value syntax line used by nearby declarations.
1678	}	Closes the current CSS rule block.
1679		Blank line used to group related style declarations.
1680	@media (max-width: 768px) {	Starts a media query block for responsive behavior.
1681	.login-container {	Opens a CSS rule block for the selector on this line.
1682	flex-direction: column;	Sets CSS property `flex-direction` to `column` for the active selector.
1683	min-height: auto;	Sets CSS property `min-height` to `auto` for the active selector.
1684	max-width: 440px;	Sets CSS property `max-width` to `440px` for the active selector.
1685	}	Closes the current CSS rule block.
1686		Blank line used to group related style declarations.
1687	.login-visual {	Opens a CSS rule block for the selector on this line.
1688	display: none;	Sets CSS property `display` to `none` for the active selector.
1689	}	Closes the current CSS rule block.
1690		Blank line used to group related style declarations.
1691	.login-form-wrapper {	Opens a CSS rule block for the selector on this line.
1692	padding: 36px 28px;	Sets CSS property `padding` to `36px 28px` for the active selector.
1693	}	Closes the current CSS rule block.

Line	Code	Explanation
1694	}	Closes the current CSS rule block.
1695		Blank line used to group related style declarations.
1696	@media (max-width: 980px) {	Starts a media query block for responsive behavior.
1697	.content {	Opens a CSS rule block for the selector on this line.
1698	padding: 24px;	Sets CSS property `padding` to `24px` for the active selector.
1699	}	Closes the current CSS rule block.
1700	}	Closes the current CSS rule block.
1701		Blank line used to group related style declarations.
1702	@media (max-width: 640px) {	Starts a media query block for responsive behavior.
1703	.content {	Opens a CSS rule block for the selector on this line.
1704	padding: 20px;	Sets CSS property `padding` to `20px` for the active selector.
1705	}	Closes the current CSS rule block.
1706		Blank line used to group related style declarations.
1707	.container-fluid > .d-flex.justify-content-between.align-items-center.mb-4,	CSS selector/value syntax line used by nearby declarations.
1708	.container-fluid > .d-sm-flex.align-items-center.justify-content-between.mb-4 {	Opens a CSS rule block for the selector on this line.
1709	padding: 14px 16px;	Sets CSS property `padding` to `14px 16px` for the active selector.
1710	}	Closes the current CSS rule block.
1711	}	Closes the current CSS rule block.
1712		Blank line used to group related style declarations.
1713	/* Confirm panels */	CSS comment used to document the stylesheet.
1714	.confirm-panel {	Opens a CSS rule block for the selector on this line.
1715	padding: 20px;	Sets CSS property `padding` to `20px` for the active selector.
1716	border-radius: var(--radius);	Sets CSS property `border-radius` to `var(--radius)` for the active selector.
1717	border: 1px solid rgba(248, 113, 113, 0.35);	Sets CSS property `border` to `1px solid rgba(248, 113, 113, 0.35)` for the active selector.
1718	background: rgba(254, 226, 226, 0.6);	Sets CSS property `background` to `rgba(254, 226, 226, 0.6)` for the active selector.
1719	margin-bottom: 20px;	Sets CSS property `margin-bottom` to `20px` for the active selector.
1720	}	Closes the current CSS rule block.
1721		Blank line used to group related style declarations.
1722	.confirm-panel ul {	Opens a CSS rule block for the selector on this line.
1723	margin: 15px 0;	Sets CSS property `margin` to `15px 0` for the active selector.
1724	}	Closes the current CSS rule block.
1725		Blank line used to group related style declarations.
1726	.confirm-actions .btn + .btn {	Opens a CSS rule block for the selector on this line.
1727	margin-left: 10px;	Sets CSS property `margin-left` to `10px` for the active selector.
1728	}	Closes the current CSS rule block.
1729		Blank line used to group related style declarations.
1730	@keyframes pulse {	Starts a keyframes animation definition.
1731	0% { box-shadow: 0 0 0 0 rgba(34, 197, 94, 0.4); }	CSS selector/value syntax line used by nearby declarations.
1732	70% { box-shadow: 0 0 0 10px rgba(34, 197, 94, 0); }	CSS selector/value syntax line used by nearby declarations.
1733	100% { box-shadow: 0 0 0 0 rgba(34, 197, 94, 0); }	CSS selector/value syntax line used by nearby declarations.

Line	Code	Explanation
1734	}	Closes the current CSS rule block.

File: templates/admin_assign.html (Lines: 102)

Line	Code	Explanation
1	{% extends 'base.html' %}	Jinja control statement ('extends'): template inheritance: chooses parent layout.
2		Blank line used to separate template sections.
3	{% block sidebar %}	Jinja control statement ('block'): starts a named block.
4	<i class="fas fa-home"></i> Dashboard	Opening HTML tag ' '.
5	<i class="fas fa-building"></i> Depts	Opening HTML tag ' '.
6	<i class="fas fa-user-graduate"></i> Students	Opening HTML tag ' '.
7	<i class="fas fa-chalkboard-teacher"></i> Teachers	Opening HTML tag ' '.
8	<i class="fas fa-book"></i> Courses	Opening HTML tag ' '.
9	<i class="fas fa-link"></i> Assign	Opening HTML tag ' '.
10	<i class="fas fa-clock"></i> Exams	Opening HTML tag ' '.
11	<i class="fas fa-money-bill"></i> Fees	Opening HTML tag ' '.
12	<i class="fas fa-bullhorn"></i> Notices	Opening HTML tag ' '.
13	<i class="fas fa-book-open"></i> Library	Opening HTML tag ' '.
14	{% endblock %}	Jinja control statement ('endblock'): ends a named block.
15		Blank line used to separate template sections.
16	{% block content %}	Jinja control statement ('block'): starts a named block.
17	<div class="container-fluid">	Opening HTML tag '<div>' and adds CSS classes.
18		Blank line used to separate template sections.
19	<div class="d-flex justify-content-between align-items-center mb-4">	Opening HTML tag '<div>' and adds CSS classes.
20	<h2 class="mb-0">Course Assignments</h2>	Opening HTML tag '<h2>' and adds CSS classes.
21	</div>	Closing HTML tag '</div>'.
22		Blank line used to separate template sections.
23	<div class="card shadow-sm mb-4">	Opening HTML tag '<div>' and adds CSS classes.
24	<div class="card-header bg-white">	Opening HTML tag '<div>' and adds CSS classes.
25	<h5 class="mb-0">Assign New Course</h5>	Opening HTML tag '<h5>' and adds CSS classes.
26	</div>	Closing HTML tag '</div>'.
27	<div class="card-body">	Opening HTML tag '<div>' and adds CSS classes.
28	<form method="POST" class="row g-3 align-items-end">	Opening HTML tag '<form>' and adds CSS classes, sets form HTTP method.
29		Blank line used to separate template sections.
30	<div class="col-md-5">	Opening HTML tag '<div>' and adds CSS classes.
31	<label class="form-label fw-bold">Select Teacher</label>	Opening HTML tag '<label>' and adds CSS classes.
32	<select name="teacher_id" class="form-select" required>	Opening HTML tag '<select>' and adds CSS classes, sets form field name, marks field as required.

Line	Code	Explanation
33	<option value="" selected disabled>Choose a teacher...</option>	Opening HTML tag `<option>`.
34	{% for t in teachers %}	Jinja control statement (`for`): starts a template loop.
35	<option value="{{ t['user_id'] }}">{{ t['full_name'] }}</option>	Opening HTML tag `<option>`.
36	{% endfor %}	Jinja control statement (`endfor`): ends a template loop.
37	</select>	Closing HTML tag `</select>`.
38	</div>	Closing HTML tag `</div>`.
39		Blank line used to separate template sections.
40	<div class="col-md-5">	Opening HTML tag `<div>` and adds CSS classes.
41	<label class="form-label fw-bold">Select Course</label>	Opening HTML tag `<label>` and adds CSS classes.
42	<select name="course_id" class="form-select" required>	Opening HTML tag `<select>` and adds CSS classes, sets form field name, marks field as required.
43	<option value="" selected disabled>Choose a course...</option>	Opening HTML tag `<option>`.
44	{% for c in courses %}	Jinja control statement (`for`): starts a template loop.
45	<option value="{{ c['course_id'] }}">{{ c['course_name'] }} ({{ c['course_code'] }})</option>	Opening HTML tag `<option>`.
46	{% endfor %}	Jinja control statement (`endfor`): ends a template loop.
47	</select>	Closing HTML tag `</select>`.
48	</div>	Closing HTML tag `</div>`.
49		Blank line used to separate template sections.
50	<div class="col-md-2">	Opening HTML tag `<div>` and adds CSS classes.
51	<button type="submit" class="btn btn-primary w-100">	Opening HTML tag `<button>` and adds CSS classes.
52	<i class="fas fa-plus-circle me-1"></i> Assign	Opening HTML tag `<i>` and adds CSS classes.
53	</button>	Closing HTML tag `</button>`.
54	</div>	Closing HTML tag `</div>`.
55		Blank line used to separate template sections.
56	</form>	Closing HTML tag `</form>`.
57	</div>	Closing HTML tag `</div>`.
58	</div>	Closing HTML tag `</div>`.
59		Blank line used to separate template sections.
60	<div class="card shadow-sm">	Opening HTML tag `<div>` and adds CSS classes.
61	<div class="card-header bg-white">	Opening HTML tag `<div>` and adds CSS classes.
62	<h5 class="mb-0">Current Assignments</h5>	Opening HTML tag `<h5>` and adds CSS classes.
63	</div>	Closing HTML tag `</div>`.
64	<div class="card-body p-0">	Opening HTML tag `<div>` and adds CSS classes.
65	<table class="table table-hover table-striped mb-0 align-middle">	Opening HTML tag `<table>` and adds CSS classes.
66	<thead class="table-dark">	Opening HTML tag `<thead>` and adds CSS classes.
67	<tr>	Opening HTML tag `<tr>`.
68	<th>Teacher Name</th>	Opening HTML tag `<th>`.
69	<th>Assigned Course</th>	Opening HTML tag `<th>`.
70	<th style="width: 150px;">Action</th>	Opening HTML tag `<th>`.
71	</tr>	Closing HTML tag `</tr>`.
72	</thead>	Closing HTML tag `</thead>`.

Line	Code	Explanation
73	<tbody>	Opening HTML tag `<tbody>`.
74	{% for a in assignments %}	Jinja control statement (`for`): starts a template loop.
75	<tr>	Opening HTML tag `<tr>`.
76	<td class="fw-bold">{{ a['full_name'] }}</td>	Opening HTML tag `<td>` and adds CSS classes.
77	<td>	Opening HTML tag `<td>`.
78	{{ a['course_code'] }}	Opening HTML tag `<span>` and adds CSS classes.
79	{{ a['course_name'] }}	Jinja expression output: injects dynamic data into HTML.
80	</td>	Closing HTML tag `</td>`.
81	<td>	Opening HTML tag `<td>`.
82	<a href="{{ url_for('delete_assignment', id=a['assign_id']) }}"	HTML line containing embedded Jinja dynamic expression(s).
83	class="btn btn-danger btn-sm"	Static HTML content rendered directly to the browser.
84	onclick="return confirm('Are you sure you want to remove this assignment?')">	Static HTML content rendered directly to the browser.
85	<i class="fas fa-trash-alt"></i> Remove	Opening HTML tag `<i>` and adds CSS classes.
86		Closing HTML tag `</a>`.
87	</td>	Closing HTML tag `</td>`.
88	</tr>	Closing HTML tag `</tr>`.
89	{% else %}	Jinja control statement (`else`): fallback conditional branch.
90	<tr>	Opening HTML tag `<tr>`.
91	<td colspan="3" class="text-center text-muted py-4">	Opening HTML tag `<td>` and adds CSS classes.
92	No courses assigned yet.	Static HTML content rendered directly to the browser.
93	</td>	Closing HTML tag `</td>`.
94	</tr>	Closing HTML tag `</tr>`.
95	{% endfor %}	Jinja control statement (`endfor`): ends a template loop.
96	</tbody>	Closing HTML tag `</tbody>`.
97	</table>	Closing HTML tag `</table>`.
98	</div>	Closing HTML tag `</div>`.
99	</div>	Closing HTML tag `</div>`.
100		Blank line used to separate template sections.
101	</div>	Closing HTML tag `</div>`.
102	{% endblock %}	Jinja control statement (`endblock`): ends a named block.

File: templates/admin_delete_student_confirm.html (Lines: 46)

Line	Code	Explanation
1	{% extends 'admin_dash.html' %}	Jinja control statement ('extends'): template inheritance: chooses parent layout.
2	{% block content %}	Jinja control statement ('block'): starts a named block.
3	<div class="container-fluid">	Opening HTML tag '<div>' and adds CSS classes.
4	<div class="d-flex justify-content-between align-items-center mb-4">	Opening HTML tag '<div>' and adds CSS classes.
5	<h3 class="mb-0">Confirm Force Delete Student</h3>	Opening HTML tag '<h3>' and adds CSS classes.
6	Back to Students	Opening HTML tag '<a>' and adds CSS classes, defines a link target.
7	</div>	Closing HTML tag '</div>'.
8		Blank line used to separate template sections.
9	<div class="card">	Opening HTML tag '<div>' and adds CSS classes.
10	<div class="card-body">	Opening HTML tag '<div>' and adds CSS classes.
11	<div class="confirm-panel">	Opening HTML tag '<div>' and adds CSS classes.
12	Warning: This action is irreversible and will permanently delete the student and ALL related data!	Opening HTML tag '<p>'.
13		Blank line used to separate template sections.
14		Opening HTML tag ' '.
15	Student: {{ student.full_name }} ({{ student.email }})	Opening HTML tag ' '.
16	Department: {{ student.dept_name or 'N/A' }}	Opening HTML tag ' '.
17	Semester: {{ student.semester or 'N/A' }}	Opening HTML tag ' '.
18		Closing HTML tag ' '.
19		Blank line used to separate template sections.
20	{% if total_related > 0 %}	Jinja control statement ('if'): starts a conditional branch.
21	Related records that will also be deleted ({{ total_related }} total):	Opening HTML tag '<p>'.
22		Opening HTML tag ' '.
23	Borrowed books: {{ counts.borrowes }}	Opening HTML tag ' '.
24	Fee records: {{ counts.fees }}	Opening HTML tag ' '.
25	Exam results: {{ counts.results }}	Opening HTML tag ' '.
26	Attendance records: {{ counts.attendance }}	Opening HTML tag ' '.
27	Course enrollments: {{ counts.enrollments }}	Opening HTML tag ' '.
28		Closing HTML tag ' '.
29	{% else %}	Jinja control statement ('else'): fallback conditional branch.
30	No related records found.	Opening HTML tag '<p>'.
31	{% endif %}	Jinja control statement ('endif'): ends conditional block.
32	</div>	Closing HTML tag '</div>'.
33		Blank line used to separate template sections.
34	<form method="POST" class="confirm-actions">	Opening HTML tag '<form>' and adds CSS classes, sets form HTTP method.
35	<button type="submit" class="btn btn-danger">	Opening HTML tag '<button>' and adds CSS classes.
36	<i class="fas fa-trash-alt me-2"></i>Confirm: Delete Student & All Related Data	Opening HTML tag '<i>' and adds CSS classes.

Line	Code	Explanation
37	</button>	Closing HTML tag `</button>`.
38		Opening HTML tag `<a>` and adds CSS classes, defines a link target.
39	Cancel	Static HTML content rendered directly to the browser.
40		Closing HTML tag `</a>`.
41	</form>	Closing HTML tag `</form>`.
42	</div>	Closing HTML tag `</div>`.
43	</div>	Closing HTML tag `</div>`.
44	</div>	Closing HTML tag `</div>`.
45		Blank line used to separate template sections.
46	{% endblock %}	Jinja control statement (`endblock`): ends a named block.

File: .env.example (Lines: 11)

Line	Code	Explanation
1	SECRET_KEY=change_me_to_a_long_random_value	Sets `SECRET_KEY` (Flask secret key used for signing session cookies) with example value `change_me_to_a_long_random_value`.
2	DB_HOST=localhost	Sets `DB_HOST` (database host name) with example value `localhost`.
3	DB_USER=root	Sets `DB_USER` (database username) with example value `root`.
4	DB_PASSWORD=	Sets `DB_PASSWORD` (database password) with empty value.
5	DB_NAME=ncit_sis	Sets `DB_NAME` (database/schema name) with example value `ncit_sis`.
6	DB_PORT=3306	Sets `DB_PORT` (database TCP port) with example value `3306`.
7		Blank line used to separate groups of environment variables.
8	# Optional seed values	Comment describing the environment variables below.
9	SEED_ADMIN_NAME=System Admin	Sets `SEED_ADMIN_NAME` (default name for seeded admin account) with example value `System Admin`.
10	SEED_ADMIN_EMAIL=admin@ncit.edu.np	Sets `SEED_ADMIN_EMAIL` (default email for seeded admin account) with example value `admin@ncit.edu.np`.
11	SEED_ADMIN_PASSWORD=admin123	Sets `SEED_ADMIN_PASSWORD` (default password for seeded admin account) with example value `admin123`.